

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of

CERTAIN WEARABLE DEVICES

Inv. No. 337-TA-1478

COMPLAINANTS' PRE-HEARING STATEMENT

[REDACTED]

Pursuant to Ground Rule 11.1, Complainants Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. (collectively, “Complainants”) respectfully submit this pre-hearing statement.

I. WITNESSES

Complainants reserve the right to call additional witnesses necessitated by any of the Administrative Law Judge’s pre-hearing or hearing rulings; additional witnesses to provide foundational testimony if any party contests the authenticity or admissibility of material proffered at the evidentiary hearing; substitute witnesses if one of Complainants’ anticipated witnesses becomes unavailable; any witness identified by Respondents in their Pre-Hearing Statement; additional witnesses to respond to issues raised after the submission of this list; and/or additional rebuttal witnesses to respond to evidence or testimony presented at the evidentiary hearing by Respondents that could not reasonably have been anticipated as of the date of this list.

Further, Complainants intend to rely on the designated deposition testimony of all witnesses deposed in this matter (including those identified in Complainants’ Pre-Hearing Brief) and who do not appear live at the hearing. Complainants reserve the right to call any deposition witness live in the event that individual’s designated deposition testimony is not admitted into the record.

A. Expert Witnesses

Complainants presently intend to call the following expert witnesses during the evidentiary hearing.

1. Dr. Gregory Francis Welch

Dr. Gregory Francis Welch has a Ph.D. in computer science from the University of North Carolina at Chapel Hill. Dr. Welch is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric

signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Dr. Welch's curriculum vitae, attached hereto as **Exhibit 1**, sets forth his qualifications.

Dr. Welch is expected to testify regarding technical matters relating to one or more of the Asserted Patents, including at least the subject matter of the Asserted Patents; the functionality, design, and operation of Respondents' products accused of infringement in this Investigation; the functionality, design, and operation of Complainants' products contended to satisfy the technical prong of the domestic industry requirement; the state of the art at the time of the invention of the Asserted Patents; the validity and enforceability of the Asserted Patents; and the construction of claims and claim terms of the Asserted Patents. Additionally, Dr. Welch may rebut expert witness testimony of Respondents' experts.

2. Dr. Steven Warren

Dr. Steven Warren has a Ph.D. in electrical engineering from the University of Texas at Austin. Dr. Warren is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Dr. Warren's curriculum vitae, attached hereto as **Exhibit 2**, sets forth his qualifications.

Dr. Warren is expected to testify regarding technical matters relating to one or more of the Asserted Patents, including at least the subject matter of the Asserted Patents; the functionality, design, and operation of Respondents' products accused of infringement in this Investigation; the

[REDACTED]

functionality, design, and operation of Complainants' products contended to satisfy the technical prong of the domestic industry requirement; the state of the art at the time of the invention of the Asserted Patents; the validity and enforceability of the Asserted Patents; and the construction of claims and claim terms of the Asserted Patents. Additionally, Dr. Warren may rebut expert witness testimony of Respondents' experts.

3. Julia Rowe

Ms. Julia Rowe has a Master's degree in accountancy from DePaul University and is a Vice President at Charles River Associates. Ms. Rowe is an expert in some of the matters at issue in this case, including but not limited to the application of economics to intellectual property and international trade and has extensive professional experience in applying financial, accounting, and economic principles to complex business disputes, including significant experience in disputes involving intellectual property and matters brought before the International Trade Commission. Ms. Rowe's curriculum vitae, attached hereto as **Exhibit 3**, sets forth her qualifications. Ms. Rowe is expected to testify regarding matters relating to proof of the economic prong of the domestic industry requirement.

B. Fact Witnesses

Complainants presently intend to call the following fact witnesses during the evidentiary hearing.

1. Christopher Martinez

Mr. Christopher Martinez is the Director of Product Marketing at Samsung Electronics America, Inc. He may testify about Samsung's domestic investments in the activities related to the Domestic Industry Products contended to practice the Asserted Patents. Mr. Martinez may be contacted through counsel for Complainants.



2. Ho Jong Kim

Mr. Ho Jong Kim is a Staff Engineer at Samsung Electronics Co., Ltd. He may testify about the Domestic Industry Products contended to practice the '789 Asserted Patent. Mr. Kim may be contacted through counsel for Complainants.

3. Dr. Hyunjun Jung

Dr. Hyunjun Jung is a Principal Engineer at Samsung Electronics Co., Ltd. He may testify about algorithms used by the Domestic Industry Products contended to practice the '677, '675, and '359 Asserted Patents. Dr. Jung may be contacted through counsel for Complainants.

4. Sukwang Lim

Mr. Sukwang Lim is a Staff Engineer at Samsung Electronics Co., Ltd. He may testify about algorithms used by the Domestic Industry Products contended to practice the '677, '675, and '359 Asserted Patents. Mr. Lim may be contacted through counsel for Complainants.

II. EXHIBIT LIST

A copy of Complainants' Amended Trial Exhibit List is attached hereto as **Exhibit 4**. Complainants reserve the right to use and rely upon any exhibit submitted by Respondents, including any exhibit identified on Respondents' exhibit lists (including exhibits that are unused or withdrawn by Respondents).

III. STIPULATIONS

The parties are continuing to discuss but have not, at present, agreed to any stipulations concerning the hearing in this investigation. The parties have otherwise entered into the following stipulation(s) in this Investigation:

- Joint Stipulation Regarding Importation and Inventory Between Complainants and Respondents (June 1, 2026) (EDIS Doc. ID 883875).

IV. PRE-HEARING CONFERENCE AGENDA

Respondents propose that the pre-hearing conference address the following items (to the extent they are not resolved or mooted before the pre-hearing conference):

- discussion of Your Honor’s rules and expectations for trial;
- discussion of the trial procedures, disclosures, and witness schedule, if necessary;
- resolution of any outstanding motions *in limine*;
- procedure for the admission of deposition designations and associated exhibits; and
- resolution of any other critical matters and pending motions.

V. ESTIMATED DATE AND APPROXIMATE LENGTH FOR EACH WITNESS

The parties have negotiated the approximate witness order and approximate length of examination of each witness.

The table below presents the witnesses in the order Complainants propose they appear and the parties’ current estimate of total time (in minutes) for witness testimony (including direct, cross, and any redirect examination). Complainants respectfully reserve the right to alter the order of their witnesses and length of time taken on any witness examination, as facts and circumstances dictate. Complainants also reserve the right to call any witnesses in Respondents’ case-in-chief as adverse witnesses.

Witness Name	Estimated Call Date	Complainant Time (mins)	Respondent Time (mins)
Opening	September 16, 2026	30	15
Complainants’ Case-in-Chief			
Christopher Martinez	September 16, 2026	60	30
Ho Jong Kim	September 16, 2026	30	20
Dr. Hyunjun Jung	September 16, 2026	30	20
Sukwang Lim	September 16, 2026	30	20
Dr. Gregory Welch	September 17, 2026	120	30
Dr. Stephen Warren	September 17, 2026	90	30
Julia Rowe	September 18, 2026	90	45
Respondents’ Case-in-Chief/Rebuttal			
Miguel Christophy	September 18, 2026	30	60

Witness Name	Estimated Call Date	Complainant Time (mins)	Respondent Time (mins)
Michael Larson	September 18, 2026	30	60
Dr. Patrick Mercier	September 18, 2026	45	60
John Jarosz	September 20, 2026	45	90
Dr. Bobak Mortazavi	September 20, 2026	120	300
Complainants' Rebuttal Case			
Dr. Gregory Welch	September 21, 2026	90	60
Dr. Stephen Warren	September 21, 2026	60	60
TOTALS		900	900

VI. CERTIFICATION OF GOOD FAITH EFFORTS TO SETTLE THE INVESTIGATION

Complainants certify that they have engaged in good faith efforts to settle this dispute with Respondents. As of the submission of this pre-hearing statement, Complainants have participated in mediation sessions with Respondents but have not reached a settlement agreement.

Dated: August 10, 2026

Respectfully submitted,

/s/ D. Sean Trainor

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*Counsel for Complainants Samsung
Electronics Co., Ltd. and Samsung
Electronics America, Inc.*

EXHIBIT 1

GREGORY FRANCIS WELCH

Curriculum Vitae — June 8, 2026

CONTACT INFORMATION

University of Central Florida

College of Nursing

12201 Research Parkway

Orlando, FL 32826-3298 (USA)

Computer Science

4000 Central Florida Boulevard

Orlando, FL 32816-2362 (USA)

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3100 Technology Parkway

Orlando, FL 32826-3281 (USA)

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PROFESSIONAL EXPERIENCE (SUMMARY)

- 2011–present **University of Central Florida**
Pegasus Professor (2020)
AdventHealth Endowed Chair in Healthcare Simulation (2013)
Professor, College of Nursing, Academic Health Sciences Center (2013)
Professor, Computer Science, College of Engineering and Computer Science (2011)
Professor, Institute for Simulation & Training (2011)
Co-Director, IST Synthetic Reality Lab (2011)
Faculty, Graduate Certificate in Mixed Reality Engineering (2020)
Faculty, Modeling & Simulation Graduate Program (2011)
Faculty, Interactive Computing Experiences Research Cluster (2014)
- 1996–2020 **University of North Carolina at Chapel Hill**
Adjunct Professor, Computer Science (2012-2020)
Research Professor (Assistant, Associate), Computer Science (1996–2012)
UNC Site Coordinator, NSF Graphics and Visualization Center (1996–1998)
- 1990–1992 **Northrop Defense Systems Division**
Senior Engineer, Airborne Electronic Countermeasures, Digital Systems Group
- 1987–1990 **NASA Jet Propulsion Laboratory (California Institute of Technology)**
Voyager Spacecraft Project, Flight Command and Data Management Section

EDUCATION

- May 1997 **Ph.D., Computer Science**
University of North Carolina at Chapel Hill, Chapel Hill, NC
Under the direction of Gary Bishop
- May 1995 **M.S., Computer Science**
University of North Carolina at Chapel Hill, Chapel Hill, NC
- May 1986 **B.S. with *Highest Distinction*, Electrical Engineering Technology**
Purdue University, West Lafayette, IN

PERSONAL AWARDS

- **IEEE Virtual Reality Service Award**, IEEE Visualization and Graphics Technical Committee (VGTC), “in recognition of his service to the AR/VR community in organizing conferences, facilitating life-long learning through workshops, seminars, and tutorials, and supporting young researchers through his role as a founder of the XR Future Faculty Forum,” March 23, 2026 during the 33rd IEEE Virtual Reality Conference in Daegu, Korea.
- **IEEE Virtual Reality Academy**, IEEE Visualization and Graphics Technical Committee, March 13, 2022 during the 28th IEEE Virtual Reality Conference in Christchurch, New Zealand.
- **Scroll & Quill Society**, the University of Central Florida, October 1, 2021. The award recognizes “sustained and outstanding achievements in research and/or creative activities” at UCF.
- **Pegasus Professor**, the University of Central Florida, April 1, 2020. The award recognizes faculty excellence in the three primary areas of academic endeavor: teaching, research and service. There were approximately 50 Pegasus Professors at UCF in 2020.
- **IEEE Virtual Reality Technical Achievement Award**, IEEE Visualization and Graphics Technical Committee (VGTC), “in recognition for his contributions to human motion tracking and to mixed reality applications in medicine and training,” March 21, 2018 during the 25th IEEE Virtual Reality Conference in Reutlingen, Germany.
- **Luminary Award**, University of Central Florida, for “exceptional faculty whose work is advancing their discipline and making a difference,” October 18, 2017.
- **Research Incentive Award (RIA)**, University of Central Florida, for “outstanding research, scholarly, or creative activity that advances the body of knowledge in a particular field, including interdisciplinary research and collaborations,” March 7, 2017.
- **Outstanding Performance as Co-Chairman of the IEEE VR 2014 Conference**, IEEE Computer Society, April 3, 2014.
- **Outstanding Performance as Co-Chairman of the IEEE VR 2013 Conference**, IEEE Computer Society, March 18, 2013.
- **Rock StAR Award**, 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), for teaching role in an introductory course on Augmented Reality (AR) for approximately 40 Mexican students interested in AR, September 20, 2016. Mérida, México.
- **Excellence in Teaching Award**, UNC-Chapel Hill Computer Science Student Association, for 3D Computer Modeling and Animation, Spring 2007.
- **Outstanding Senior Design Project Award**, “The Easy Chair: A Microprocessor-Controlled Wheelchair for Children With Muscular Disorders,” E.T., Purdue University, 1986

JOINT AWARDS

- **Virtual Simulation Innovation Award** to the UCF CON STIM Center, January 13, 2025.
- **Best Paper Award**, “Intuitive user interfaces for real-time magnification in augmented reality,” with Ryan Schubert and Gerd Bruder, 29th ACM Symposium on Virtual Reality Software and Technology (VRST), October 9, 2023.
- **TechConnect Innovation Award**, “Intelligent Object Magnification for Augmented Reality Displays,” with Dr. Gerd Bruder, Zubin Choudhary, and Dr. Kangsoo Kim. 2021 TechConnect Innovation Showcase, October 18-21, 2021, Washington, DC. USA
- **Best Demo Award**, “Real Time Magnification in Augmented Reality,” with Zubin Choudhary, Jesus Ugarte, and Gerd Bruder. ACM Spatial User Interaction 2021, November 9–10, 2021 (Virtual).

- **Best Demo Audience Choice Award**, “Towards Interactive Virtual Dogs as a Pervasive Social Companion in Augmented Reality,” with Nahal Norouzi, Kangsoo Kim, and Gerd Bruder. International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE 2020).
- **Best Long Paper**, “Effects of shared gaze parameters on visual target identification task performance in augmented reality,” with Nahal Norouzi, Austin Erickson, Kangsoo Kim, Ryan Schubert, Joseph J. LaViola Jr., Gerd Bruder, in *Proceedings of the ACM Symposium on Spatial User Interaction (SUI)*, page 11, New Orleans, LA USA, October 20 2019. Association of Computing Machinery.
- **Best Paper**, “The physical-virtual table: Exploring the effects of a virtual human’s physical influence on social interaction,” with Myungho Lee, Nahal Norouzi, Gerd Bruder, and Pamela J. Wisniewski. 24th ACM Symposium on Virtual Reality Software and Technology; Tokyo, Japan; December 1, 2018.
- **Best Paper Honorable Mention**, “Blowing in the Wind: Increasing Copresence With a Virtual Human via Airflow Influence in Augmented Reality,” with Kangsoo KIM and Gerd BRUDER. ICAT-EGVE 2018 - International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments; Limassol, Cyprus; November 9 2018.
- **Best Student Paper**, 2017 ACM Symposium on Virtual Reality Software (VRST 2017), “Exploring the effects of observed physicality conflicts on real-virtual human interaction in augmented reality,” by Kangsoo Kim (student advisee), Gerd Bruder, and Greg Welch; Gothenberg, Sweden; November 10, 2017.
- **Long Lasting Impact Paper**, 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), for a pair of 1998 papers on “Spatial Augmented Reality” (SAR), with co-authors Prof. Ramesh Raskar, Prof. Henry Fuchs, and Deepak Bandyopadhyay; Mérida, México; September 20, 2016.
- **Second Best Paper** at 5th ACM/IEEE International Conference on Distributed Smart Cameras; Ghent, Belgium; August 25, 2011.
- **Best Paper**, ACM Symposium on Virtual Reality Software and Technology 1999 (VRST 99); London, England, United Kingdom; December 22, 1999.

ADVISEE AWARDS (WHEN ADVISEES)

- Jonathan Jules (undergraduate advisee), **National Action Council for Minorities in Engineering (NACME) Scholarship** for 2019–2020, September 30, 2019.
- Salam Daher (Ph.D. advisee), **Collegiate Award** from the National Center for Women in Technology (NCWIT) — four recipients total, May 15, 2018, Grapevine, TX USA.
- Salam Daher (Ph.D. advisee), **RADM Fred Lewis Scholarship Award**, Interservice/Industry Training, Simulation and Education Conference (I/ITSEC), August 5, 2017.
- Salam Daher (Ph.D. advisee), **Link Foundation Fellowship in Modeling, Simulation, and Training** for the 2016–2017 academic year, March 16, 2016.
- Joshua Morton (undergraduate advisee), **Rodney F. Hood Undergraduate Research Award** in support of “Tremor Sensing and Quantification” (November 15, 2010–June 30, 2011).

OTHER HONORS (SELECTED)

- **Keynote Talks** and **Other Invited Talks** as indicated below.
- **Fellow**, Institute of Electrical and Electronics Engineers (IEEE), January 1, 2022.
- **Fellow**, National Academy of Inventors (NAI), December 7, 2021.
- **Member**, Academy of Science, Engineering and Medicine of Florida (ASEMFL), November 7, 2025.
- Internationally-recognized **Kalman filter** web site (<http://www.cs.unc.edu/~welch/kalman/>) and associated article “An Introduction to the Kalman Filter,” UNC Technical Report TR95-041, **cited 12,990 times** according to Google as of 08 JUN 2026.
- **Prof. Rudolph Kalman personally invited Prof. Welch** to attend the National Academy of Engineering ceremony where he was awarded the 2008 Charles Stark Draper Prize for the development of what became known as the **Kalman Filter**.
- **IEEE Technical Expert** for Virtual, Augmented and Mixed Reality, IEEE Public Visibility Initiative, 2018–present.
- **Member**, World Economic Forum (WEF), Metaverse Governance Working Group (2022–2025).
- **Member**, World Economic Forum (WEF), Global Future Councils, Council on Virtual Reality and Augmented Reality (2018–2022).
- **Member**, National Council of Architectural Registration Boards (NCARB), Futures Task Force (2017–2019).
- **Certificate of Appreciation** for two-day course on an “Introduction to the Kalman Filter” for Autonomy Incubator Short Course Series (AISCs), NASA Langley Research Center (LaRC), Hampton, Virginia, November 5-6, 2014,
- **Featured exhibit**, Collaborations: Humanities, Arts & Technology (CHAT) Festival, “The Bathysphere: Motion Capture as Art,” with Prof. Francesca Talenti, February 16–20, 2010, UNC-Chapel Hill.

PROFESSIONAL SERVICE

- Professional Associations
 - Fellow, Institute of Electrical and Electronics Engineers (IEEE)
 - Fellow, National Academy of Inventors (NAI)
 - Vice Chair for Conferences, IEEE Visualization and Graphics Technical Community (VGTC) Executive Committee
 - Member, Association for Computing Machinery (ACM)
 - Pioneer, ACM Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH)
 - Member, European Association for Extended Reality (EuroXR)
 - Member, European Association for Computer Graphics (Eurographics)
 - Member, International Virtual Reality Healthcare Association (Founding Member)
 - Member, Academy of Science, Engineering and Medicine of Florida (ASEMFL)
 - Member, Society for Simulation in Healthcare (SSIH)
 - Member, International Nursing Association of Clinical Simulation & Learning (INACSL)
 - Member, Southern Nursing Research Society (SNRS)

- Event Organization

- Dagstuhl Seminar on “Beyond XR” (Seminar 27151), co-organized with Misha Sra (University of California - Santa Barbara, US), Jeanine Stefanucci (University of Utah - Salt Lake City, US), and Frank Steinicke (Universität Hamburg, DE), April 11–16, 2027, The International Conference and Research Center for Computer Science, Dagstuhl, Germany.
- IEEE Virtual Reality 2026 (VR 2026), Extended Reality for the Operating Room (XR4OR), with Gabriel Zachmann, Peter Haddawy, Ron Kikinis, Sudanthi Wijewickrema, and Torsten Kuhlen. March 22, 2026, Daegu, Korea.
- IEEE Virtual Reality 2026 (VR 2026), 2nd International Workshop on Real And Virtual Spaces Influences (REVISI), with Laura Battistel, Kangsoo Kim, Riccardo Parin, Massimiliano Zampini, and Gerd Bruder. March 21, 2026, Daegu, Korea.
- AIXVR 2027, General Co-Chair with Wolfgang Hürst and Phil Sheu, January–February 2027, Vancouver, Canada.
- SIBGRAPI 2025, with Gustavo Carneiro, João Paulo Papa, and Manuel M. Oliveira. September 30 – October 3, 2025, Salvador, Bahia, Brazil.
- IEEE Virtual Reality 2025 (VR 2025), 1st Workshop on XR Accessibility (XRACCESS), with Bobby Bodenheimer, Gerd Bruder, Dylan Fox, Anne-Hélène Olivier, Jeanine Stefanucci, and Daniel Zielasko. March 9, 2025, Saint-Malo, France.
- IEEE Virtual Reality 2025 (VR 2025), 1st International Workshop on Real And Virtual Spaces Influences (REVISI), with Laura Battistel, Kangsoo Kim, Riccardo Parin, Massimiliano Zampini, and Gerd Bruder. March 8, 2025, Saint-Malo, France.
- IEEE Virtual Reality 2024 (VR 2024), Co-General Chair with Carolina Cruz-Neira (University of Central Florida), March 16–21, 2024, Orlando, FL, USA.
- EuroXR 2023 General Co-Chair with Patrick Bourdot (University Paris-Saclay, France) and Benjamin de Witt (Utrecht, The Netherlands), November 28–30, 2023, Rotterdam, The Netherlands.
- Workshop on Inclusive and Generalizable Human Subject Research in XR, ISMAR 2022, with Stevie Carnell, Gerd Bruder, Jeanine Stefanucci, Sun Joo (Grace) Ahn, Valerie Taylor, Tabitha Peck, and Jeremy Bailenson. October 17, 2022, Singapore, Singapore (and virtual).
- Workshop on Considerations for Research Studies in XR, with Nahal Norouzi (University of Central Florida), Advertising in the Metaverse, 2022 American Academy of Advertising Annual Conference, March 24, 2022, St. Petersburg, FL USA.
- IEEE Virtual Reality 2020, Best Paper awards committee, November 22–26, 2020. Atlanta, Georgia, USA. (Held on-line due to Covid-19.)
- Extended Reality for Good (XR4Good) workshop, 18th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2019), co-organized with Arindam Dey (University of Queensland), Mark Billingham (University of South Australia), Gun Lee (University of South Australia), and Stephan Lukosch (TU Delft). October 2019, Beijing, China.
- VR 2019 Program Committee Co-Chair, IEEE Virtual Reality 2019 (Osaka, Japan), with Bruce Thomas (University of South Australia in Adelaide, Australia), Torsten Kuhlen (RWTH Aachen University, Aachen, Germany), and Rob Teather (Carleton University, Ottawa, Canada).
- Augmented Reality Visioning Workshop (ARVW 2018), with Gerd Bruder (University of Central Florida) and Peter Squire (Office of Naval Research), July 30–August 3, 2018, Warrenton, VA, USA.

- VR 2018 Program Committee Co-Chair, IEEE Virtual Reality 2018 (Germany), with Kiyoshi Kiyokawa (Osaka University, Japan), Frank Steinicke (University of Hamburg, Germany), and Bruce Thomas (University of South Australia in Adelaide, Australia).
- Virtual and Augmented Reality for Good (VAR4Good) workshop, 25th IEEE Conference on Virtual Reality and 3D User Interfaces (VR 2018), co-organized with Bruce Thomas, Arindam Dey, and Mark Billinghurst (all University of South Australia in Adelaide, Australia), March 18, 2018, Reutlingen, Germany.
- Augmented Reality for Good (AR4Good) workshop, 16th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2017), co-organized with Arindam Dey and Mark Billinghurst (both University of South Australia in Adelaide, Australia), October 9, 2017, Nantes, France.
- Combined 26th International Conference on Artificial Reality and Telexistence (ICAT 2016) and 21st Eurographics Symposium on Virtual Environments (ICAT-EGVE 2016) *Best Paper Award* committee, December 7–9, 2016. Little Rock, Arkansas, USA.
- ACM Symposium on Virtual Reality Software and Technology (VRST 2016) *Best Paper Award* committee, November 2–4, 2016. Munich, Germany.
- 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), co-organized “AR101,” an introductory course on Augmented Reality (AR) for a group of approx. 40 Mexican students interested in AR. September 20, 2016. Mérida, México.
- 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), **Best Paper Award** committee, September 19–23, 2016. Mérida, México.
- ISMAR 2016 Outreach Co-Chair, IEEE International Symposium on Mixed and Augmented Reality (ISMAR2016), Mérida, México, September 19–23, 2016.
- Workshop “Human Perception and Psychology in Augmented Reality (HPPAR),” with Bruce H. Thomas (The University of South Australia Mawson Lakes) and James Baumeister (University of South Australia), IEEE International Symposium on Mixed and Augmented Reality (ISMAR2015), September 29, 2015, Fukuoka, Japan.
- Dagstuhl Seminar on “Virtual Realities” (Seminar 13241), co-organized with Guido Brunnett (TU Chemnitz, Germany), Sabine Coquillart (INRIA Rhône-Alpes, France), and Robert van Liere (Center for Mathematics and Computer Science, Netherlands), June 10–14, 2013, The International Conference and Research Center for Computer Science, Dagstuhl, Germany.
- IEEE Virtual Reality 2013 (VR 2013), Co-General Chair with Ben Lok (University of Florida), March 16–23, 2013, Orlando, FL, USA.
- 11th International Symposium on Mixed and Augmented Reality (ISMAR 2012), Co-General Chair with Blair MacIntyre (Georgia Tech), November 5–8, 2012, Atlanta, Georgia, USA.
- Joint Virtual Reality Conference 2011 (JVRC 2011), International Program Committee Co-chair with Sabine Coquillart (INRIA Rhône-Alpes, France) and Anthony Steed (University College of London, UK).
- 10th International Symposium on Mixed and Augmented Reality (ISMAR 2011), Program Committee Co-Chair with Gerhard Reitmayr (Graz University of Technology) and Jun Park (Hongik University).
- VR 2011 Research Demos Co-Chair, IEEE Virtual Reality 2011 (Singapore), with Dirk Reiners (University of Louisiana at Lafayette, USA), Henry Duh (National University of Singapore), and Jiung-Yao Huang (National Taipei University, Taiwan).
- VR 2010 Research Demos Co-Chair, IEEE Virtual Reality 2010 (Waltham, MA, USA), with Yoshifumi Kitamura (Osaka University, Japan), Laura Monroe (Los Alamos National

Laboratories, USA), Dirk Reiners (University of Louisiana at Lafayette, USA), and Simon Richir (Laval Virtual, France).

- Dagstuhl Seminar on “Virtual Realities” (Seminar 08231), co-organized with Guido Brunnett (TU Chemnitz, Germany) and Sabine Coquillart (INRIA Rhône-Alpes, France), June 1–6, 2008, The International Conference and Research Center for Computer Science, Dagstuhl, Germany.
- VR 2007 Local Arrangements Co-Chair, IEEE Virtual Reality 2007 (Charlotte, NC, USA), with Zachary Wartell (UNC-Charlotte, USA), Sabarish Babu (UNC-Charlotte, USA), and Regis Kopper (Virginia Tech, USA).
- Workshops/Tutorials Chair, 5th IEEE and ACM International Symposium on Mixed and Augmented Reality (Santa Barbara, CA, USA).
- Co-Chair with Chris Jaynes (University of Kentucky) of IEEE CVPR 2006 International Workshop on Projector-Camera Systems, (ProCams 2006), New York, NY, USA.
- 3DPVT 2006 Organizing Committee, Third International Symposium on 3D Data Processing, Visualization and Transmission (Chapel Hill, NC, USA).
- EDT 2006 Co-chair, IEEE Virtual Reality 2006 International Workshop on Emerging Display Technologies (Alexandria, VA, USA), with Mark Bolas (USC and Fakespace Labs) and Andreas Simon (FH-Aargau).
- ICAT 2005 Program Co-Chair, 15th International Conference on Artificial Reality and Telexistence (Christchurch, New Zealand), with Sang Chul Ahn (KIST, Korea), and Haruo Noma (ATR, Japan).
- ProCams 2005 Posters Chair, IEEE CVPR 2005 International Workshop on Projector-Camera Systems (San Diego, CA, USA).
- EDT 2005 Co-chair, IEEE Virtual Reality 2005 International Workshop on Emerging Display Technologies (Bonn, Germany), with Mark Bolas (USC and Fakespace Labs) and Andreas Simon (Fraunhofer IMK).
- I3D 2001 Registration Chair, ACM Symposium on Interactive 3D Graphics (Chapel Hill, NC, USA).

- Boards and Committees

- Member, World Economic Forum (WEF), Metaverse Governance Working Group (2022–present).
- Member, World Economic Forum (WEF), Global Future Councils, Council on Virtual Reality and Augmented Reality (2018–2022)
- Founding Member, International Virtual Reality Healthcare Association, Advisory Board (2019–present)
- Emeritus Member, International Symposium for Mixed & Augmented Reality, Steering Committee (2012–2014, Emeritus 2014–present)
- Member, UCF IST Modeling & Simulation Graduate Program, Advisory Committee (2012–present)
- Member, UCF Information Technology Resource, Advisory Committee (2014–present)
- Member, UCF Office of Research, COVID-19 Lab Reopening Task Force (2020–2021)
- Member, National Council of Architectural Registration Boards (NCARB), Futures Task Force (2017–2019)
- Senior Advisor, TrakMark Working Group — SIG MR and VR Society of Japan (past)
- Member, Board of Directors, HiBall Tracker, Inc. (past)
- Member, UNC Committee on Student Conduct (past)
- Member, UNC Computer Science Committee on Curriculum and Planning (past)

- Reviewer, Committee, and Editorial Activity
 - Reviewer/panelist, National Science Foundation (NSF), U.S. Department of Energy (DOE), European Research Council, and other funding organizations/units
 - Reviewer, NSF Computing Innovation Fellows
 - Associate Editor, PRESENCE: Virtual and Augmented Reality (2002–present)
 - Associate Editor, Frontiers in Virtual Reality (2014–present)
 - Editorial Board, Virtual Reality & Intelligent Hardware (2019–present)
 - Editorial Board, Journal of Virtual Reality (past)
 - Reviewer, ACM: International Conference on Graphics and Interactive Techniques (SIGGRAPH); Symposium on Virtual Reality Software and Technology (VRST); Symposium on Interactive 3D Graphics; Multimedia; User Interface Software and Technology (UIST); Transactions on Mathematical Software;
 - IEEE: Computer Graphics and Applications (CG&A); International Symposium on Mixed and Augmented Reality (ISMAR) — Area Chair; Virtual Reality (VR); Transactions on Visualization and Computer Graphics; Transactions on Aerospace and Electronic Systems; Conference on Computer Vision and Pattern Recognition (CVPR)
 - Guest Editor, Special Issue, Special Section on Pattern Recognition Letters with the Best Papers from SIBGRAPI 2025, Elsevier, 2025.
 - International Symposium on 3D Data Processing, Visualization, and Transmission (3DPVT)
 - International Symposium on Ubiquitous Virtual Reality (ISUVR)
 - IEEE/ACM International Symposium on Wearable Computers
 - The combined International Conference on Artificial Reality & Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE)
 - EURASIP Journal of Applied Signal Processing; Transactions on Automatic Control
 - Journal of Optimal Control Applications and Methods
 - Eurographics Workshop on Virtual Environments
 - New Immersive Displays for the Near Future (Laval Virtual 2011 Symposium)
- Developer of popular free Mac OS X Mail plugins MailFollowUp and MailRecent
- Testifying intellectual property expert witness / consultant in areas including human-computer interaction, virtual and augmented reality, handheld devices, position and motion estimation, sensor fusion, interactive electronic systems, and related technology.

KEYNOTE/PLENARY TALKS

- International Virtual Reality Healthcare Association (IVRHA) Global Symposium 2026, Tampa, FL, USA, February 27, 2026.
- SIBGRAPI 2025, Opening Keynote, Salvador, Brazil, October 1, 2025.
- Purdue University, Purdue Polytechnic, Dean's Lecture Series, West Lafayette, IN, April 17, 2025.
- ISTEc Distinguished Lecture, Colorado State University, Fort Collins, CO, November 13, 2023.
- IEEE Life Member Affinity Group Inaugural Meeting, Orlando, FL, December 21, 2022.
- 2nd Workshop on Replication in Extended Reality (WoR XR), ISMAR 2022, Singapore, Singapore, October 17, 2022.
- AI Colloquium, Pusan National University, December 2, 2021.
- Hoag Advances in Clinical Virtual Reality Symposium, October 1, 2021.

- Training XR: Workshop on 3D Content Creation for Simulated Training in eXtended Reality, IEEE Virtual Reality 2020, on line (originally Atlanta, GA USA), March 22, 2020.
- ChinaVR 2019, Shenzhen, China, November 24, 2019.
- 6th ACM Symposium on Spatial User Interaction, Berlin, Germany, October 13–14, 2018.
- New Knowledge and Innovations, Sarasota Memorial Health Care System, Sarasota, Florida, May 4, 2018.
- International Conference on Computer-Aided Design and Computer Graphics (CAD/Graphics), Zhangjiajie, China, August 26, 2017.
- Society for Design and Process Science conference (SDPS), Orlando, Florida, December 5, 2016.
- Annual Meeting of the Association for Information Science and Technology (ASIS&T), Copenhagen, Denmark, October 16, 2016.
- 22nd Annual Research Day, Sigma Theta Tau International Honor Society of Nursing, Theta Epsilon chapter, Winter Park, Florida, USA, April 8, 2014.
- Society for Computer Science, Workshop on Virtual Reality-Augmented Reality (GI VR-AR 2013), Würzburg, Germany, September 19, 2013.
- International Symposium on Ubiquitous Virtual Reality (ISUVR), Daejeon, South Korea, August 23, 2012.
- Joint Virtual Reality Conference (JVRC), Fellbach, Germany, September 29, 2010.
- Winter Augmented Reality Meeting (WARM), Graz, Austria, February 24–25, 2010.
- Real Action, Virtual Environments (RAVE), Barcelona, Spain, March 4, 2009.
- Digital City Monitoring and Emergency Management (D2D), Shenzhen, China, October 25, 2008.
- Eurographics Symposium on Virtual Environments (EGVE), Eindhoven, The Netherlands, May 29, 2008.

OTHER INVITED TALKS (SELECTED)

- “Participant Data Considerations for the Virtual Experience Research Accelerator (VERA),” University of Maryland, Department of Computer Science, Guest Lecture, May 9, 2024.
- “The Virtual Experience Research Accelerator (VERA),” ISTeC Distinguished Lecture, Colorado State University, Fort Collins, CO, November 13, 2023.
- “Universal User Data (UUD) for Virtual Experience Research,” Computer Science Department, Colorado State University, Fort Collins, CO, November 13, 2023.
- “The Virtual Experience Research Accelerator (VERA) and Universal User Data (UUD),” the 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 3rd Workshop on Replication in Extended Reality (WoRXR), Sydney, Australia, October 16, 2023.
- “The Virtual Experience Research Accelerator (VERA) and Universal User Data (UUD),” the 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2nd Workshop on Inclusion, Diversity, Equity, Accessibility, Transparency, and Ethics in XR (IDEA-TE XR), Sydney, Australia, October 20, 2023.
- “The Virtual Experience Research Accelerator (VERA),” The 5th XR Access Symposium, New York City, USA, June 15, 2023.

- “Embodied Agents for Simulating, Comforting, and Assisting Humans,” 7th Annual IVRHA Virtual Reality and Healthcare Global Symposium, University of Pennsylvania Medicine, Philadelphia, PA. March 4, 2023.
- “How to XR: Hardware & Technology,” World Economic Forum (WEF), Knowledge Lab with the Global Future Council on AR/VR 2021, April 29, 2021.
- “Personal Agents for Workplace (Nursing) Skills and Emotional Support,” HCI and the Future of Work and Wellbeing: A Series of Conversations. May 7, 2020, online (Longwood, FL USA).
- “Physical-Virtual Training for Healthcare,” International Women’s Club of Gothenburg, May 26, 2020, Gothenburg, Sweden (via video during pandemic).
- “Aware and Influential Virtual Humans and Animals,” Naval Postgraduate School, MOVES lecture, April 13, 2020, Monterey, CA (via video during pandemic).
- “Vera Real: Stroke assessment using a Physical Virtual Patient (PVP)” with Laura Gonzalez and Salam Daher (both UCF), 2019 Conference of the International Nursing Association for Clinical Simulation and Learning (INACSL), Phoenix, AZ, June 21, 2019.
- “Enhanced Perception & Cognition in Augmented Reality (EPiC AR),” Naval Postgraduate School, MOVES Open House, May 23, 2019, Monterey, CA.
- “Identifying User Physical States,” XR Advance Webinar Series, Iowa State University, April 22, 2019. Lecture series was a follow-up to a June 2017 VR/AR Workshop sponsored by NSF.
- “Physical-Virtual Patients: Human Shape, Appearance, Awareness, and Influence,” Virtual Reality & Health, Tucson, AZ, March 7, 2019.
- “Human Surrogates: Real, Virtual, and Augmented Shape, Appearance, and Intelligence,” Rubin Center for Healthy Aging, Sarasota, Florida, March 3, 2019.
- “Awareness and Influence of Human Surrogates in Augmented Reality,” 2017 International Conference on Computer-Aided Design and Computer Graphics (CAD/Graphics), Zhangjiajie, China, August 26, 2017.
- “Bridging the Telepresence Valley,” 2016 Annual Meeting of the Association for Information Science and Technology (ASIS&T), Copenhagen, Denmark, October 16, 2016.
- “Human Surrogates and their Effects on Humans,” Naval Postgraduate School, MOVES Institute, Simulation and Training Course, Monterey, CA, September 8, 2016.
- “Human Surrogates and their Effects on Humans,” Technische Universität München (TU Munich), Garching b. Munich, Germany, June 28, 2016.
- “Interactive Rear-Projection Physical-Virtual Patient Simulators,” Mini-Lecture at NextMed/MMVR22 (Medicine Meets Virtual Reality), Los Angeles, CA, April 7, 2016.
- “Making the Virtual Real: Virtual Reality, Augmented Reality, Physical-Virtual Reality,” Examination Development & Research committee of the National Council of Architectural Registration Boards (NCARB), Phoenix, AZ, January 30, 2016.
- “My patient is a dummy — so why should I care?” classroom lecture, Rollins College (Winter Park, FL), hosted by Darin Hughes, November 11, 2015.
- “Making the Virtual Real: Physical Manifestations of Dynamic Virtual People, Objects, and Places,” Qualcomm/Vienna University of Technology (Vienna, Austria), September 15, 2014.
- “Making the Virtual Real: Physical Manifestations of Dynamic Virtual People, Objects, and Places,” Graz University of Technology (Graz, Austria), September 17, 2014.
- “Past and Current Research in Modeling and Reproducing Humans for Warfighter Training,” Orlando VA Medical Center (Orlando, FL), August 26, 2014.
- “Making the Virtual Real: Physical Manifestations of Dynamic Virtual People, Objects, and Places,” Harris Corporation, Melbourne, FL, February 18, 2014.

- “Face-to-Face: Interactive Simulation Using a Mobile Physical-Virtual Patient,” work with Steven Talbert, Mary Lou Sole, and Karen J. Aroian, SNRS 28th Annual Conference, San Antonio, TX, February 15, 2014.
- “Physical-Virtual Humans for Healthcare,” the University of Texas at San Antonio, San Antonio, TX, February 13, 2014.
- “Physical-Virtual Humans for Training and Teleportation,” Max Planck Institute, Tübingen, Germany, September 17, 2013, and Bonn-Rhein-Sieg University of Applied Sciences, Bonn, Germany, September 20, 2013.
- “Interactive Shader Lamps: Projector-Based Graphics in Motion,” GI VR-AR (Würzburg, Germany), September 19, 2013.
- “Physical-Virtual Humans for Training and Teleportation,” TU Chemnitz (Chemnitz, Germany), June 6, 2013.
- “Physical-Virtual Humans: Challenges and Opportunities,” KIST (Seoul, South Korea) and KAIST (ISUVR 2012, Daejeon, South Korea), August 22–23, 2012.
- “Physical Manifestations of Virtual Humans,” Technische Universität München (TU Munich); Garching b. Munich, Germany; April 11, 2011.
- “Crossing Realities: The Convergence of Physical and Virtual Worlds,” Microsoft Research; Redmond, WA; May 25, 2010.
- “Crossing Realities: An Overview of Related Work at UNC-Chapel Hill,” Joint Virtual Reality Conference (JVRC); Lyon, France; December 7, 2009.
- “Virtual Artifacts and Experiences in Real Environments,” second RAVE 2009 workshop (Real Action, Virtual Environments); Barcelona, Spain; March 4, 2009.
- “A Living, Breathing, Dynamic Digital City,” D2D 2008 workshop (Digital City Monitoring and Emergency Management); Shenzhen, China; October 25, 2008.
- “Immersive Display Research at UNC-CH: A Sampling,” with Henry Fuchs; Max Planck Institute, Tübingen, Germany; June 9, 2008..
- “Motion Tracking as an Epic Battle Between Information and Uncertainty,” 14th Eurographics Symposium on Virtual Environments (EGVE 2008); Eindhoven, The Netherlands; May 29, 2008..
- “Improving, Expanding and Extending 3D Telepresence,” International Workshop on Advanced Information Processing for Ubiquitous Networks, ICAT 2005; Christchurch, New Zealand; December 8, 2005..
- “3D Telepresence for Off-Line Surgical Training and On-Line Remote Consultation,” 2nd CREST Symposium on Telecommunication, Teleimmersion and Telexistence; University of Tokyo; December 9–10, 2004..
- “Immersive Telepresence for Surgical Teaching and Remote Consultation: Extending Medical Expertise Over Time and Space,” University of Florida, Department of Computer and Information Science and Engineering; “Geometry, Graphics, Vision, Visualization/ Simulation (G2V2)” seminar series; October 17, 2003. .
- “Sensor Information Efficiency for Pose Estimation,” University of Rochester, Department of Computer Science; April 13, 2001..
- “Inside the HiBall Tracking System,” Siemens Corporate Research, Imaging and Visualization Department; Princeton, NJ; October 25, 2000..
- “Tracking for Interactive Computer Graphics: To the Hallway and Beyond,” the Boeing Corporation, Virtual Reality Group, Seattle, WA, April 21, 1999. (Also presented at Microsoft Research; Seattle, WA; April 21, 1999.).
- “Spatially-Augmented Visualization,” Walt Disney Imagineering, Virtual Reality Studio; Burbank, CA; September 1998..

INVITED WORKSHOP PARTICIPATION

- The 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 3rd Workshop on Replication in Extended Reality (WoR XR), Sydney, Australia, October 16, 2023.
- The 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2nd Workshop on Inclusion, Diversity, Equity, Accessibility, Transparency, and Ethics in XR (IDEATE XR), Sydney, Australia, October 20, 2023.
- ACM SIGGRAPH Frontiers Workshop: Digital Avatars: Risks, Harms, Barriers, Opportunities, with Eakta Jain (Co-Organizer), Apu Kapadia (Co-Organizer), Theodore Kim, and Taslima Akter. August 6, 2023, Los Angeles, CA.
- Anticipating the Metaverse: The Future Of Mixed Reality, invitation-only Workshop at the University of Southern California Institute for Creative Technologies, Los Angeles, CA, March 24, 2023.
- ACM SIGGRAPH Frontiers Workshop: Content Generation for Workforce Training, with Holly Rushmeier (Organizer), Carolina Cruz-Neira, Eakta Jain, Rajesh Jha, Jerome Solomon, and Louise Yarnall. July 28, 2019, Los Angeles, CA.
- National Science Foundation (NSF) Principal Investigator workshop on the Future of Work at the Human-Technology Frontier (FW-HTF), “Panel: Augmented Reality in the Future of Work,” with Joe Gabbard (moderator), Virginia Tech; Doug Bowman, Virginia Tech; Steve Feiner, Columbia University; and Henry Fuchs, University of North Carolina at Chapel Hill. April 5, 2019, NSF Headquarters (Alexandria, VA).
- Computing Community Consortium (CCC) Visioning Workshop: Content Generation for Workforce Training, March 14–15, 2019, Atlanta, GA.
- Computing Community Consortium (CCC) Computing Research Symposium: Addressing National Priorities and Societal Needs 2017, October 23–24, 2017, Washington, DC.
- National Science Foundation (NSF) Virtual Reality (VR)/Augmented Reality (AR) Workshop, July 17–18, 2017, Washington, DC.
- Computing Community Consortium (CCC) Visioning Workshop on Discovery & Innovation in Smart & Pervasive Health, December 5–6, 2016, Washington, DC.
- National Research Council (NRC) Mapping Science Committee meeting, October 3, 2012, National Academies Keck Center, Washington, DC, USA.
- Computing Community Consortium (CCC) Visioning Workshop on Spatial Computing, September 10–11, 2012, National Academies Keck Center, Washington, DC, USA.
- IMMERSCOM 2009 (2nd International Conference on Immersive Telecommunications), Panel with Jaron Lanier (Microsoft), Randy Harrell (CISCO), Frantz Lohier (Logitech), William C. Wickes (Hewlett Packard), and Zhengyou Zhang (Microsoft), May 29, 2009.

PANEL PARTICIPATION

- Rob Lindeman, Ferran Argelaguet, **Greg Welch**, Solène Lambert, Kyle Johnsen, and Anthony Steed. “Future Conference Formats: Balancing Knowledge Dissemination, Socialising and Environmental Impacts,” The 33rd IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2026), March 24, 2026, Daegu, Korea.
- **Greg Welch**, Mindi Anderson, Desiree Díaz, Frank Guido-Sanz, and Elizabeth Wells-Beede, “Patents for Healthcare Simulation Technology: A Multidisciplinary Perspective,” the 2024 International Meeting for Simulation in Healthcare, San Diego, CA, January 23, 2024.

- **Greg Welch**, Jeni Paay, Tim Chen, Bruce Thomas, Kim Vincs, Mark Billingham, and Aaron Quigley. “Why is XR still so hard to use?” The 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2023), October 16, 2023, Sydney, Australia.
- **Greg Welch**, Tabitha Peck, Jonathan Beever, and Christine Hemphill. “The Virtual Experience Research Accelerator (VERA),” The 5th XR Access Symposium, New York City, USA, June 15, 2023.
- **Greg Welch**, Nahal Norouzii, Rosario Casas. “Approaching the Metaverse — An Overview & the State of the Art,” Advertising in the Metaverse, 2022 American Academy of Advertising Annual Conference, March 24, 2022, St. Petersburg, FL USA.
- **Greg Welch**, Salam Daher, Laura Gonzalez, Jacob Stuart. “Workshop on Collaborations with Domain Experts for Virtual Agent Research (CoVAR),” Intelligent Virtual Agents (IVA 2020), October 18, 2020, online workshop (Longwood, FL USA).
- **Greg Welch**, Laura Gonzalez, Juan Cendan, Mindi Anderson. “Simulating Patients for Healthcare Training: Appearance, Shape, Influence, and Awareness,” Serious Play Conference, Healthcare Track, July 25, 2019, Orlando, FL USA.
- Joe Gabbard (Virginia Tech), Doug Bowman (Virginia Tech), Steve Feiner (Columbia University), Henry Fuchs (University of North Carolina at Chapel Hill), and **Greg Welch** (UCF). “Augmented Reality in the Future of Work,” National Science Foundation (NSF) Principal Investigator workshop on the Future of Work at the Human-Technology Frontier (FW-HTF), April 5, 2019, NSF Headquarters (Alexandria, VA).
- **Greg Welch** (UCF), Albert “Skip” Rizzo (University of Southern California), Maria Sanchez-Vives (Institute of Biomedical Research August Pi i Sunyer), and Petr Legkov (University of Osnabrück), “The Future Impact of Neuroscience and Cognitive Psychology on Virtual Environments,” IEEE Conference on Virtual Reality and 3D User Interfaces (VR 2018), Reutlingen, Germany.
- Anthony Steed (University College London), Doron Friedman (IDC Herzliya), **Greg Welch** (UCF), Susumu Tachi (The University of Tokyo), and moderator Mel Slater (University of Barcelona). “Instantaneous Beaming to Distance Places — A Possible and Desirable Future?” IEEE Virtual Reality 2017, March 21, Los Angeles, CA USA.
- **Greg Welch** (UCF), Charles Hughes (UCF), Linda Gibson-Young (UCF), Chait Renduchintala (UCF), Peter Kincaid (UCF), Sarah Matthews (FL DOH), Tom Herring (FL DOH), “Technology & Public Health Use Now and in the Future,” 2014 Florida Public Health Educational Meeting, Orlando, FL, August 1, 2014.
- Betty Mohler (Max Planck Institute for Biological Cybernetics), **Greg Welch** (UNC-CH), Daniel Thalmann (Nanyang Technological University and EPFL), and Victoria Interrante (University of Minnesota), “Avatars in Virtual Environments,” IEEE Virtual Reality 2011 (VR 2011), Singapore, March 21, 2011.
- Randy Harrell (CISCO), Jaron Lanier (Microsoft), Frantz Lohier (Logitech), **Greg Welch** (UNC-CH), William C. Wickes (Hewlett Packard), and Zhengyou Zhang (Microsoft Research), IMMERSCOM 2009 (2nd International Conference on Immersive Telecommunications), May 29, 2009.
- Henry Fuchs, Bernd Fröhlich (Bauhaus-Universität Weimar), and **Greg Welch**, “Display Technology,” 14th Eurographics Symposium on Virtual Environments (EGVE 2008), Eindhoven, The Netherlands, May 30, 2008.
- Barbara Hayes-Roth, Austin Henderson, Ramesh Jain, Lev Manovich, **Greg Welch**, and Gopal Pingali, “Experiential Telepresence: How Can Telepresence Research be Guided Towards Better End User Experience?” ACM SIGMM 2003 Workshop on Experiential Telepresence 2003 (ETP 2003), November 7, 2003, Berkeley, CA USA.

 GRADUATED PH.D. ADVISEES

- Austin Erickson, 2023: Understanding, Modeling, and Simulating the Discrepancy Between Intended and Perceived Image Appearance On Optical See-Through Augmented Reality Displays
- Nahal Norouzi, 2021: The Social and Behavioral Influences of Interactions with Virtual Dogs as Embodied Agents in Augmented and Virtual Reality
- Jason Hochreiter, 2019: Multi-Touch Detection and Semantic Response on Non-Parametric Rear-Projection Surfaces
- Myungho Lee, 2019: Mediated Physicality: Inducing Illusory Physicality of Virtual Humans Via Their Interactions With Physical Objects
- Salam Daher, 2018: Physical-Virtual Patient Simulators: Bringing Tangible Humanity to Simulated Patients
- Kangsoo Kim, 2018: Environmental Physical-Virtual Interactivity to Improve Social Presence With a Virtual Human in Mixed Reality
- Peter Lincoln, 2017 (UNC Chapel Hill, with Henry Fuchs): Low Latency Displays for Augmented Reality
- Christoph Resch, 2016 (TU München, with Gudrun Klinker): Enhancing Projective Spatial Augmented Reality in Industry: A Model Based Approach for Registration and Calibration
- Feng Zheng, 2015: Spatio-Temporal Registration in Augmented Reality
- Jinghe Zhang, 2013: Uncertainty-Driven Adaptive Estimation with Applications in Electrical Power Systems
- Peter Keitler, 2011 (TU München, with Gudrun Klinker): Management of Tracking and Tracking Accuracy in Industrial Augmented Reality Environments
- Adrian Ilie, 2010: On-Line Control of Active Camera Networks
- Hua Yang, 2008: Differential Tracking Through Sampling and Linearizing the Local Appearance Manifold
- B. Danette Allen: 2007, Hardware Design Optimization for Human Motion Tracking Systems
- Aditi Majumder, 2003: A Practical Framework To Achieve Perceptually Seamless Multi-Projector Displays
- Ruigang Yang, 2003: View-Dependent Pixel Coloring — A Physically-Based Approach for 2D View Synthesis
- Ramesh Raskar, 2001: Projector-Based Three Dimensional Graphics

 SUPERVISED POSTDOCTORAL SCHOLARS

- Dr. Nahal Norouzi, University of Central Florida, January 2022–May 2022.
- Dr. Kangsoo Kim, University of Central Florida, January 2019–December 2020.
- Dr. Salam Daher, University of Central Florida, January 2019–July 2019.
- Dr. Arjun Nagendran, University of Central Florida, August 2011–January 2013, with Dr. Charles Hughes (primary advisor/mentor).

GRADUATED M.S. AND B.S. HONORS THESIS ADVISEES

- Camila Acevedo Carillo, 2023 (UCF B.S. Computer Science, Advisor)
 - Angelica Garcia, 2017 (UCF M.S. Modeling & Simulation, Advisor)
 - Justin Grace, 2016: Recognizing Pain Using Novel Simulation Technology (UCF B.S.N. Honors in the Major, Committee Member)
 - Krishna Kollu, 2012: Modeling the Neural Pathology of Parkinson's Disease (UNC B.S. C.S. Honors, Advisor)
 - Caitlyn Losee, 2011: The Bathysphere: Motion Capture and Immersive Projection (UNC B.S. C.S. Honors, Advisor)
-

STUDENT ADVISEES (ALL)

Ahinya Alwin (current); Jasmine Joyce DeGuzman (current); Stephanie Liebel (current); Hiroshi Furuya (current); Zubin Choudhary (current); Matthew Gottsacker (current); Tiffany Losekamp (current, College of Nursing, with Mindi Anderson); Aden Thompson (2026) Armin Delmo (2026); Corey Clements (2025); Lam Nguyen (2025); Akhil Pavan Sai Machavaram (2024, MS CS); Nicholas Piazza (2024); Juanita Benjamin (2023); Austin Erickson (2023, Ph.D.); Camila Acevedo Carillo (2023); Nahal Norouzi (2021, Ph.D.); Faith Sauber (2021); Indira Avendano (2021); Nafisa Mostofa (2021); Alyssa Feagans (2020); Jonathan Jules (2020); Alexis Lambert (2020); Yazdan Jamshidi (2019, M.S.); Jason Hochreiter (2019, Ph.D.); Brandon Belna (2018); Ryan Schubert (2018); Salam Daher (2018, Ph.D.); Kangsoo Kim (2018, Ph.D.); Stephanie Mutter (JUN–AUG 2018, NSF REU); Brandon Belna (JUN–AUG 2017, NSF REU); Peter Lincoln (2017, Ph.D., with Henry Fuchs); Christoph Resch (2016, Ph.D., TU München, with Gudrun Klinker); Kiran Pandit (JUN–AUG 2016, NSF REU); Feng Zheng (2015, Ph.D.); Ryan Patrick (2013, M.S.); Jinghe Zhang (2013, Ph.D.); Krishna Kollu (2012, B.S. Undergraduate Honors); Caitlyn Losee (2011, B.S. Undergraduate Honors); Peter Keitler (2011, Ph.D., TU München, with Gudrun Klinker); Adrian Ilie (2010, Ph.D.); Tianren Wang (2009–2011); Hua Yang (2008, Ph.D.); Danette Allen (2007, Ph.D.); Vincent Noel (2006, M.S.); Michael Noland (2006, M.S.); Aditi Majumder (2003, Ph.D.); Ruigang Yang (2003, Ph.D.); and Ramesh Raskar (2001, Ph.D.).

STUDENT COMMITTEES

Samuel Cosgrove (current, M.S., Advisor Joe LaViola); Xinyu Hu (current, Ph.D., Advisor Ryan McMahan); Kyle Martin (current, Ph.D., Advisor Joe LaViola); Alec Moore (2022, Ph.D., Advisor Ryan McMahan); Kevin Pfeil (2022, Ph.D., Advisor Joe LaViola); Andres Vargas (2022, Ph.D., Advisor Joe LaViola); Sarah Braeger (2021, Ph.D., Advisor Hassan Foroosh); Corey Pittman (2021, Ph.D., Advisor Joe LaViola); Min Wang (2020, Ph.D., Advisor Hassan Foroosh); Alex Blate (2019, Ph.D., UNC, Advisor Henry Fuchs); Baoyuan Liu (2016, Ph.D., Amazon, Advisor Hassan Foroosh); Rick Skarbez (2016, Ph.D., UNC, Advisor Frederick P. Brooks and Mary Whitton); Alexandra Carson (2016, BSN Honors in the Major); Justin Grace (2016, BSN Honors in the Major, Advisor Kelly Allred); Monika Schaeffer (2011, M.S., Duke University, Advisor Ron Parr); Brian Clipp (2010, Ph.D., UNC, Advisor Marc Pollefeys); David Gallup (2010, Ph.D., UNC, Advisor Marc Pollefeys); Andrew Nashel (2010, Ph.D., UNC, Advisor Henry Fuchs);

Tyler Johnson (2009, Ph.D., UNC, Advisor Henry Fuchs; Seon Joo Kim (2008, Ph.D., UNC, Advisor Marc Pollefeys); Philip McLeod (2008, Ph.D., University of Otago, Advisor Geoff Wyvill); Patrick Quirk (2006, M.S., UNC, Advisor Henry Fuchs); Kok-Lim Low (2005, Ph.D., UNC, Advisor Anselmo Lastra); Lindsey McCarthy (2011, M.S. UNC Dentistry, Advisor Lucia Cevidanes); Michael Rosenthal (2005, Ph.D., UNC, Advisor Henry Fuchs); Benjamin Lok (2002, Ph.D., UNC, Advisor Frederick Brooks); Nicholas Vallidis (2002, Ph.D., UNC, Advisor Gary Bishop); Gopi Meenakshisundaram (2001, Ph.D., UNC, Advisor Dinesh Manocha); Mark Livingston (1998, Ph.D., UNC, Advisor Henry Fuchs); and Hans Weber (UNC, Advisor Gary Bishop).

PH.D. THESIS EXAMINER (INTERNATIONAL)

- Ylva Hansdotter, “The Affordances of Immersive Virtual Reality for Stimulating Prosocial Behaviour: A Mixed-Methods Pro-Environmental Intervention Study,” University College Dublin; Advisor: Prof. Lizbeth Goodman; 2022.
- Susanne Schmidt, “Blended Spaces: Perception and Interaction in Projection-Based Spatial Augmented Reality Environments,” Universität Hamburg; Advisor: Dr. Simone Frintrop; 2020.
- Iana Podkosova, “Walkable Multi-User VR: The Effects of Physical and Virtual Colocation,” Technische Universität Wien; Advisor: Dr. techn. Hannes Kaufmann, Priv.-Doz.; Second advisor: Dr. techn. Peter Ferschin, Ass. Prof; 2019.
- Michael Robert Marner, “Physical-Virtual Tools for Interactive Spatial Augmented Reality,” The University of South Australia; Principal Supervisor: Prof. Bruce Hunter Thomas; Associate Supervisor Prof. Christian Sandor; 2013.
- Philip Grant McLeod, “Fast, Accurate Pitch Detection Tools for Music Analysis,” the University of Otago; 2008.

CLASSROOM TEACHING

- *Augmented, Virtual, and Mixed Realities in Healthcare*, UCF, 2025 (Spring), with Mindi Anderson.
- *3D Computer Modeling and Animation*, First Year Seminar, UNC-CH, 2003–2007. Created the course, which was chosen by UNC from among competing First Year Seminar proposals. Awarded Computer Science Student Association *Excellence in Teaching* award, Spring 2007
- *Introduction to Programming (Java)*, UNC-CH, 2002–2003
- *Team Software Engineering*, UNC-CH, 2001 (Spring)
- *Exploring Virtual Worlds*, UNC-CH, 1997–1998, with Henry Fuchs.
- NSF STC Center-Wide Lecture Series, coordinated within the Center, 1997
- *Computers: Power Tools for the Mind*, UNC-CH, 1995

TUTORIALS/SHORT COURSES

- VR Research at Scale with the Virtual Experience Research Accelerator (VERA) ? An Early-Access Hands-On Tutorial, IEEE Virtual Reality 2026 (VR 2026), with Ali Haskins, Corey Clements, John T. Murray, Gerd Bruder, and Chloe Beato. March 22, 2026, Daegu, Korea.
- Participating in the Testing of VERA, IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2024), with Ali Haskins, John T. Murray, Gerd Bruder, and Tabitha Peck. October 21, 2024, Seattle, Washington.
- AR101 (An Introduction to Augmented Reality), 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), September 18, 2016. Mérida, México. **Greg Welch** with Mark Billinghurst, Tobias Höllerer, Dieter Schmalstieg, Gerhard Reitmayr, Kiyoshi Kiyokawa, and Steve Feiner.
- **Greg Welch**. “An Introduction to the Kalman Filter,” NASA Langley Research Center, National Institute of Aerospace, Hampton, VA USA, November 5–6, 2014.
- **Greg Welch**. “Tracking: Beyond 15 Minutes of Thought” and “An Introduction to the Kalman Filter,” University of Würzburg, Interactive Intelligent Spaces Summer School, Würzburg, Germany, September 13–14, 2013.
- **Greg Welch** and Gary Bishop. “An Introduction to the Kalman Filter,” ACM SIGGRAPH 2001 tutorial, August 12, 2001, Los Angeles, CA USA.
- **Greg Welch**, Gerhard Reitmayr (Graz Univ. of Tech.), Vincent Lepetit (Swiss Federal Institute of Tech.), and Brian Clipp (UNC-Chapel Hill), “Tracking for AR Tracking Researchers,” International Symposium on Mixed and Augmented Reality (ISMAR 2009), October 19, 2009.
- **Greg Welch** and Gary Bishop. “An Introduction to the Kalman Filter,” ACM SIGGRAPH 2001 tutorial, August 12, 2001, Los Angeles, CA USA.
- B. Danette Allen, Gary Bishop and **Greg Welch**. “Tracking: Beyond 15 Minutes of Thought,” ACM SIGGRAPH 2001 short course, August 12, 2001, Los Angeles, CA USA.
- **Greg Welch**. “An Introduction to the Kalman Filter,” multi-day seminars offered periodically at UNC Chapel Hill, 1996–2009.

PATENTS

- US 12,464,309, “Spatially Explicit Auditory Cues for Enhanced Situational Awareness,” with Gerd Bruder and Zubin Choudhary.
- US 12,008,917, “Physical-Virtual Patient System,” with Gerd Bruder, Salam Daher, Jason Hochreiter, Mindi Anderson, Laura Gonzalez, and Desiree A. Diaz.
- US 11,954,900, “Audiovisual Detection of Expectation Violations in Disparate Home Automation Systems,” with Gerd Bruder, Ryan Schubert, and Austin Erickson.
- US 11,798,127, “Spatial Positioning of Targeted Object Magnification,” with Gerd Bruder, Kangsoo Kim, and Zubin Choudhary.
- US 11,729,448, “Intelligent Digital Interruption Management,” with Matthew Gottsacker, Nahal Norouzi, and Gerd Bruder.
- US 11,557,216, “Adaptive Visual Overlay for Anatomical Simulation,” with Joseph LaViola II, Francisco Guido-Sanz, Gerd Bruder, Mindi Anderson, and Ryan Schubert.
- US 11,550,470, “Grammar Dependent Tactile Pattern Invocation,” with Gerd Bruder and Ryan McMahan.

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- [3] **Greg Welch**. The infrared touch-pad. Purdue University, E.E.T. 421 Report, February 26, 1986.
- [2] **Greg Welch** and James P. Williams. The easy chair: A microprocessor-controlled wheelchair for children with muscular disorders. Purdue University, E.E.T. 490/491 Senior Design Project, Preliminary Report, December 1985.
- [1] James Williams and **Greg Welch**. The pressure sensitive touch-pad. Purdue University, E.E.T. 454 Project Report, April 30, 1985.

PH.D. DISSERTATION

Title: SCAAT: Incremental Tracking with Incomplete Information
 Date: October 1996
 Advisor: Gary Bishop
 Committee: Gary Bishop, Henry Fuchs, Leandra Vicci (née Vernon Chi), Anselmo Lastra, Russell Taylor, and John Poulton
 PDF: http://www.cs.unc.edu/~welch/media/pdf/scaat_dissertation.pdf

ACTIVE FUNDING

- ONR Awards# N00014-25-1-2245 for “Visualization Enhancements for Augmented Reality (VEAR),” **Greg Welch** (PI) and Gerd Bruder (Co-PI). Dr. Peter Squire, Program Manager. \$700,000 total for April 2025–March 2027.
 - ONR DURIP Award# N00014-25-1-2159 for “Hybrid Visualization Testbed for Warfighter Augmented Reality Head-Worn Display Research with High-Ambient Photometric Environmental Properties,” **Greg Welch** (PI) and Gerd Bruder (Co-PI). Dr. Peter Squire, Program Manager. \$233,247 for February 2025–January 2027.
 - NSF Award# 2235066 for “Collaborative Research: CCRI: Grand: Virtual Experience Research Accelerator (VERA),” **Greg Welch** (Lead PI) and Gerd Bruder (PI) at the University of Central Florida; Shiri Azenkot (PI, NSF Award# 2235067) at Cornell Tech; Jeremy Bailenson at Stanford University (PI, NSF Award# 2235070); Tabitha Peck at Davidson College (PI, NSF Award# 2235068); and Valerie Jones Taylor at Rutgers University (PI, NSF Award# 2350377). NSF PO: Ephraim P. Glinert (IIS CHS), \$4,900,000 total (\$4,192,490 to UCF and \$707,510 to collaborative partners) for April 2023–March 2027. <https://veraccel.org>
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PAST FUNDING

- ONR Awards# N00014-21-1-2578 for “Enhanced Information Comprehension via Augmented Reality (EICAR),” **Greg Welch** (PI) and Gerd Bruder (Co-PI). Dr. Peter Squire, Program Manager. \$1,290,291 total for June 2021–May 2025.
- NSF Award# 1800961 for “CHS: Medium: Collaborative Research: Augmented Reality Agents with Pervasive Awareness, Appearance, and Abilities,” **Greg Welch** (PI), Gerd Bruder (Co-PI), and Damla Turgut (Co-PI) at UCF; Jeremy Bailenson (Co-PI) at Stanford University; and Benjamin Lok (Co-PI) at the University of Florida. NSF PO: Ephraim P. Glinert (IIS CHS), \$674,264 for August 15, 2018–July 31, 2024
- Vision Products, Inc., Trust and Situational Awareness in Augmented Reality Soldier Training, SBIR-Phase II. PIs Greg Welch and Gerd Bruder. PM: Ted Harmer. \$299,318. 2022–2023.
- ONR DURIP Award# N00014-21-1-2882 for “Testbed for Enhanced Perception and Cognition via Augmented Reality (EICAR),” **Greg Welch** (PI). Dr. Peter Squire, Program Manager. \$177,484 for August 2021–May 2024.
- NSF Award# 1852002 for “REU Site: Research Experiences for Undergraduates Site on Internet of Things (IoT),” Damla Turgut (PI), Student Mentors: Damla Turgut, Ladislau Boloni, **Greg Welch**, Gerd Bruder, Samiul Hasan, Qun Zhou, Hyoung Jin Cho, and Andrew Dickerson at the University of Central Florida. NSF PO: Jonathan Sprinkle, \$323,945 for May 1, 2019–April 30, 2023

- NSF Award# 1564065 for “CHS: Medium: Physical-Virtual Patient Bed for Healthcare Training and Assessment,” **Greg Welch** (PI), Juan Cendan (Co-PI), and Laura Gonzalez (Co-PI), NSF PO: Ephraim P. Glinert (IIS CHS), \$894,431 for August 1, 2016–July 31, 2022
- Soar Technology, Inc., “Handoff Training for Combat Casualty Care (HTC3) CCHAT System PHASE II” **Greg Welch** (PI), Gerd Bruder (Co-PI), and Francisco Guido-Sanz (Co-PI). SoarTech PI: Alyssa Tanaka. \$463,949 for June 15, 2019–April 30, 2022
- Florida High Tech Corridor Council (FHTCC), Handoff Training for Combat Casualty Care (HTC3), PIs: **Greg Welch** (PI), Gerd Bruder (Co-PI), and Francisco Guido-Sanz (Co-PI). SoarTech PI: Alyssa Tanaka. \$248,928 for December 16, 2019–April 30, 2022.
- University of Central Florida, Exploratory Effort Aimed at a Center for Virtual and Augmented Reality for Healthcare, PIs **Greg Welch** and Carolina Cruz-Neira (CS), Co-PIs Mindi Anderson (CON), Gerd Bruder (IST), Juan Cendan (COM), Lisa Dieker (CCIE), Charlie Hughes (CS), and Ben Noel (FIEA). \$40K for 2020–2021.
- ONR Award# N00014-18-1-2927 for “Enhanced Perception and Cognition in Augmented Reality (EPiC AR),” **Greg Welch** (PI) and Gerd Bruder (Co-PI). Dr. Peter Squire, Program Manager. \$1,175,000 total for January 2018–December 2021.
- SA Photonics, Inc. “Blended Reality Solution for Live, Virtual, and Constructive Field Training PHASE II.” **Greg Welch** (PI) and Gerd Bruder (Co-PI). SA Photonics PI: Michael Browne. \$156,138.50 for December 21, 2018–May 22, 2021.
- Soar Technology, Inc., “Intelligent Diagnostic Trauma Algorithms for a Ruggedized Autonomous Combat Casualty Care Capability.” Francisco Guido-Sanz (PI) and **Greg Welch** (Co-PI). SoarTech PI: Alyssa Tanaka. \$7,500 for September 4, 2018–April 3, 2019
- Soar Technology, Inc., “Combat Casualty Care Augmented Reality Intelligent Training System - C3ARESIS Phase II,” **Greg Welch** (PI), Gerd Bruder (Co-PI), and Francisco Guido-Sanz (Co-PI). SoarTech PI: Glenn Taylor. \$145,900 for April 6, 2018–April 5, 2019.
- Florida High Tech Corridor Council, “FHTCC: Combat Casualty Care Augmented Reality Intelligent Training System (C3ARESIS),” **Greg Welch** (PI), Gerd Bruder (Co-PI). \$49,760 for April 6, 2018–April 5, 2019.
- REU Grant (Award# 1560302) for “REU Site: Research Experiences in the Internet of Things,” led by Damla Turgut (PI) and Yier Jin (Co-PI), **Greg Welch** (Investigator) et al. \$359,912.00 for June 2016–May 2019
- ONR DURIP grant for “Transportable Human-Surrogate Interaction System (THuSIS),” **Greg Welch** (PI). Dr. Peter Squire, Program Manager. \$148,216 for September 15, 2016–March 14, 2017.
- Lockheed Martin Corporation grant for “Improving Augmented Reality Technology,” Joseph LaViola (PI), **Greg Welch** (Co-PI), and Malcolm Butler (Co-PI), \$100,000, September 1, 2016–August 29, 2017.
- UCF Office of Research and Commercialization, Mentoring Program Award, “Fear of Falling in Older Adults,” Ladda Thiamwong (PI) and **Greg Welch** (Mentor), \$3,000, July 1, 2017–June 30, 2018.
- Soar Technology, Inc., “Handoffs for Joint Service Casualty Care (HJSCC),” Francisco Guido-Sanz (PI) and **Greg Welch** (Co-PI). SoarTech PI: Alyssa Tanaka. \$59,763.00 for November 15, 2017–June 14, 2018.
- Soar Technology, Inc., “Handoff Training for Combat Casualty Care (HTC3),” **Greg Welch** (PI), Francisco Guido-Sanz (Co-PI), and Gerd Bruder (Co-PI). SoarTech PI: Alyssa Tanaka. \$60,000.00 for December 17, 2017–July 17, 2018.
- SA Photonics, Inc., “Blended Reality Solution for Live, Virtual, and Constructive Field Training,” **Greg Welch** (PI) and Gerd Bruder (Co-PI). SoarTech PI: Michael Browne. \$31,000.00 for November 16, 2017–June 1, 2018.

- Soar Technology, Inc., “A16-076 Augmented/Mixed Reality for Force-on-Force Combat Casualty Care Training- Phase 1 Option,” **Greg Welch** (PI). SoarTech PI: Glenn Taylor. \$6,000.00 for November 27, 2017–March 26, 2018.
- ONR grant for “Human-Surrogate Interaction,” **Greg Welch** (PI) at UCF; and Jeremy Bailenson at Stanford University. Dr. Peter Squire, Program Manager. \$2,312,188 total for March 2014–December 2017.
- SoarTech/Department of the Army SBIR grant for “Combat Casualty Care Augmented Reality Intelligent Training System (C3ARESIS),” **Greg Welch** (UCF PI) and Frank Guido-Sanz (UCF), \$18,000 + \$6,000 option, July 1, 2016–December 31, 2016
- ONR DURIP grant for “A Testbed for Evaluating Human Surrogates for Live-Virtual Training,” **Greg Welch** (PI) and Arjun Nagendran (Co-PI). Dr. Peter Squire, Program Manager. \$178,437 for November 2014–April 2016.
- ONR DURIP grant for “A Physical-Virtual Human-Robot Interaction System for Training, Education, and Rehabilitation,” **Greg Welch** (PI), Charlie Hughes (Co-PI), and Arjun Nagendran (Investigator). Dr. Peter Squire, Program Manager. \$268,598. July 2012–July 2014.
- ONR grant for “3D Display and Capture of Humans for Live-Virtual Training,” led by **Greg Welch** (PI) and Charlie Hughes (Co-PI) at UCF, Jeremy Bailenson at Stanford University, and Mel Slater at the University of Barcelona. Dr. Peter Squire, Program Manager. \$1,316,334 total for October 2011–September 2014.
- University of Central Florida, Major Research Equipment grant for “Comparative Evaluations of Display and Control Paradigms for Surrogate Humans,” **Greg Welch** and Charlie Hughes, \$57,500, February 2013–June 2013.
- University of Central Florida, Institute for Simulation & Training, Simulation for Healthcare Education and Learning Laboratory (SHELL) grant for “Simulating Subtle Clinical Changes and Realistic Conversation: Going Beyond Mannequins,” Kelly Allred (PI), Patricia Weinstein, **Greg Welch**, Steve Talbert, Erica Hoyt, Karen J. Aroian, Anne E. Norris, Mary Lou Sole, Susan K. Chase. \$125,000 in October 2012.
- DOE grant for “Advanced Kalman Filter for Real-Time Responsiveness in Complex Systems,” led by **Greg Welch** (PI) at UNC and Zhenyu Huang (PI) at the Pacific Northwest National Laboratory. \$320,502 total for September 2009–August 2013.
- ARA/DARPA contract for “ULTRA-Vis Phase 3: Sensor Fusion Methods/Software to Achieve Robust Pose Estimation,” **Greg Welch** (PI). Alberico Menozzi, TPOC (ARA). \$73,438 total for June 2012–June 2013.
- ONR grant for “3D Display and Capture of Humans for Live-Virtual Training,” led by **Greg Welch** (lead PI) and Henry Fuchs (PI) at UNC, and Amela Sadagic (PI) at the Naval Postgraduate School. Roy Stripling and Clarke Lethin, Program Managers. \$2.3M total for May 2009–September 2012. Welch led on the proposal and leads the subsequent research at UNC.
- NSF CRI:IAD grant for “Integrated Projector-Camera Modules for the Capture and Creation of Wide-Area Immersive Experiences,” led by Henry Fuchs (PI), **Greg Welch** (Co-PI), Leonard McMillan (Co-PI), Mary Whitton (Co-PI), and Svetlana Lazebnik (Co-PI). \$310K total for April 2008–March 2012. Welch led on the proposal at UNC.
- ONR contract for “Behavior Analysis and Synthesis for Intelligent Training (BASE-IT),” led by **Greg Welch** (PI) at UNC, Amela Sadagic (PI) at the Naval Postgraduate School, and Rakesh Kumar (PI) and Hui Cheng (Co-PI) at Sarnoff. Roy Stripling, Ph.D., Program Manager. \$2.2M total for February 2008–September 2011. Welch led on the proposal and leads the subsequent research at UNC.
- ONR SBIR Phase II contract for “Deployable Intelligent Projection Systems for Training,” led by Henry Fuchs (PI) and **Greg Welch** (PI), sub-contract from Renaissance Sciences Corporation

- (RSC), Jeff Clark, CEO, \$374,920 for March 2009–February 2011. Welch led on the proposal and leads the subsequent research at UNC.
- UNC-Chapel Hill “Scholarship, Creative Activity or Research in the Humanities and Fine Arts” led by Francesca Talenti (PI) and **Greg Welch** (Co-PI), \$10K for June 2009–May 2010.
 - IARPA A-SpaceX contract for “Mockup Future Analyst Workspace (A-Desk),” led by **Greg Welch** (lead PI) and Henry Fuchs (PI). Jeff Morrison, Ph.D., Program Manager. \$260K for April 2008–December 2008. Welch led the proposal and the subsequent research at UNC.
 - ONR SBIR contract for “Deployable Intelligent Projection Systems for Training,” led by **Greg Welch** (PI) at UNC and Karl Matias (PI) of Renaissance Sciences Corporation, sub-contract to UNC-Chapel Hill (Henry Fuchs, PI), \$80K for Phase 1, June 2007–September 2008. Welch led on the proposal and led the subsequent research at UNC.
 - IARPA VACE contract for “3D Content Extraction from Video Streams,” led by Marc Pollefeys (PI) and **Greg Welch** (Co-PI), with Jan-Michael Frahm. Dan Aldridge, Program Manager. \$660K October 2006–September 2008.
 - Cisco Systems grant for “Telepresence Wall: Research Exhibit,” with Henry Fuchs. \$439K for August 2007–July 2008. Welch co-led the proposal and the subsequent research at UNC.
 - ONR STTR contract for “Deployable Intelligent Projection Systems for Training: Enhanced Integrated Pose Estimation Technologies,” led by **Greg Welch** (PI) at UNC and Karl Matias (PI) of Renaissance Sciences Corporation, sub-contract to UNC-Chapel Hill (Henry Fuchs, PI), \$70K for Phase 1, August 2007–February 2008. Welch led on the proposal and led the subsequent research at UNC.
 - Cisco Systems grant for “Prototype for Two-station, Four-Person, Proper Eye-Gaze Telepresence System,” with Henry Fuchs. \$376K total for August 2006–July 2007. Welch co-led on the proposal and co-led the subsequent research at UNC.
 - National Library of Medicine contract for “3D Telepresence for Medical Consultation” led by **Greg Welch** (lead PI) and Prof. Henry Fuchs (PI) at UNC, Prof. Bruce Cairns, M.D. (Co-PI) at UNC, Prof. Ketan Mayer-Patel (Co-PI) at UNC, and Prof. Diane Sonnenwald (Co-PI) at Göteborg University and the University College of Borås. \$2.5M total for October 2003–December 2007. Welch led on the proposal and led the subsequent research at UNC.
 - ONR VIRTE contract for “Front-Projective Display For Virtual Environments: Phase 2,” with Henry Fuchs and Herman Towles. Dylan Schmorrow, Ph.D. CDR MSC USN, Program Manager. \$560K total for October 2004–December 2007.
 - Office of Naval Research DURIP 2006 grant for “Computing for Real World Acquisition, Display and Immersive Training,” with Henry Fuchs and Marc Pollefeys. \$136K total.
 - DARPA DSO contract for “Wide Area Visuals for a Simulator in a Box,” with Henry Fuchs and Herman Towles. Ralph Chatham, DARWARS Program Manager. \$1.2M total for 2003–2006.
 - NSF ITR grant “Electronic Books for the Tele-Immersion Age: A New Paradigm for Teaching Surgical Procedures,” led by **Greg Welch** (PI) at UNC and Andy van Dam (PI) at Brown University, \$609K (UNC) for Sep 2001–Aug 2005. Welch led on the proposal and led the subsequent research at UNC.
 - DOE contract “Front-Projection Display Wall, Group Tele-Immersion, and Tracking,” with Henry Fuchs and Herman Towles, \$1.8M for September 2001–August 2004.
 - “3D Tele-Immersion Over Next Generation Internet,” UNC/UPenn DARPA contract with Henry Fuchs, Herman Towles, Kostas Danillidis, and Ruzena Bajcsy, \$550K (UNC) for June 2001–December 2002.
 - Argonne National Laboratories contract “Compensating for Color Variations Across Multi-Projector Displays,” \$61K for September 2001–August 2002. Led by **Greg Welch** (PI).

- NSF grant “High-Fidelity Tele-Immersion for Advanced Surgical Training,” UNC/UPenn/Brown, with Henry Fuchs, Kostas Danillidis, and Andy van Dam, \$750K (UNC) for January 2001–December 2001.
- Naval Research Lab contract “Technology for Full-Body Tracking,” with Gary Bishop. Larry Rosenblum, NRL program manager. \$100,271 over October 2000–September 2001.
- NSF Research Infrastructure grant for the NSF Graphics and Visualization Science and Technology Center. The entire Center participated in the proposal process, **Greg Welch** played a significant role in the proposal content/realization. This grant supported equipment for Center-wide televideo and several related research projects. Support under this grant totaled \$1.3M for the Center over October 1998–September 2001. The UNC portion was \$304K.
- “The National Tele-Immersion Initiative,” with Henry Fuchs. Gift from Advanced Network and Services, Inc., Al Weiss, Terry Rogers, and Jaron Lanier. \$750K per year over January 1998–December 2000.
- NSF Graphics and Visualization Science and Technology Center, 5-year Renewal. The entire Center participated in the renewal proposal process, including a major site visit at the University of Utah, summer 1997. In addition to contributing to the written proposal, **Greg Welch** played a significant role in the Utah site visit. Total Center support under this renewal totals approximately \$13.7M for the entire Center for the final four years (February 1998–January 2002). The UNC portion totaled approximately \$2.7M over those four years.
- DARPA contract “Geospatially Registered Information for Dismounted Infantry” (GRIDS), with Gary Bishop and Vernon Chi. Joint effort with Raytheon Defense Systems, Hughes Research Labs, UNC, and the University of Southern California. Total support over two years (May 1997–April 1999) was approximately \$3.1M, UNC portion was approximately \$1.2M.

ACTIVITIES AS A STUDENT

- President, UNC-Chapel Hill Computer Science Student Association, 1994–1995
- Graduated with *Highest Distinction* from Purdue University (among three-tenths of the baccalaureate graduates having the highest graduation indexes)
- *Outstanding Senior Design Project*, “The Easy Chair: A Microprocessor-Controlled Wheelchair for Children With Muscular Disorders,” E.T., Purdue University, 1986
- *Distinguished Student*, Purdue University, 1982–1986 (all semesters)
- Residence Hall Counselor, Purdue University, 1985–1986
- Tau Alpha Pi National Honor Society, Pi Alpha Chapter, 1986
- Golden Key National Honor Society, 1984
- Phi Eta Sigma National Honor Society, 1983

PROFESSIONAL EXPERIENCE (MORE DETAIL)

University of Central Florida, Orlando, Florida*AdventHealth Endowed Chair in Healthcare Simulation*, October 2013–present

Professor, College of Nursing, October 2013–present

Professor, Computer Science Division of CECS, November 2013–present

Professor, Institute for Simulation & Training, November 2013–present

Co-Director, IST Synthetic Reality Lab (SREAL), August 2011–present

Co-Director, Interactive Computing Experiences Research Cluster of Excellence (2014–2017)

Research Professor, Institute for Simulation & Training (IST), August 2011–October 2013

Research Professor, Department of Computer Science, CECS, October 2011–October 2013

University of North Carolina at Chapel Hill, Chapel Hill, North Carolina

Adjunct Research Professor, Computer Science, September 2012–September 2020

Research Professor, Computer Science, July 2010–August 2012

Research Associate Professor, Computer Science, September 2001–June 2010

Adjunct Associate Professor, Applied & Materials Sciences, 2005–2008

Adjunct Assistant Professor, Applied & Materials Sciences, 2000–2003

Research Assistant Professor, Computer Science, September 1996–August 2001

UNC Site Coordinator, NSF Graphics and Visualization STC, September 1996–August 1998

- Coordinate UNC research for the National Science Foundation's Science and Technology Center for Graphics and Visualization.
- STC members included Brown University (PI Andy van Dam, Co-PI John Hughes), The California Institute of Technology (PI Al Barr), Cornell University (PI Don Greenberg), The University of Utah (PI Rich Riesenfeld, Co-PI Elaine Cohen), and UNC (PI Henry Fuchs).
- Primary goals of the STC, which existed from 1991–2002, included build a stronger scientific foundation for computer graphics and scientific visualization, and to helping create a basic framework for future interactive graphical environments.

University College Dublin (2011–2016)

Visiting Professor, School of Information and Library Studies

Keio University (2010–2012)

Visiting Professor, Graduate School of Media Design

Renaissance Sciences Corporation, Chandler, Arizona

Principal Investigator II, June 2007–2008

- Part-time research and development efforts related to “intelligent projector units” for deployable training systems.
- Work with Jeff Clark and Karl Mathias at RSC to coordinate UNC-RSC joint efforts.

Northrop Defense Systems Division, Rolling Meadows, Illinois

Senior Software Engineer, Digital Systems, June 1990–June 1992 (Secret Clearance)

- Developed, integrated, and maintained embedded software for the AN/ALQ-135, the electronic (radar) countermeasures system currently deployed on the USAF F-15 Eagle.
- Responsible for the AN/ALQ-135 Receiver-Transmitter Compatibility Feature which ensures that radar countermeasures do not interfere with radar or threat acquisitions.

NASA Jet Propulsion Laboratory (Caltech), Pasadena, California

CCS/COMSIM Programmer and Analyst, Voyager Sequence Team, January 1987–May 1990

- Enhanced and maintained the software Command Simulator for the Voyager spacecraft. This simulator is (still) used to validate all instructions sent to the spacecraft.
- Programmed and maintained the Voyager spacecraft Computer Command Subsystem (CCS), the master processor on both Voyager spacecraft.
- Simulated and validated Voyager spacecraft command sequence activity. Responsible for the simulation and validation of all sequenced commanding during Voyager II closest approach to the planet Neptune.

Significant Projects (developed under own initiative)

- Developed MEMMAN, a ground-based software system that optimizes the allocation of the limited Voyager spacecraft memory. Savings during the Voyager II Neptune encounter (August, 1989) enabled additional planetary imaging.
- Developed specialized IBM PC/AT telecommunications software for the Voyager Flight Team. Software submitted to NASA for distribution under COSMIC. Still used today.

Purdue University, West Lafayette, Indiana

Undergraduate, Electrical Engineering Technology, August 1982–May 1986

- Co-developed “The Easy Chair: A Microprocessor-Controlled Wheelchair for Children With Muscular Disorders.” The wheelchair included an ultrasonic “bumper” system, a custom infrared touch-pad, motor controllers, and a user-programmable 8085-based control unit.
- The Easy Chair project was awarded *Outstanding Senior Design Project* by the Purdue University Department of Electrical Engineering Technology, May 1986.
- Residence Hall Counselor (Resident Assistant) at Cary Quadrangle, August 1985–May 1986.

EXHIBIT 2



EDUCATION

The University of Texas at Austin, Austin, TX 78712

Ph.D. in Electrical Engineering, May 1994, G.P.A. 3.833/4.0

Kansas State University, Manhattan, KS 66506

M.S. in Electrical Engineering, May 1991, G.P.A. 4.0/4.0

B.S. in Electrical Engineering, May 1989, G.P.A. 3.934/4.0 (Summa Cum Laude)

DOCTORAL DISSERTATION

Combined Intravascular Ultrasound and Fluorescence Spectroscopy for the Diagnosis of Atherosclerosis

- Developed a catheter-based diagnostic system for atherosclerosis by interfacing a laser-induced fluorescence system to commercial intravascular ultrasound imaging equipment
- Wrote a Monte Carlo radiation transport code to simulate photon transport and fluorescence emission in tissue
- Initiated a clinical study to correlate restenosis rates of patients undergoing arterial atherectomy with the chemical composition of the atherosclerotic tissue removed

MASTER'S THESIS

An Empirical Study of the Dynamic Hysteresis Characteristics of Ferromagnets

- Developed a frequency-dependent, wide-bandwidth ferromagnetic hysteresis model suitable for circuit simulation programs
- Accurately simulated low and high-frequency hysteresis in ferrite toroidal cores
- Created an exhaustive annotated bibliography on mathematical hysteresis modeling

EXPERIENCE

Professor

February 2017 – Present

Associate Professor

February 2002 – January 2017

Assistant Professor

August 1999 – January 2002

Kansas State University, Department of Electrical and Computer Engineering

- Received ABET accreditation for the new KSU Biomedical Engineering Program
- Served as the Program Director for the new KSU Biomedical Engineering Program
- Designed and resourced a new KSU Biomedical Education and Innovation Laboratory
- Established a new KSU undergraduate program in Biomedical Engineering
- Established and directed the KSU Medical Component Design Laboratory
- Received grant support for biomedical component research/education
- Initiated multi-disciplinary teaching and research relationships with industry and academia
- Represented Kansas State through invited presentations, conferences, and short courses
- Supported and advised graduate students
- Integrated undergraduate honors students into a medical component research program
- Remained active in professional societies
- Served on KSU Faculty Senate and University-level academic/administrative committees
- Founded a new Midwest Professional Chapter of the IEEE EMBS
- Founded and advised a new KSU Student Chapter of the IEEE EMBS
- Instructed courses in the Biomedical, Electrical, and Computer Engineering curricula
- Helped to redesign the EE Bioengineering Option
- Designed real-world course projects that often required industry input

- Obtained internships for capable undergraduate students
- Advised students regarding academic and career goals

Professor**Associate Professor**

Kansas State University, Department of Gerontology

February 2017 – Present

July 2006 – January 2017

Principal Member of the Technical Staff

1998 – July 1999

Sandia National Laboratories, Albuquerque, NM

TELEMEDICINE PROJECT (Principal Investigator - \$3.2M effort) – “*The Role of Technology in Reducing Health Care Costs*”

- Designed a secure, plug-and-play information architecture for telemedicine
- Assembled a high-performance, multi-disciplinary team to modify a commercial telemedicine system and demonstrate its effectiveness within a secure framework
- Empowered team members to pursue innovative ideas
- Participated in invitation-only workshops dedicated to advanced home care technology

MONTE CARLO RADIATION TRANSPORT PROJECT – “*Monte Carlo Radiation Transport in Computer-Aided-Design (CAD) Geometries*”

- Presented ITS-CAD, a CAD-based Monte Carlo radiation transport code, to the high-performance computing community for the first time
- Prepared material to market this capability to the scientific community
- Completed initial algorithms for electron trapping within solid bodies

Senior Member of the Technical Staff

March 1996 – October 1998

Sandia National Laboratories, Albuquerque, NM

TELEMEDICINE PROJECT (Principal Investigator)

- Performed an industry-wide technology survey on modular telemedicine systems
- Received an individual performance award for the phase I final report
- Assisted with the definition of a clinical study for determining the cost-effectiveness and diagnostic feasibility of telemedicine applied to hypertension monitoring
- Established relationships with leaders in the telemedicine community

MONTE CARLO RADIATION TRANSPORT PROJECT

- Designed a method to incorporate the ACIS boundary representation modeler into the Integrated Tiger Series of legacy Monte Carlo radiation transport codes
- Incorporated C++ classes for analyzing complex hierarchical assemblies
- Developed efficient algorithms to track particles in these geometries
- Implemented a heterogeneous grid structure to increase code efficiency and make simulation times less dependent on geometrical complexity
- Applied the ACIS-based transport code to a neutron generator dose deposition problem

OTHER

- Studied the feasibility of using infrared imaging technology to remotely monitor vital signs and physiological indicators of malevolent intent
- Presented proposals to various customers regarding light-based, wearable sensors to determine human physiological status and to ascertain a wearer's identity
- Surveyed burn depth measurement technologies for a military customer
- Promoted high-school science in New Mexico through regional science competitions

Post-Doctoral Member of the Technical Staff
Sandia National Laboratories, Albuquerque, NM

July 1994 – February 1996

PERSONAL STATUS MONITOR PROJECT

- Completed an extensive literature search on pulse oximetry, motion artifact, and non-invasive vital sign measurement via light
- Performed Monte Carlo simulations to determine ideal light-based sensor configurations for minimizing motion artifact
- Produced a prototype of an ideal reflectance sensor incorporating a vertical cavity laser array and a large area photodiode
- Developed a MATLAB-based graphical user interface for time-frequency analysis
- Received an individual performance award for the final report
- Designed instrumentation, defined measurement parameters, and proposed operational criteria for the Emergency/Distress Monitor

MODEL BASED DESIGN PROJECT

- Electromagnetic Subproject Leader
 - Planned the budget and technical approach for magnetic field analyses on the W88 magnetic stronglink
 - Conducted an extensive review of commercial, state-of-the-art signal integrity and 3D electromagnetic modeling tools
- Demonstrated model-based design principles to multiple outside organizations, including the Technologies Enabling Agile Manufacturing thrust leaders, Boeing Aerospace, Proctor & Gamble, Motorola, Allied Signal, Schlumberger, and Ansoft
- Co-designed Sandia's Engineering Integration Facility
- Wrote the statement of work and budget proposal for the Sandia/Boeing virtual prototyping cooperative research and development agreement

Graduate Research Assistant

June 1992 – June 1994

The University of Texas at Austin, Department of Electrical and Computer Engineering

- Designed a system to integrate laser-induced fluorescence data from coronary artery cross-sections with two-dimensional intravascular ultrasound images
- Conducted a clinical study with the University of Texas Medical Branch in Galveston to determine the efficacy of this approach
- Performed experiments to characterize the laser-induced fluorescence of human coronary artery and aorta at 476 nm excitation
- Developed a Monte Carlo simulation code to predict excitation and emission profiles (absorption, reflection, and transmission) given an incident laser beam
- Created an algorithm to correct fluorescence spectra obtained with non-contact excitation/emission fiber bundles
- Prepared and published my own histology specimens
- Optimized an arterial tissue fluorescence model that determines arterial chemical composition based on laser-induced fluorescence spectra at 476 nm excitation
- Initiated a clinical effort with Ochsner Clinic in New Orleans to assess likelihood of restenosis based on chemical composition of atherectomy specimens
- Mentored undergraduate students performing honors projects

Graduate Teaching Assistant**August 1991 – May 1992***The University of Texas at Austin, Department of Electrical and Computer Engineering*

- Taught *Pascal Programming* (Spring 1992)
- Monitored the computer laboratory for *Pascal Programming* and *Introduction to Computation*
- Taught one *Electrical Engineering Lab I* section and monitored a second (Fall 1991)
- Graded midterm exams, final exams, and oral reports for *Electrical Engineering Lab I*

Research Assistant**June 1991 – August 1991***Lawrence Livermore National Laboratory, Human Genome Flow Cytometry Unit, Livermore, CA*

- Developed a Pascal interface between a Hewlett-Packard workstation and a Summagraphics digitizer board to determine dyed chromosome separation in a DNA mapping procedure
- Developed post-processing software for the high-speed cell sorter

Graduate Teaching Assistant**August 1989 – May 1991***Kansas State University, Department of Electrical Engineering, Manhattan, KS*

- Taught *Electrical Engineering Lab I* for two semesters
- Graded assignments for *Electromagnetic Theory I* and *Electronics I*
- Lectured during the instructor's absence

Research Assistant**June 1989 – August 1989***Lawrence Livermore National Laboratory, Nuclear Safety Systems Division, Livermore, CA*

- Experimentally determined the high-frequency hysteresis behavior of ferrite cores used in pulse transformer technology
- Obtained a government security clearance

Teaching Assistant**August 1988 – May 1989***Kansas State University, Department of Electrical Engineering, Manhattan, KS*

- Graded assignments for *Electromagnetic Theory I* and *Intro. to Computer Engineering*

JOURNAL PUBLICATIONS

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- Warren, Steve, Jeffrey Lebak, and Jianchu Yao. “Lessons Learned from Applying Interoperability and Information Exchange Standards to a Wearable Point-of-Care System,” *Transdisciplinary Conference on Distributed Diagnosis and Home Healthcare (D2H2)*, April 2–4, 2006, Marriott Crystal Gateway, Arlington, VA, pp. 101–104. Available at IEEE Xplore: <http://ieeexplore.ieee.org>. Poster award.
- Yao, Jianchu, Scott Simmons, and Steve Warren. “Ease of Use Considerations for Wearable Point-of-Care Devices in Home Environments,” *Transdisciplinary Conference on Distributed Diagnosis and Home Healthcare (D2H2)*, April 2–4, 2006, Marriott Crystal Gateway, Arlington, VA, pp. 8–11. Available at IEEE Xplore: <http://ieeexplore.ieee.org>.
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- Warren, Steve, Jeffrey Lebak, Jianchu Yao, Jonathan Creekmore, Aleksandar Milenkovic, and Emil Jovanov. “Interoperability and Security in Wireless Body Area Network Infrastructures,”

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- Martin, Matthew and Steve Warren, “Multiple Geometry Representations in Monte Carlo Radiation Transport,” *Transactions of the American Nuclear Society*, 2004, Vol. 91, pp. 183–184. ANS Winter Meeting, November 14–18, 2004, Omni Shoreham Hotel, Washington, D.C.
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- Warren, Steve, Jianchu Yao, Ryan Schmitz, and Lule Nagl. “Wearable Telemonitoring Systems Designed with Interoperability in Mind,” *25th Annual Conference of the IEEE EMBS*, Fiesta Americana Grand Coral Beach Hotel, Cancun, Quintana Roo, Mexico, September 17–21, 2003, pp. 3736–3739. Available at IEEE Xplore: <http://ieeexplore.ieee.org>.
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- Yao, Jianchu, Ryan Schmitz, and Steve Warren. “A Wearable Standards-Based Point-of-Care System for Home Use,” *25th Annual Conference of the IEEE EMBS*, Fiesta Americana Grand Coral Beach Hotel, Cancun, Quintana Roo, Mexico, September 17–21, 2003, pp. 3732–3735. Available at IEEE Xplore: <http://ieeexplore.ieee.org>.
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EMBS, Fiesta Americana Grand Coral Beach Hotel, Cancun, Quintana Roo, Mexico, September 17–21, 2003, pp. 3196–3198. Poster presentation. Available at IEEE Xplore:

<http://ieeexplore.ieee.org>.

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- Lutjemeier, B.J., D.K. Townsend, S. Koga, S. Warren, C.A. Harms, and T.J. Barstow. “Oscillations in Femoral Venous Blood Flow During Dynamic Exercise,” *American College of Sports Medicine 50th Annual Meeting*, San Francisco, CA, May 28-31, 2003, <http://www.acsm.org>. Poster presentation.
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- Kensek, Ronald P., Steve Warren, John A. Halbleib, Wesley C. Fan, and L.J. Lorence. “ITS Monte Carlo Radiation Transport Codes: Status and Future Plans,” *4th Topical Meeting on Industrial Radiation and Radioisotope Measurement Applications (IRRMA '99)*, October 3-7, 1999. Available from Sandia National Laboratories as SAND99-0682A and SAND99-2560C.
- Warren, Steve and Richard L. Craft. “Designing Smart Health Care Technology into the Home of the Future,” *First Joint Meeting of BMES and EMBS*, Atlanta, GA, October 13-16, 1999, p. 677. Available at IEEE Xplore: <http://ieeexplore.ieee.org>.
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- Craft, Richard L., Steve Warren, Raymond C. Parks, Linda K. Gallagher, Rudy J. Garcia, and Donald R. Funkhouser. "High-Surety Telemedicine in a Distributed, 'Plug-and-Play' Environment," *Toward An Electronic Patient Record '99 (TEPR '99)*, Orange County Convention Center, Orlando, FL, May 1-6, 1999, Session W9, Wed. May 5, pp. 176-186. Proceedings are available on CD-ROM through the Medical Records Institute, <http://www.medrecinst.com>.
- Warren, Steve, Richard L. Craft, and Raymond C. Parks. "High-Surety Telemedicine in a Distributed, 'Plug-and-Play' Environment," *4th Annual Conference - HII99: Improving Health in a Digital World*, Renaissance Hotel, Washington, D.C., April 26-28, 1999.
- Parks, Raymond C., Steve Warren, and Richard L. Craft. "Security Tradeoffs in Telemedicine," *4th Annual Conference - HII99: Improving Health in a Digital World*, Renaissance Hotel, Washington, D.C., April 26-28, 1999.
- Craft, Richard L., Steve Warren, Raymond C. Parks, and Anthony E. Sill, "Telemedicine Device Architectures and Telemedicine System Security," *American Telemedicine Association (ATA) Conference*, Salt Lake City, Utah, April 18-21, 1999.
- Warren, Steve, Ronald P. Kensek, and Leonard J. Lorence. "Monte Carlo Transport in Computer-Aided Design (CAD) Geometries," Display presentation. *Supercomputing '98: High Performance Networking and Computing Conference*, Orlando, FL, November 7-13, 1998.
- Warren, Steve, Jerry T. Love, Timothy J. Miller, and George R. Laguna. "Sensor and Signal Issues in the Design of a Light-Based Ambulatory Vital Signs Monitor," *ATACCC-97: Advanced Technology Applications to Combat Casualty Care*, Fort Walton Beach, FL, May 19-22, 1997, 31 pages. Proceedings are available on CD-ROM through Unconventional Concepts, Inc. at <http://www.unconventional-inc.com/>.
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- Carpenter, Kenneth H. and Steve Warren. "Modeling the Response of Ferrites to Fast Pulse Excitations," *14th Annual Conference on the Properties and Applications of Magnetic Materials*, Illinois Institute of Technology, Chicago, IL, May 15, 1995. Carpenter, Warren.
- Warren, Steve, Susan Chudnow, Alfred Johnston, Michael Davis, and Rebecca Richards-Kortum. "Fluorescence Spectroscopy and Intravascular Ultrasound for Imaging Coronary Artery Disease In Vitro," *31st Annual Rocky Mountain Bioengineering Symposium*, Kansas State University, Manhattan, KS, April 22-23, 1994.
- Warren, Steve, Susan Chudnow, Sharon Thomsen, Alfred L. Johnston, Michael J. Davis, and Rebecca R. Richards-Kortum. "Combined Ultrasonic and Spectroscopic Imaging of Coronary Artery Disease," *12th Annual Houston Conference on Biomedical Engineering Research*, University Hilton Hotel, University of Houston Campus, Houston, TX, February 10-11, 1994.
- Warren, Steve, Karl Pope, Youseph Yazdi, Alfred L. Johnston, Michael J. Davis, and Rebecca R. Richards-Kortum. "Combined Ultrasonic and Spectroscopic Imaging of Coronary Artery Disease," *Thirteenth Annual Meeting, American Society for Lasers in Surgery and Medicine*, New Orleans, LA, April 18-20, 1993.
- Warren, Steve, Karl Pope, Youseph Yazdi, Alfred L. Johnston, Michael J. Davis, and Rebecca R. Richards-Kortum. "Monte Carlo Fluorescence Verification of Experimental Results for the Combined Ultrasonic and Spectroscopic Imaging of Coronary Artery Disease," *30th Annual Rocky Mountain Bioengineering Symposium, Holiday Inn, Market Square*, San Antonio, TX, April 2-3, 1993. In *Biomedical Sciences Instrumentation*, Instrument Society of America, John Enderle, Ed., ©1993.
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- Pope, Karl, Steve Warren, Youseph Yazdi, Alfred Johnston, Michael Davis, and Rebecca Richards-Kortum. “Dual Imaging of Arterial Walls: Intravascular Ultrasound and Fluorescence Spectroscopy,” *SPIE: Biomedical Optics '93*, Los Angeles, CA, Jan. 1993. *Proceedings: Biomedical Optics '93: Diagnostic and Therapeutic Cardiovascular Interventions III*, SPIE 1878-06.
- Warren, Steve, Karl Pope, Youseph Yazdi, Alfred L. Johnston, Michael J. Davis, and Rebecca R. Richards-Kortum. “Combined Intravascular Ultrasound and Fluorescence for the Real-Time Diagnosis of Atherosclerosis,” *International Conference on Lasers '92*, Wyndham Greenspoint Hotel, Houston, TX, December 7-10, 1992. *Proceedings: International Conference on Lasers '92*.
- Warren, Steve. “Combining Intravascular Ultrasound and Fluorescence Spectroscopy for the Real-Time Diagnosis of Atherosclerosis,” *Poster Presentation: Gordon Research Conference on Lasers in Medicine and Surgery*, Colby-Sawyer College, New London, New Hampshire, June 6-10, 1992.
- Carpenter, Kenneth H., Steve Warren, and Ronald S. Lee. “An Empirical Study of Magnetization in Ferrites Excited by Fast Rise Time Pulses,” *Joint XUSC/LFSC Meeting*, Los Alamos National Laboratory, October 31 – November 2, 1989.

INVITED PRESENTATIONS

- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering: Wellness Tools for Children with Autism Spectrum Disorder,” *AAI 860 – Systems and Data in Animal Health*, Kansas State University, Olathe Campus, Via Zoom from Kansas State University, December 11, 2025. (Invited)
- Warren, Steve, Charles Carlson, Don Gruenbacher, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “New KSU Undergraduate Biomedical Engineering Degree Program,” Guest Lecture, Kansas State University, Engineering Hall, Room DUE 0093, *DEN 160 – Introduction to Engineering*, April 22, 2025. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering: Wellness Tools for Children with Autism Spectrum Disorder,” *AAI 860 – Systems and Data in Animal Health*, Kansas State University, Olathe Campus, Via Zoom from Kansas State University, November 21, 2024. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering: Wellness Tools for Children with Autism Spectrum Disorder,” *AAI 860 – Systems and Data in Animal Health*, Kansas State University, Olathe Campus, Via Zoom from Kansas State University, November 15, 2023. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Applied Telehealth Tools for Neurotypical and Special Needs Populations,” MGH/Harvard Medical School Biomedical Engineering Seminar Series, Via Zoom from Kansas State University, January 13, 2023. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering,” Guest Lecture, Kansas State University, *DEN 160 – Introduction to Engineering*, October 25, 2022. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering and Public Health,” Guest Lecture, Kansas State University, *KIN 110 – Introduction to Public Health*, October 20, 2022. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering,” Guest Lecture, Kansas State University, *DEN 160 – Introduction to Engineering*, October 22, 2021. (Invited)

- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering and Public Health,” Guest Lecture, Kansas State University, *KIN 110 – Introduction to Public Health*, October 14, 2021. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering,” Kansas State University, Pre-Health Ambassadors, Zoom, April 27, 2021. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering and Public Health,” Guest Lecture, Kansas State University, *KIN 110 – Introduction to Public Health*, April 22, 2021. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Bed-Based Systems to Track the Nighttime Wellness of Children with Autism Spectrum Disorder,” *2020 One Health Research Symposium: Biosensors*, Virtual Event Sponsored by BioNexus KC, Kansas City, MO, November 16-18, 2020, <https://bionexuskc.org/event/2020-one-health-research-symposium-biosensors/>. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Applied Telehealth Tools for Neurotypical and Special Needs Populations,” University of North Florida, College of Computing, Engineering, and Construction, Jacksonville, FL, October 6, 2020. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, Bala Natarajan, Taehee Lee, Kelsey Warren, Sophie Braynock, and Dallas Rice. “Biomedical Engineering Endeavors at Kansas State University,” University of North Florida, College of Computing, Engineering, and Construction, Jacksonville, FL, November 12, 2019. (Invited)
- Warren, Steve and Charles Carlson. “Biomedical Engineering (BME) Program Overview,” Kansas State University, Electrical and Computer Engineering, College of Engineering Advisory Council, October 25, 2019. (Invited)
- Warren, Steve. “Plans for the Reichenberger Cornerstone Teaching Scholar Award,” Kansas State University, College of Engineering Named-Award Celebration with the Carl R. Ice Family, October 18, 2019. (Invited)
- Warren, Steve. “Biomedical Engineering (BME) Program Overview,” Kansas State University, College of Engineering, Engineering Ambassadors, November 5, 2018. (Invited)
- Warren, Steve. “Biomedical Engineering (BME) Program Overview,” Kansas State University, Electrical and Computer Engineering, ECE Advisory Council Exchange, October 12, 2018. (Invited)
- Warren, Steve. “Tools to Quantify Well-Being in Children with Severe Disabilities,” University of North Florida, College of Computing, Engineering, and Construction, Jacksonville, FL, October 9, 2018. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected Wireless World,” Biomedical Science Seminar, Derby High School, Derby, KS, November 1, 2017. (Invited)
- Warren, Steve. “Research and Technology to Quantify Human and Animal Health,” University of North Florida, College of Computing, Engineering, and Construction, Jacksonville, FL, October 10, 2017. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected Wireless World,” *Future City Leadership Retreat*, Lawrence College and Career Center, Lawrence, KS, September 22, 2017. Middle school student presentations. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected Wireless World,” *ID 545 – Interior Design Studio 7, ID 651 – Design for Supportive Environments*, Kansas State University, College of Human Ecology, Manhattan, KS, September 15, 2017. (Invited)
- Warren, Steve. “Medical Device Design in the KSU Medical Component Design Laboratory,” KSU College of Engineering, Netsmart Executives, November 14, 2016. (Invited.)

- Warren, Steve, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, Ahmad Suliman, Shangxian Wang, Taishan Li, Kim Fowler, Ed Brokesh, Jack Xin, Wayne Piersel, Janine Kesterson, and Steve Stoffregen. “Experiences and Lessons Learned: Design Projects Intended to Aid and Quantify Children with Severe Disabilities,” University of North Florida, Department of Electrical Engineering, October 25, 2016. Interactions with Dr. Juan Aceros, Dr. Mary Lundy, and other UNF faculty. Supported by the National Institutes of Health. (Invited)
- Steve Warren, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, Ahmad Suliman, Tianyu Lin, and Shangxian Wang. “NSF GARDE Project: Overview and Next Steps,” KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society, KSU College of Engineering, March 7, 2016. (Invited)
- Warren, Steve, Punit Prakash, and David Thompson. “Technology for Successful Aging,” Meadowlark Hills Advisory Board, Meadowlark Hills Retirement Community, Manhattan, KS, November 11, 2014. (Invited)
- Warren, Steve. “Biomedical Research in the K-State Department of Electrical & Computer Engineering: Wearable Wireless Medical Devices and Applications,” Combined Grand Rounds, Kansas Medical Education Foundation, Topeka, KS, October 18, 2014. (Invited)
- Warren, Steve. “Bioengineering Option in Electrical Engineering,” ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, October 14, 2014. (Invited)
- Larsen, Jace, Job Shiach, and Steve Warren. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society – Top 3 Reasons to Join EMBS,” ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, September 16, 2014. (Invited)
- Warren, Steve. “Med/Bio Projects Supported by the KSU Medical Component Design Laboratory,” Manhattan High School, Medical Explorers Club, Mercy Hospital Conference Room, Manhattan, KS, February 6, 2014. (Invited)
- Warren, Steve. “Med/Bio Projects Supported by the KSU Medical Component Design Laboratory,” *CIS 450 – Computer Architecture and Organization*, Department of Computing & Information Sciences, Kansas State University, November 4, 2013. (Invited)
- Warren, Steve. “Med/Bio Projects Supported by the KSU Medical Component Design Laboratory,” KSU Student Chapter of the IEEE EMBS, Kansas State University, September 19, 2013. (Invited)
- Warren, Steve, Morgan Hoffman, and Kaleb Anderson. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society – Top 3 Reasons to Join EMBS,” ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, September 17, 2013. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected World,” KSU Presidential Lecture Series, Delta Kappa Gamma, Meadowlark Hills, Manhattan, KS, April 17, 2013. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected World,” KSU Presidential Lecture Series, Haskell Indian Nations University, Lawrence, KS, April 3, 2013. (Invited)
- Warren, Steve. “NASA-Funded KSU Efforts to Assess/Predict Astronaut Performance,” ECE 015 – New Student Assembly, Kansas State University, Manhattan, KS, November 14, 2012. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected World,” Meadowlark Hills Lecture Series, Sponsored by the KSU Center on Aging, Manhattan, KS, November 8, 2012. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected World,” KSU Presidential Lecture Series, Sports Medicine, Athletic Training, and Health Careers Class, Olathe North High School, Olathe, KS, September 27, 2012. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected World,” KSU Presidential Lecture Series, Life Enrichment Group, Butler Community College, El Dorado, KS, September 18, 2012. (Invited)
- Warren, Steve. “Biomedical Device Design for a Connected Wireless World,” *2012 IEEE International Conference on Computer Science and Automation Engineering (CSAE2012)*,

Zhangjiajie, China, May 25-27, 2012, <http://www.ieee-csae.org/keynote.asp>. Invited keynote presentation.

- Warren, Steve. “Biomedical Device Design for a Connected Wireless World,” Kansas State University Medical Component Design Laboratory, Research Overview and Recruiting Discussion, Beihang University, Beijing, China, May 23, 2012. (Invited)
- Warren, Steve. “Biomedical Device Design for a Connected Wireless World,” Kansas State University Medical Component Design Laboratory, Research Overview and Recruiting Discussion, Jiaotong University, Dalian, China, May 22, 2012. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected World,” KSU Presidential Lecture Series, 7th–12th Grade Gifted Student Seminar, Community Church of God, Phillipsburg, KS, March 28, 2012. (Invited)
- Warren, Steve. “Bioengineering Option in Electrical Engineering,” ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, November 8, 2011. (Invited)
- Warren, Steve. “Wearable/Ingestible Sensors to Acquire Animal-Level State of Health in Cattle Herds,” Frontiers in Multiscale Computational Modeling for Zoonotic Epidemics, Embassy Suites Kansas City Plaza, Kansas City, MO, October 11, 2011. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected World,” KSU Presidential Lecture Series, Gifted and AP Student Seminar, Topeka West High School, Topeka, KS, September 30, 2011. (Invited)
- Warren, Steve. “Update on the Newly Funded NSF RAPD (GARDE) Project: KSU Senior Design Projects to Aid Heartspring Children,” Heartspring, Wichita, KS, Heartspring Board Meeting, September 28, 2011. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Kansas State University, Electrical and Computer Engineering, ECE Advisory Council Exchange, April 8, 2011. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” UMKC Student Chapter of the IEEE Engineering in Medicine and Biology Society, Department of Computer Science and Electrical Engineering, School of Computing and Engineering, University of Missouri at Kansas City, 546 Robert H. Flarsheim Hall, Kansas City, MO, 64110, March 11, 2011. (Invited)
- Warren, Steve. “Biomedical Research and Technology for the Future,” 2011 Future Cities Competition, K-State Union Forum Hall, Kansas State University, Manhattan, KS, January 15, 2011. (Invited Keynote Speaker)
- Warren, Steve. “KSU Medical Component Design Laboratory: Biomedical Research and Veterinary Opportunities,” Information exchange with Hill’s Pet Nutrition, ECE Garmin Conference Room, Kansas State University, Manhattan, KS, November 29, 2010. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Information exchange with Robert Bestgen, ECE Garmin Conference Room, Kansas State University, Manhattan, KS, November 19, 2010. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Industrial and Manufacturing Systems Engineering Graduate Seminar, Kansas State University, Manhattan, KS, November 3, 2010. (Invited)
- Warren, Steve. “Bioengineering Option in Electrical Engineering,” ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, October 12, 2010. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Eta Kappa Nu Student Body Meeting, Kansas State University, Manhattan, KS, September 15, 2010. (Invited)
- Warren, Steve and Jason Orr. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society – Top 3 Reasons to Join EMBS,” ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, September 14, 2010. (Invited)
- Warren, Steve. “Focus Areas in the KSU Medical Component Design Laboratory,” Information exchange with Robert Bestgen, Information Exchange with Honeywell Federal Manufacturing

and Technologies, Kansas City Plant, 2000 East 95th St., Kansas City, MO 64131-6159, August 19, 2010. (Invited)

- Warren, Steve. “Thoughts from a Former Student, Mentee, and Colleague,” Richard G. Gallagher Retirement Dinner, KSU Alumni Center Ballroom, Kansas State University, Manhattan, KS, May 7, 2010. (Invited)
- Warren, Steve. “Biomedical Research and Education Focus Areas at Kansas State University,” NSF CCLI Information Exchange, East Carolina University, Greenville, NC, December 17, 2009. (Invited)
- Warren, Steve. “Physical Device Opportunities for the KSU/Cerner Partnership,” Discussion with Adam Cross, Cerner Corporation, Kansas State University, Manhattan, KS, December 15, 2009. (Invited)
- Warren, Steve. “Physical Device Opportunities for the KSU/Cerner Partnership,” Cerner Innovation Campus, Kansas City, MO, November 20, 2009. (Invited)
- Warren, Steve. “Intelligent Pervasive Technology Applied to Home Health Care,” Discussion with Cerner Executives, Cerner Health Conference 2009, Kansas City, Missouri, October 5, 2009. (Invited)
- Warren, Steve. “Plug-and-Play Monitoring Systems: Thought Progression and Prototype Implementation,” NSF CPS Team Information Exchange, Cerner Health Conference 2009, Kansas City, Missouri, October 5, 2009. (Invited)
- Warren, Steve. “KSU Research in Wearable Telemonitoring Technology,” Aging related research presentation for the Center on Aging, Justin Hall Room 254, Kansas State University, Manhattan, KS, March 27, 2009. (Invited)
- Warren, Steve, Daniel Andresen, Donald Gruenbacher, and Bala Natarajan. “KSU Research in Wearable Telemonitoring Technology,” Information Exchange with Abhi Ray, Digital Rays, Kansas State University, Manhattan, KS, August 5, 2008. (Invited)
- Warren, Steve, Daniel Andresen, Donald Gruenbacher, and Bala Natarajan. “KSU Research in Wearable Telemonitoring Technology,” Information Exchange with Colonel Wayne Parks, Great Plains Cyber Electronic University Consortium, Kansas State University, Manhattan, KS, July 9, 2008. (Invited)
- Warren, Steve. “A Partnership Toward Pervasive Health Care Delivery Environments,” Program Development Exchange with Cerner Corporation, Kansas State University, May 5, 2008. Presentation and white paper. (Invited)
- Warren, Steve. “Pervasive Home Health Care Technology,” KSU Epicenter Information Exchange with Dr. Tim Study, IBM Consulting Client Executive, America's Healthcare and Life Sciences, Kansas State University, January 31, 2008. (Invited)
- Warren, Steve. “Plug-and-Play Monitoring Systems: Thought Progression and Prototype Implementation,” Program Development Exchange with Cerner Corporation, Kansas State University, January 23, 2008. (Invited)
- Warren, Steve. “Intelligent Pervasive Technology Applied to Home Health Care,” Information Exchange with Cerner Corporation, Kansas City, MO, December 17, 2007. (Invited)
- Warren, Steve. “Trends in Personalized Health Care Technology,” Graduate Seminar, Department of Industrial and Manufacturing Systems Engineering, Kansas State University, November 16, 2007. (Invited)
- Warren, Steve. “Pervasive Home Health Care Technology,” Health Care Advisory Committee, Department of Industrial and Manufacturing Systems Engineering, Kansas State University, October 25, 2007. (Invited)
- Warren, Steve. “Intelligent Care Systems for the Home,” KSU Student Chapter of the IEEE EMBS, Kansas State University, March 8, 2007. (Invited)
- Warren, Steve. “Home Health Monitoring Devices,” UMKC Student Chapter of the IEEE, Department of Computer Science and Electrical Engineering, School of Computing and Engineering, University of Missouri at Kansas City, 546 Robert H. Flarsheim Hall, Kansas City, MO, 64110, March 6, 2007. (Invited)

- Warren, Steve. "In Pursuit of Intelligent Wearable Health Monitoring Systems," Kansas City Section of the IEEE, Wyndham Garden Hotel, Overland Park, KS, November 16, 2006. (Invited)
- Warren, Steve, Daniel Andresen, Kevin Smith, Angel Martinez, Roland Craddolph, Patrick York, Howard Erickson, Tammi Hildreth, and David Poole. "Wearable and Wireless: Distributed, Sensor-Based Telemonitoring Systems for State of Health Determination in Cattle," Information Exchange with Greatbatch, Inc., Tadtman Board Room, Alumni Center, Kansas State University, Feb. 7, 2006. (Invited)
- Warren, Steve and Emil Jovanov. "The Need for Rules of Engagement Applied to Wireless Body Area Networks," *IEEE Consumer Communications and Networking Conference*, Harrah's Las Vegas Hotel & Casino, Las Vegas, NV, January 8–10, 2006. (Invited paper/presentation.) (16% acceptance)
- Warren, Steve, Daniel Andresen, Luke Nagl, Scott Schoenig, Balakumar Krishnamurthi, Howard Erickson, Tammi Hildreth, David Poole, and Mark Spire. "Wearable and Wireless: Distributed, Sensor-Based Telemonitoring Systems for State of Health Determination in Cattle," Anatomy & Physiology Department Seminar, Kansas State University, Practice Management Center, Trotter Hall, September 27, 2004.
- Warren, Steve, Jianchu Yao, Ryan Schmitz, and Jeff Lebak. "Reconfigurable Point-of-Care Systems Designed with Interoperability Standards," *26th Annual Conference of the IEEE EMBS*, Westin St. Francis Hotel, San Francisco, CA, September 1–5, 2004, pp. 3270–3273. (Invited)
- Warren, Steve. "Wearable Medical Device Networks Designed with Interoperability in Mind," U.S. Food and Drug Administration Staff College, 9200 Corporate Blvd., Rockville, MD, November 3, 2003.
- Warren, Steve. "Beyond Telemedicine: Infrastructures for Intelligent Home Care Technology," *Pre-ICADI Workshop on Technology for Aging, Disability, and Independence*, The Royal Academy of Engineering, 29 Great Peter Street, Westminster, London SW1P 3LW (England), June 26–27, 2003, <http://www.cise.ufl.edu/~helal/preICADI>.
- Warren, Steve. "Commodity Building Blocks for Home Health Care," Community Health Council, Mercy Hospital, Manhattan, KS, September 11, 2002. Summarized in the September 2002 Community Health Council bulletin, <http://www.chcnet.info>.
- Warren, Steve and Ronald P. Kensek. "Monte Carlo Transport in Computer-Aided Design (CAD) Geometries," *Spatial Insider's Summit 2002*, Westminster, CO, May 13–15, 2002.
- Warren, Steve, Ruth Miller, Daniel Marcus, Thomas Barstow, Howard Erickson, David Poole, and Daniel Andresen. "Biomedical Sensor Webs for Human and Animal Monitoring," Kansas State University, February 13, 2002. Meeting with Daniel L. Schmoldt, National Program Leader for the U.S. Department of Agriculture Instrumentation and Sensors Program.
- Warren, Steve. "Bioengineering Option in Electrical Engineering," IEEE Student Chapter Meeting, Kansas State University, Manhattan, KS, October 9, 2001.
- Warren, Steve. "Beyond Telemedicine: Home Care Technologies," *Informational Meeting: Research to Aid Persons with Disabilities (RAPD)*, National Science Foundation, Arlington, VA, June 28, 2001.
- Warren, Steve. "Medical Component Confederacies and Home-Based Sensor Webs," *FDA Workshop on Home Care Technology Trends*, U.S. Food and Drug Administration, Rockville, MD, January 25, 2001.
- Warren, Steve. "Smart Health Care Technology and the Home of the Future," *Biomedical Engineering Seminar*, Department of Biomedical Engineering, University of Minnesota, Twin Cities Campus, November 30, 1999. Broadcast via UNITE instructional television.
- Warren, Steve. "The Health Care Facelift: How Telecommunications Technology, Distributed Computing, and Intelligent Medical Devices will Redefine the Patient-Physician Relationship," *Fall 1999 Seminar Series*, Department of Electrical and Computer Engineering, Wichita State University, October 22, 1999.
- Warren, Steve. "The Vital Signs of a Dynamic Engineering Curriculum," Kansas State University, Department of Electrical and Computer Engineering, April 2, 1999.

- Warren, Steve. "Radiation Transport in Complex CAD/CAM/CAE Geometries," Sandia National Laboratories, Center 6500 Management Team Briefing, March 2, 1998.
- Love, Jerry T. and Steve Warren. "The Personal Status Monitor Project," for the Warfighter Personal Status Monitor project team, Combat Casualty Care Research Program, Sandia National Laboratories, August 16, 1996.
- Warren, Steve. "Photo-Plethysmographic Signal Acquisition and Motion Artifact," Sandia National Laboratories, Center 9400 Technology Exchange Meeting, August 22, 1996.
- Warren, Steve. "Pulse Oximetry Measurement and Motion Artifact," for Colonel William P. Wiesmann, Director, Combat Casualty Care Research Program (RAD II), U.S. Army Medical and Research Material Command, June 17, 1996.
- Warren, Steve. "Burn Depth Measurement," for Colonel William P. Wiesmann, Director, Combat Casualty Care Research Program (RAD II), U.S. Army Medical and Research Material Command, June 17, 1996.
- Warren, Steve. "Model Based Design at Sandia National Laboratories," The University of Texas at El Paso, *MFG 390 – Special Topics in Manufacturing*. El Paso, TX, October 25, 1994.
- Warren, Steve. "Agile Product Realization at Sandia National Laboratories," *Chesapeake Electronics Show*, Holiday Inn, Greenbelt, MD. October 13, 1994.
- Warren, Steve. "Combined Ultrasound and Fluorescence Spectroscopy for Physico-Chemical Imaging of Atherosclerosis," Los Alamos National Laboratory, Chemical and Laser Sciences Division. Los Alamos, New Mexico, February 24, 1994.
- Warren, Steve. "Combined Ultrasound and Fluorescence Spectroscopy for Physico-Chemical Imaging of Atherosclerosis," Sandia National Laboratories, Program Development Department. Albuquerque, New Mexico, January 23, 1994.

OTHER PRESENTATIONS

- Warren, Steve, Don Gruenbacher, Crystal Strauss, Elizabeth Goetz, John Baybutt, and Charles Carlson. "KSU Undergraduate Biomedical Engineering Degree Program," Women in Engineering Spring Event, DUE 3099, Carl R. Ice College of Engineering, Kansas State University, February 28, 2025.
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. "The New Kansas State University Biomedical Engineering Degree Program," 2024 Kansas HOSA Fall Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, October 17, 2024.
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. "The New Kansas State University Biomedical Engineering Degree Program," 2024 Kansas HOSA Spring Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, March 27, 2024.
- Warren, Steve, and Charles Carlson. "New KSU Undergraduate Biomedical Engineering Degree Program," 2023 Health Professions Summer Program, DUE 3099, Carl R. Ice College of Engineering, Kansas State University, July 25, 2023.
- Garcia, Damian and Steve Warren. "The Potential for EEG Waves to Examine Attentiveness in Children With Special Needs," Undergraduate Research Showcase, Office of Undergraduate Research and Creative Inquiry, Kansas State University, Student Union Ballroom, April 13, 2023. Poster session.
- Garcia, Damian, Charles Carlson, and Steve Warren. "Needs Exploration at the No Stone Unturned Therapeutic Learning Center," Undergrad Research Showcase, Office of Undergraduate Research and Creative Inquiry, Kansas State University, Student Union Ballroom, July 28, 2022. Poster session.
- Warren, Steve, and Charles Carlson. "New KSU Undergraduate Biomedical Engineering Degree Program," 2022 Health Professions Summer Program, DUE 3099, Carl R. Ice College of Engineering, Kansas State University, July 11 and July 28, 2022.

- Warren, Steve, Charles Carlson, Don Gruenbacher, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, Bala Natarajan, Thomas Barstow, and Carl Ade. "KSU Undergraduate Biomedical Engineering Degree Program, Plus Potential Joint KSU/Garmin Endeavors," Discussion with Garmin Representatives, Kansas State University, Carlo R. Ice College of Engineering, April 20, 2022.
- Warren, Kelsey, Charles Carlson, and Steve Warren. "Hands-On Undergraduate BME Design Projects," *Engineering Undergraduate Research and Creative Inquiry Showcase*, Kansas State University, Carl R. Ice College of Engineering, April 17-27, 2021.
- Rice, Dallas and Steve Warren. "Autonomous Sanitation/Disinfection Drone," *Engineering Undergraduate Research and Creative Inquiry Showcase*, Kansas State University, Carl R. Ice College of Engineering, April 17-27, 2021.
- Warren, Steve, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, Daniel Andresen, John Hatcliff, Thomas Barstow, Carl Ade, and Howard Erickson. "Human and Animal Biomedical Research in the KSU Medical Component Design Laboratory," KSU/NBAF Biomedical Collaboration Meeting, Kansas State University, College of Engineering, Manhattan, KS, March 29, 2019.
- Steve Warren, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, and Don Gruenbacher. "A Snapshot of KSU ECE Endeavors with National Instruments Toolsets," Igor Alvarado, National Instruments, Kansas State University, College of Engineering, Manhattan, KS, February 21, 2019.
- Carlson, Charles, Ahmad Suliman, and Dong Ren. "Bed-Based Unobtrusive Sleep Quality Monitoring," General Meeting, Kansas State University Student Chapter of the IEEE Engineering in Medicine and Biology Society, KSU College of Engineering, February 12, 2019.
- Ren, Dong, Charles Carlson, and Steve Warren. "Algorithm to Differentiate the Pulse Rates of Two individuals that Share a Sensor-Laden Bed," *Research and the State*, Kansas State university, November 8, 2018.
- Steve Warren, David Thompson, Charles Carlson, and Dong Ren. "KSU Biomedical Engineering Program Overview," Shawnee Mission School District, Biotechnology Signature Program STEM Visit, Kansas State University, Department of Electrical & Computer Engineering, Manhattan, KS, November 1, 2018.
- Carlson, Charles and Ahmad Suliman. "Overview of the GARDE Project," General Meeting, Kansas State University Student Chapter of the IEEE Engineering in Medicine and Biology Society, KSU College of Engineering, October 8, 2018
- Carlson, Charles, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, Bala Natarajan, and Steve Warren. "A Pilot Study of an Unobtrusive Bed-Based Sleep Quality Monitor for Severely Disabled Autistic Children," K-State GRAD Forum, Kansas State University, Regnier Hall, College of Architecture, March 29, 2018.
- McKittrick, Andrew, Shangxian Wang, Charles Carlson, Punit Prakash, David Thompson, Bala Natarajan, Wayne Piersel, Janine Kesterson, Steve Stoffregen, and Steve Warren. "Enuresis Detection Using Thermal Sensing Methods," Kansas State University, College of Engineering, Undergraduate Research Poster Forum, April 26, 2018. Poster presentation.
- Washington, Derrius, Charles Carlson, and Steve Warren. "Improving Sleep Quality for Children with Autism Spectrum Disorder," Developing Scholars Poster Forum, Kansas State University, K-State Union Ballroom, April 15, 2018.
- Carlson, Charles, Alexandra Lyle, Jake Chappell, Mariah Brown, Brie Phillips, and Steve Warren. "The KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society," Engineering Scholars Day, KSU College of Engineering, Manhattan, KS, October 24, 2017.
- Wang, Shangxian, Charles Carlson, Punit Prakash, David Thompson, Bala Natarajan, Wayne Piersel, Rachel Boulevard, Steve Stoffregen, and Steve Warren. "Enuresis Detection Via Thermocouple Data Analysis," Kansas State University, College of Engineering, Undergraduate Research Poster Forum, April 27, 2017. Poster presentation.

- Carlson, Charles and Ahmad Suliman. “Unobtrusive Bed Based Ballistocardiogram Monitor for Severely Disabled Autistic Children,” Kansas State University, College of Engineering, Open House, March 31 – April 1, 2017.
- Warren, Steve. “Biomedical Research in the K-State Department of Electrical and Computer Engineering: Wearable Wireless Medical Devices and Applications,” Kansas State University, Electrical & Computer Engineering, August 29, 2016. Information exchange with KSU Animal Sciences.
- Ortega, Martin, Steve Warren, and Charles Carlson. “Ambient Sensor Research for Sleep Quality Environmental Assessment,” Research Immersion: Pathways to STEM (RiPS) Poster Session, KS-LSAMP Program, -Kansas State University, Union Ballroom, July 27, 2016.
- Wang, Shangxian, Taishan Li, Charles Carlson, Punit Prakash, David Thompson, Bala Natarajan, Wayne Piersel, Janine Kesterson, Steve Stoffregen, and Steve Warren. “Enuresis Detection Using Thermal Sensing Methods,” Kansas State University, College of Engineering, Undergraduate Research Poster Forum, April 15, 2016. Poster presentation.
- White, Austin, Charles Carlson, and Steve Warren. “Electrodermal Activity,” Kansas State University, College of Engineering, Undergraduate Research Poster Forum, April 15, 2016. Poster presentation.
- Carlson, Charles, Ahmad Suliman, Punit Prakash, David Thompson, and Steve Warren. “Sensor Beds for Tracking Sleep in Severely Disabled Children,” Kansas State University, Electrical & Computer Engineering, General Engineering Course, March 8, 2016.
- Wang, Shangxian, Charles Carlson, Punit Prakash, David Thompson, Bala Natarajan, Wayne Piersel, Janine Kesterson, and Steve Warren. “Enuresis Detection by Way of Thermocouple Data Analysis,” Kansas State University, College of Engineering, Undergraduate Research Poster Forum, December 3, 2015. Poster presentation.
- Warren, Steve, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, Ahmad Suliman, Tianyu Lin, and Shangxian Wang. “NSF GARDE Project: Overview and Next Steps,” Heartspring, Wichita, KS, November 13, 2015.
- Warren, Steve. “American Society for Engineering Education: Overview and KSU Involvement,” Kansas State University, College of Engineering, ECE Garmin Conference Room, May 7, 2015.
- Shiach, Job, Julia Nyiro, Megan Richards, and Steve Warren. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society,” ECE Advisory Council, Kansas State University, March 27, 2015.
- Warren, Steve. “Trip Debrief: IEEE EMBC 2014, Chicago, IL,” Bio/Edu Seminar, Kansas State University, Electrical & Computer Engineering, Manhattan, KS, October 15, 2014.
- Warren, Steve. “Med/Bio Projects Supported by the KSU Medical Component Design Laboratory,” Meadowlark Hills Administration, KSU ECE, Manhattan, KS, August 18, 2014.
- Warren, Steve. “Med/Bio Projects Supported by the KSU Medical Component Design Laboratory,” Dimensional Innovations, KSU ECE, Manhattan, KS, July 14, 2014.
- White, Austin, Sergio Ortiz, Punit Prakash, and Steve Warren. “Monitoring Nighttime Well Being of Children with Disabilities,” *14th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 13, 2014, K-State Union Ballroom, Manhattan, KS.
- Ortiz, Sergio, Austin White, Punit Prakash, and Steve Warren. “Using Commercial Tools to Validate a Bed Sensor Suite for Nighttime Monitoring of Severely Disabled Children,” *14th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 13, 2014, K-State Union Ballroom, Manhattan, KS.
- Luan, Shiwei and Steve Warren. “Modularized PCA Pump Design,” KSU Engineering Scholars Day, KSU College of Engineering, Manhattan, KS, October 30, 2013.
- Anderson, Kaleb and Steve Warren. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society,” ECE Advisory Council, Kansas State University, April 5, 2013.

- Warren, Steve, Xiongjie Dong, Chen Jia, and Tianyu Lin. “Biomedical Devices for Space Applications,” Bioengineering Session, E3 Workshop for Science Teachers Hosted by the KSU College of Engineering, Manhattan, KS, Bette Grauer, Organizer, May 30, 2013.
- College of Engineering Open House, Kansas State University, Manhattan, KS, April 19–20, 2013. NASA EPSCoR and NASA Crew Performance Projects.
- White, Austin, Hannah Gray, Steve Warren, Wen Song, Kaleb Anderson, Gregory Nitcher, and Neelou Hadavandifard. “Monitoring Night-Time Well Being of Disabled Children at Heartspring,” *13th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 21, 2013, K-State Union Ballroom, Manhattan, KS.
- Gray, Hannah, Steve Warren, Punit Prakash, Wen Song, Kaleb Anderson, Gregory Nitcher, Austin White, and Neelou Hadavandifard. “Bed Sensor Suite Project: Nighttime Wellness Monitoring for Children at Heartspring,” *13th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 21, 2013, K-State Union Ballroom, Manhattan, KS.
- Garman, Kevin, Timothy Gustafson, James Thoennes, Punit Prakash, and Steve Warren. “Weight Distribution Analysis for Patients Targeted with Unilateral Lower-Limb Orthotics or Prosthetics,” *13th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 21, 2013, K-State Union Ballroom, Manhattan, KS.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Suman Saripalli, KalScott Engineering, Intellispeak, Kansas State University, Electrical & Computer Engineering, April 9, 2013.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Jim Baxendale, Whiteboard to Boardroom Regional Commercialization Program, UMKC Innovation Center, Kansas State University, Electrical & Computer Engineering, March 7, 2013.
- Kuhn, Bill, Amelia Hodges, Xiongjie Dong, and Steve Warren. NASA Project Overview, ECE Department, Engineering Scholarship Day, February 15, 2013.
- Li, Kejia Li, Xiongjie Dong, Dana Gude, Chenyu Zheng, Devon Krenzel, Steve Warren, and Don Gruenbacher. Bioengineering Session, E3 Workshop for Science Teachers Hosted by the KSU College of Engineering, Bette Grauer, Organizer, May 31, 2012.
- Barstow, Tom, Carl Ade, Ryan Broxterman, Steve Warren, Dana Gude, Wen Song, Tommy Nelson, Dale Schinstock, and Tim Mourlam. “KSU Research Update: Standardized “Pre-Flight” Exercise Tests to Predict Performance During Extravehicular Activities in a Lunar Environment,” KSU Student Chapter of the IEEE EMBS, KSU College of Engineering, April 5, 2012.
- Warren, Steve and Timothy Gustafson. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society,” EECE Advisory Council, Kansas State University, March 30, 2012.
- Warren, Steve. “Kansas State University Department of Electrical & Computer Engineering,” KSU Electrical & Computer Engineering, March 27, 2012. Information exchange with National Instruments (Robert Canik, Brian Platt, Mark Troutfetter, and Jim Schwartz).
- Krenzel, Devon, Steve Warren, Bala Natarajan, and Gurdip Singh. “Cerner Slips and Falls Project,” Manhattan, KS, March 1, 2012.
- Warren, Steve, Dana Gude, Tommy Nelson, Wen Song, Devon Krenzel, and Kejia Li. “Standardized “Pre-Flight” Exercise Tests to Predict Performance During Extravehicular Activities in a Lunar Environment - KSU Electrical & Computer Engineering Activity Summary,” NASA HRP Mid-Contract Review, Kansas State University, Electrical & Computer Engineering, January 26, 2012.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Kim Fowler, Kansas State University, Electrical & Computer Engineering, November 14, 2011.

- Warren, Steve. “Service Design Projects Sponsored by the IEEE EMBS Student Chapter at Kansas State University”, Gerontology Alumni Open House, Meadowlark Hills, Manhattan, KS, October 28, 2011.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Carl Mayer and Chris Justice, RBC Medical Innovations, Kansas State University, Electrical & Computer Engineering, October 27, 2011.
- Warren, Steve, Caterina Scoglio, Bala Natarajan, Noel Schulz, and Sanjoy Das. “Kansas State University Department of Electrical & Computer Engineering,” The University of Texas at Austin, Department of Electrical Engineering, Engineering and Science Building, October 3, 2011. Recruiting presentation for the ECE open position.
- “Biosensor Networks and Telecommunication Subsystems for Long-Duration Missions, EVA Suits, and Robotic Precursor Scout Missions - Coordinated Research Spanning Early-Stage Innovations through Game Changing Technologies,” Kansas State University, Electrical & Computer Engineering, September 16, 2011. NASA EPSCoR project presentation/discussion with Steve Bowen, NASA astronaut.
- Li, Kejia, Steve Warren, and John Hatcliff. CIMIT Open House Demonstrations: “MDCF Demo 1: Platform-Oriented Device” and “MDCF Demo 2: Online Reconfigurable Device,” Boston, MA, Sept. 6-8, 2011. “Medical Platform Oriented Device Demo” (video) at <http://www.youtube.com/watch?v=fj-rOFNy6kk> created in association with the 3rd Annual Medical Device Connectivity Conference and Exhibition, Joseph B. Martin Conference Center at Harvard Medical School, Boston, MA, Sept. 8–9, 2011.
- Warren, Steve. “NSF RAPD Proposal Summary & Early Thoughts Regarding KSU Senior Design Projects to Aid Heartspring Children,” Heartspring, Wichita, KS, August 26, 2011.
- Mendenhall, Kekoa, Xiongjie Dong, and Steve Warren. “Controlling Future Devices ... With Your Eyes?,” Developing Scholars Program, College of Engineering, Kansas State University, April 17, 2011.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” William Graveman, Biomedical Devices of Kansas, March 29, 2011.
- Warren, Steve. “KSU Medical Component Design Laboratory Focus Areas,” Heartspring Visit, Kansas State University, Electrical & Computer Engineering, September 24, 2009.
- Warren, Steve. “KSU Medical Component Design Laboratory Focus Areas,” Shulz Family Tour of Engineering Laboratories, Kansas State University, Electrical & Computer Engineering, September 23, 2009.
- *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 17–18, 2009.
 - Warren, Steve. “The Bioengineering Option in Electrical Engineering”
 - Moratto, Zachary, Tyler Kreybiel, Paul Edgington, Steve Taylor, Zane Pauls, and Zachary Maier, and Steve Warren. “Introduction to Computer Graphics”
 - Hoskins, Seth and Steve Warren. “Wireless Magnetic Link for a Cattle Physiology Monitoring Pill”
 - Li, Kejia and Steve Warren. “Wireless Pulse Oximeter”
 - Nagaraja, Ashvin, Kejia Li, Travis Upah, Kentaro Takamatsu, Chen Jia, Grant Damas, and Steve Warren. “Vital Sign Sensing Pad”
 - Gruber, Lucinda, Adriann Sullivan, Connor Griffith, Ethan Young, and Steve Warren. “The KSU Student Chapter of the IEEE EMBS”
 - Takamatsu, Kentaro and Steve Warren. “Greenhouse Monitoring Project”
- Warren, Steve, Connor Griffith, Lucinda Gruber, and Ethan Young. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society,” EECE Advisory Council, Kansas State University, April 3, 2009.
- Hatcliff, John, Daniel Andresen, Andrew King, Sam Proctor, and Steve Warren. “A Publish-Subscribe-Based Device Integration Framework,” Food & Drug Administration, Rockville, MD, July 29, 2008.

- Bala Natarajan and Steve Warren. “Applied Biosecurity Research in the KSU Electrical & Computer Engineering Department,” Kansas State University, Biosecurity Research Institute, Sceptor Industries, April 30, 2008.
- Rome, Nicholas. “Reflective Pulse Oximetry: Creation of a Recording VI,” Poster Session, Campus Student Internship Program, Kansas State University, Fiedler Auditorium, April 23, 2008.
- Warren, Steve. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society,” EECE Advisory Council, Kansas State University, April 4, 2008.
- Gruenbacher, Don, Bala Natarajan, and Steve Warren. “Applied Biosecurity Research in the KSU Electrical & Computer Engineering Department,” Kansas State University, Biosecurity Research Institute, Raytheon Corporation, March 26, 2008.
- Warren, Steve. “Electrical & Computer Engineering,” *DEN 160 – Engineering Concepts*, Paslay Lecture Hall, Kansas State University, March 25, 2008.
- Warren, Steve. “The Biomedical Option in Electrical Engineering,” *EECE 015 – New Student Assembly*, Paslay Lecture Hall, Kansas State University, November 7, 2007.
- Warren, Steve. “Pulse Oximeter Designs in the Medical Component Design Laboratory,” Medical Component Design Laboratory, Kansas State University, October 9, 2007.
- Kevin Smith, Angel Martinez, Patrick York, Roland Craddolph, Steve Warren, Daniel Andresen, Tammi Hildreth, and Howard Erickson. “Veterinary Telemonitoring for Disease Prevention in Cattle Herds,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 7–8, 2006.
- Steve Warren. “The Bioengineering Option in Electrical Engineering,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 7–8, 2006.
- Steve Warren. “The Multimedia Option in Computer Engineering,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 7–8, 2006.
- Warren, Steve and Daniel Andresen. “Human and Animal Health Monitoring,” High School Programming Contest, Computing & Information Sciences, KSU Union Ballroom, Kansas State University, November 4, 2005.
- Warren, Steve. “Health Monitoring Efforts in the Medical Component Design Laboratory,” Graduate Student Recruitment Fair, Kansas State University Graduate School, Oct. 14–15, 2005.
- Warren, Steve, Jianchu Yao, Kevin Smith, Jeffrey Lebak, Balakumar Krishnamurthi, and Daniel Andresen. “Embedded Monitoring Systems for Human & Animal Health Assessment,” Research Experiences for Teachers Workshop, Medical Component Design Laboratory, Kansas State University, June 30, 2005.
- Steve Warren. “The Bioengineering Option in Electrical Engineering,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 8–9, 2005.
- Kevin Smith, Angel Martinez, Scott Schoenig, Patrick York, Roland Craddolph, Dominic Gelinas, Steve Warren, Daniel Andresen, Tammi Hildreth, David Poole, and Howard Erickson. “Veterinary Telemonitoring for Disease Prevention in Cattle Herds,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 8–9, 2005.
- Wareing, Austin and Steve Warren. “Optimization of Reflectance-Mode Pulse Oximeter Sensors,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 8–9, 2005.
- Schoenig, Scott, Kevin Smith, Dominic Gelinas, Balakumar Krishnamurthi, Tammi Hildreth, Howard Erickson, Mark Spire, David Poole, Daniel Andresen, and Steve Warren. “Veterinary Telemonitoring for Disease Prevention in Cattle,” *Capitol Research Forum*, Capitol Building, Topeka, KS, March 1, 2005. Poster presentation.
- Yang, Yong, and Steve Warren. “Automated Homework Generation and Assessment for Engineering Courses,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 2–3, 2004. Poster presentation.
- Luke Nagl, Scott Schoenig, Balakumar Krishnamurthi, Roland Craddolph, Steve Warren, Daniel Andresen, Tammi Hildreth, David Poole, Mark Spire, and Howard Erickson. “Veterinary

Telemonitoring for Disease Prevention in Cattle Herds,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 2–3, 2004. Poster presentation and equipment demonstration.

- Yao, Jianchu and Steve Warren. “Wearable and Wireless Medical Devices Designed with Interoperability Standards,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 2–3, 2004. Poster presentation and equipment demonstration.
- Luke Nagl and Tammi Hildreth. “BMOO: Bovine Mobile Observation Operation,” Research Experience for Teachers seminar, Mitch Nielsen, Organizer, June 25, 2003.
- Luke Nagl, Ryan Schmitz, Jianchu Yao, Steve Warren, Ravi Kollarapu, Daniel Andresen, Tammi Hildreth, David Poole, and Howard Erickson. “Veterinary Telemonitoring for Disease Prevention in Cattle Herds,” *College of Engineering Open House*, Kansas State University, Manhattan, KS, March 28–29, 2003. Poster presentation.
- Warren, Steve. “Telemedicine Project Overview,” Telemedicine and Advanced Technology Research Center, U.S. Army Medical Research and Materiel Command, Fort Detrick, Frederick, MD, March 19, 1999.
- Warren, Steve. “Telemedicine Vision and Project Summary,” Sandia National Laboratories, For Gerry Yonas, Vice President - Systems, Science, and Technology Division, February 15, 1999.
- Warren, Steve. “Portable Owner-Aware Biometric Access Control Device,” Sandia National Laboratories LDRD Surety Review, July 22, 1998.
- Warren, Steve, Ronald P. Kensek, and Leonard J. Lorence. “ITS-ACIS: Radiation Transport in CAD/CAM/CAE Geometries,” Sandia National Laboratories, Quarterly Meeting of the Neutron Generator Hostile Environment Project, April 22, 1998.
- Sill, Anthony E. and Steve Warren. “The Role of Technology in Reducing Health Care Costs,” Sandia National Laboratories, February 19, 1998.
- Warren, Steve. “Model Based Design in Defense Systems: High-Fidelity Electromagnetic Analysis,” Boeing Aerospace. Satellite link video conference, March 28, 1995.
- Warren, Steve. “Model-Based Design at Sandia National Laboratories,” Ansoft Corporation, Pittsburgh, PA, February 21, 1995.
- Warren, Steve. “Distress Monitor Conceptual Review: Requirements and Theory,” Sandia National Laboratories, September 26, 1994.
- Warren, Steve, and Randall B. Smith. “Virtual Prototyping at Sandia National Laboratories,” Workshop on U.S. Electronics Manufacturing, National Electronics Manufacturing Initiative, Georgetown University, Washington, D.C., July 25, 1994.

INVITED WORKSHOPS/MEETINGS

- **Information Exchange with Matt Dodge**, Biosense Webster, Kansas City, MO, October 25, 2022, January 20, 2023
- **Information Exchange with Rachelle White**, TRIMEDX, Salina, KS January 20, 2023
- **Information Exchange with IEEE EMBS Student Chapter**, University of North Florida, Jacksonville, FL, November 7, 2022
- **Information Exchange with Ceres Tag Representatives**, Kansas State University, Research Park Circle, Manhattan, KS, August 9, 2022.
- **Information Exchange with Karl McCombs**, Children’s Mercy Hospital, Kansas City, MO, July 7, 2022
- **Information Exchange with Garmin Representatives**, Kansas State University, Carl R. Ice College of Engineering, Manhattan, KS, April 20, 2022.
- **Information Exchange with Ron Pellegrini**, Vice President, Head of Field Service LD/POC, Siemens Healthineers, Kansas State University, College of Engineering, Manhattan, KS, October 14, 2021.
- **Information Exchange with Nick Fowler**, Mechanical Engineering, Engenious Design, Zoom Meeting, KS, October 13, 2021.

- **Information Exchange with Lee G. Willis**, Senior Project Manager and Safety System Engineer, Johnson Space Center, NASA, Kansas State University, College of Engineering, Manhattan, KS, October 7, 2021.
- **Information Exchange with Mike Guillaume**, Site Manager, TRIMEDX, Ascension Via Christi, Manhattan, KS, September 23, 2021.
- **Information Exchange with John Hrynuk**, U.S. Army Research Laboratory, Kansas State University, College of Engineering, Manhattan, KS, April 23, 2019.
- **Information Exchange with Dr. Steve Kappes, Dr. Beth Lautner, Diana Whipple, and Dr. Pat Glas**, U.S. Department of Agriculture, KSU/NBAF Biomedical Collaboration Opportunities, Kansas State University, College of Engineering, Manhattan, KS, March 29, 2019.
- **Information Exchange with Igor Alvarado**, National Instruments, KSU College of Engineering, Manhattan, KS, February 21, 2019.
- **Information Exchange with Dr. Steve Eckels**, KSU Mechanical & Nuclear Engineering, Manhattan, KS, February 6, 2017.
- **Information Exchange with Dr. Tim Rozell and Dr. Ashley Rhodes**, KSU Animal Sciences, Manhattan, KS, January 30, 2017.
- **Information Exchange with Netsmart**, KSU College of Engineering, November 14, 2016.
- **Information Exchange with Dr. Lindsey Hulbert**, KSU Animal Sciences, Manhattan, KS, August 29, 2016.
- **Information Exchange with SORA Medical**, Kansas State University, Olathe Campus, Olathe, KS, April 27, 2016. Opportunities related to the SORA Medical ApolloMED clinician app.
- **Information Exchange with i3D Plus**, Salina, KS, June 7, 2015. True 3D imaging technology and biomedical applications.
- **Information Exchange with Dennis and Malinda Dillman**, KSU ECE Department, September 30, 2014. KSU NASA projects.
- **Healthcare Robotics Expert Panel**, Kansas City Chapter of the IEEE EMBS, Embassy Suites, Overland Park, KS, September 11, 2014. K-State Salina, K-State Manhattan, Pittsburg State University.
- **Information Exchange with Meadowlark Hills**, Meadowlark Hills Retirement Community, July 16, 2014; August 4, 2014; August 18, 2014.
- **Information Exchange with Kevin Rooney, Dimensional Innovations**, KSU ECE Department, July 14, 2014.
- **Information Exchange with National Instruments**, KSU ECE Department, September 24, 2013.
- **Information Exchange with KalScott Engineering**, KSU ECE Department, April 9, 2013.
- **Information Exchange with National Instruments**, KSU ECE Department, April 2, 2013.
- **Information Exchange with Jim Jaax**, KSU ECE Department, March 29, 2013.
- **Information Exchange with UMKC Innovation Center**, KSU ECE Department, March 7, 2013.
- **NSF RAPD Design Project Reviews**, KSU ECE Department, October 9, 2012.
- **Information Exchange with National Instruments**, KSU ECE Department, March 27, 2012.
- **Information Exchange with SANY Corporation**, Changsha, China, May 29, 2012.
- **Information Exchange with Beihang University**, Beijing, China, May 23, 2012.
- **Information Exchange with Jiaotong University**, Dalian, China, May 22, 2012.
- **Mid-Contract Review**, NASA Johnson Space Center (Linda Loerch, Lori Ploutz-Snyder, and Jeffrey Ryder), KSU ECE Department, January 24, 2012.
- **Information Exchange with Kim Fowler**, KSU ECE Department, November 14, 2011.
- **Information Exchange with Carl Mayer and Chris Justice, RBC Medical Innovations**, KSU ECE Department, October 27, 2011.
- **Information Exchange with Michael Duncan, NASA Flight Surgeon**, KSU ECE Department, September 30, 2011.

- **Information Exchange with Steve Bowen, NASA Astronaut**, KSU ECE Department, September 16, 2011.
- **Information Exchange with Colorado Engineering**, KSU ECE Department, May 23, 2011.
- **Information Exchange with William Graveman, Biomedical Devices of Kansas**, KSU ECE Department, March 29, 2011.
- **Information Exchange with Hill's Pet Nutrition**, KSU ECE Department, November 29, 2010.
- **Information Exchange with Robert Bestgen**, KSU ECE Department, November 19, 2010.
- **Information Exchange with Black & Veatch**, KSU ECE Department, September 29, 2010.
- **Information Exchange with Honeywell**, Honeywell Federal Manufacturing and Technologies, Kansas City Plant, 2000 East 95th St., Kansas City, MO 64131-6159, August 19, 2010.
- **Funding Opportunities with Cerner Corporation**, Cerner Innovation Campus, Kansas City, MO, November 20, 2009.
- **Information Exchange with Heartspring Corporation**, KSU ECE Department, September 24, 2009.
- **Medical Component Design Laboratory Overview**, Heartspring, KSU ECE Department, September 24, 2009.
- **Medical Component Design Laboratory Overview**, Kirk Shulz Family Tour, KSU Engineering Laboratories, September 23, 2009.
- **Advisory Council Exchange with Heartspring**, Heartspring Campus, Wichita, KS, April 6, 2009.
- **Information Exchange with Rick McMullen**, Kansas University Director of Research Computing, Medical Component Design Laboratory, January 21, 2009.
- **MDBus Dialogue with Cerner Corporation**, 236 Nichols Hall, Kansas State University, December 17-18, 2008.
- **KSU Engineering Research Focus Meeting**, Edwards Conference Room, Fiedler Hall, Kansas State University, Manhattan, KS, September 2, 2008. Research investment areas for the KSU College of Engineering.
- **Opportunity Discussion with Digital Rays** (Abhi Ray, CEO), KSU Health IT Working Group, KSU Lafene Health Center, and Meadowlark Hills, Manhattan, KS, August 5, 2008. Opportunities related to Manhattan as a home base for the Digital Rays company.
- **Project Discussion with Meadowlark Hills Retirement Community** (Steve Shields, CEO) **and Elite Care**, Manhattan, KS, July 29, 2008. Discussion regarding the requirements for the implementation of a home care and medical information system at Meadowlark Hills.
- **Information Exchange with Colonel Wayne Parks**, Great Plains Cyber Electronic University Consortium, Kansas State University, Manhattan, KS, July 9, 2008.
- **Information Exchange with Meadowlark Hills Retirement Community** (Steve Shields, CEO), Manhattan, KS, May 19, 2008.
- **Information Exchange with the Kansas Foundation for Medical Care** (KFMC, Larry Pitmann, President and CEO) **and Lead Horse Technologies**, (Junction City, KS), Kansas State University, May 8, 2008.
- **Information Exchange with Cerner Corporation and Meadowlark Hills**, Kansas State University, Health IT Working Group, May 5, 2008.
- **Information Exchange with Sceptor Industries**, Kansas State University, Biosecurity Research Institute, April 30, 2008.
- **Information Exchange with Raytheon Corporation**, Kansas State University, Biosecurity Research Institute, March 26, 2008.
- Meeting with **HeartSpring** in Wichita, KS, March 14, 2008. Discussions regarding sensor fusion and the assessment of health and behavior in severely disabled children.
- **Architecture Dialogue with Cerner Corporation**, 236 Nichols Hall, Kansas State University, January 22-24, 2008.
- **Information Exchange for Possible Health Care Joint Ventures**, Rebecca S. MacKinnon and Michie P. Slaughter, 102 Fairchild Hall, Kansas State University, January 22, 2008.

- **Information Exchange with Cerner Corporation**, Kansas City, MO, December 17, 2007. Potential KSU-Cerner collaborations.
- **Health Care Advisory Committee Meeting**, Department of Industrial and Manufacturing Systems Engineering, Edwards Conference Room, Kansas State University, October 25, 2007.
- Host for Robert Wald, Professional Performance Award, **Seaton Society Banquet**, October 21–22, 2006.
- Meeting with **HeartSpring** in Wichita, KS, June 13, 2006. Discussions regarding how technology can help children that have severe mental and physical disabilities.
- **Meeting with the Provost and Associate Provost**, February 28, 2006. Lunch meeting to discuss research opportunities and barriers.
- **Leadership Appreciation Reception**, Blue Key Honor Society, Beach Museum of Art, January 31, 2006.
- **NSF IGERT Workshop**, Wichita State University, November 14, 2005.
- Trip to **Midwest Research Institute**, November 5, 2005. Day-long meeting to discuss biosecurity work between MRI and KSU.
- **Pre-ICADI Workshop on Technology for Aging, Disability, and Independence**, The Royal Academy of Engineering, 29 Great Peter Street, Westminster, London SW1P 3LW (England), June 26-27, 2003, <http://www.cise.ufl.edu/~helal/preICADI>.
- **Information Meeting. Research to Aid Persons with Disabilities**, National Science Foundation, Arlington, VA, June 28-29, 2001.
- **FDA Workshop on Home Care Technology Trends**, U.S. Food and Drug Administration, Rockville, MD, January 25, 2001.
- Warren, Steve. “The Health Care Facelift: How Telecommunications Technology, Distributed Computing, and Intelligent Medical Devices will Redefine the Patient-Physician Relationship,” **Four-hour short course on smart health care technology** prepared for the Department of Biomedical Engineering, University of Minnesota, Twin Cities Campus, December 1, 1999.
- **Workshops on Future Medical Devices: Home Care Technologies for the 21st Century**. Sponsored by the National Science Foundation, the Catholic University of America, and the Food and Drug Administration. Mini-Workshop: Center for Devices and Radiological Health (CDRH), Rockville, MD, December 6-7, 1998. Full Workshop: Department of Health and Human Services (DHHS), Rockville, MD, April 7-9, 1999. **Committee Chair: Smart Devices and the Home of the Future**. Smart devices and advanced technologies (civilian and government) that the National Science Foundation should fund through 2010.
- **Texas A&M Program Development Meetings** with Upper Management at Sandia National Laboratories, December 11, 1998. Discussions with Jim Wall, Ph.D., Director of Computing and Information Technology, TEES Texas Center for Applied Technology. Information architecture and surety needs for telemedicine systems used for home health and emergency response.
- **High-Performance Healthcare**. HOLON ATP Program Review sponsored by the Koop Foundation, University of Maryland Shady Grove Conference Center, Rockville, MD, June 17-18, 1998. Internet and security technology dedicated to remote health care delivery.
- **Computational Science Initiative Workshop**, Department of Energy, Office of Energy Research, Lawrence Berkeley National Laboratory Office in Washington, D.C., Jan. 28-29, 1998. Fiscal year 2000 initiative for advanced computation in the Department of Energy.
- **Army Natick Research, Development & Engineering Center Program Summary**, Sandia National Laboratories, Albuquerque, NM, February 26, 1998. Battlefield monitoring of soldier state-of-health and chemical-biological preparedness.
- Dr. John Rather, Director of Innovative Development, **Karmanos Cancer Institute**, Meeting with Upper Management of Sandia National Laboratories, August 28-29, 1997. The need for the national laboratories to apply substantial computational resources to high-resolution body imaging and subsequent cancer prevention.

- **Workshop on Lower Limb Prosthetics**, Co-sponsored by the National Center for Medical Rehabilitation Research (NCMRR), and National Institute of Child Health & Human Development, the National Institute of Health (NIH), and Sandia National Laboratories, Sandia National Laboratories, Albuquerque, NM, November 8-10, 1995. Technologies for assisted living.

INVITED SPEAKERS

- Leet, Corey. “NAMSA,” NAMSA, Minneapolis, MN, Kansas State University, KSU IEEE EMBS Student Chapter, April 9, 2026.
- Edwards, Nicholas. “BD. Advancing the World of Health,” Becton Dickinson, Columbus, NE, Kansas State University, KSU IEEE EMBS Student Chapter, March 26, 2026.
- Armstrong, Lauren and Ashley Hawkins. “Vantage MedTech,” Vantage MedTech, Lenexa, KS, Kansas State University, KSU IEEE EMBS Student Chapter, March 12, 2026.
- Goetz, Elizabeth. “Aesthetics Biomedical Innovation Center,” Aesthetics Biomedical, Scottsdale, AZ, Kansas State University, KSU IEEE EMBS Student Chapter, February 12, 2026.
- Dear, Adam. “Biosense Webster Electrophysiology Support,” Biosense Webster, Johnson & Johnson Medtech, Kansas City, KS, Kansas State University, KSU IEEE EMBS Student Chapter, December 4, 2025.
- Ryckman, Brian. “IEEE Education and Career Roles,” IEEE, East Lansing, MI, Kansas State University, KSU IEEE EMBS Student Chapter, November 17, 2025.
- Fischer, Kenneth. “KU Bioengineering Program,” University of Kansas, Lawrence, KS, Kansas State University, KSU IEEE EMBS Student Chapter, October 23, 2025.
- Braynock, Sophie. “BD: Advancing the World of Health,” Becton Dickinson, Kansas State University, KSU IEEE EMBS Student Chapter, October 16, 2025.
- Brandt, Tegan. “KLS Martin Surgical Innovations,” KLS Martin, Jacksonville, FL, Kansas State University, KSU IEEE EMBS Student Chapter, October 8, 2025.
- Leet, Corey. “NAMSA,” NAMSA, Minneapolis, MN, Kansas State University, KSU IEEE EMBS Student Chapter, April 10, 2025.
- Eastling, Marissa. “Integrating Language Acquisition Motor Planning (LAMP) into Literacy Education,” Special Education Teacher, Farmington School District, Farmington, MN, *BME 200 – Introduction to Biomedical Engineering*, Kansas State University, March 31, 2025.
- Armstrong, Lauren and Mason Diebold. “Vantage MedTech,” Vantage MedTech, Lenexa, KS, Kansas State University, KSU IEEE EMBS Student Chapter, March 27, 2025.
- Cerutti McGinnis, Allison Kathleen. “Orthotics and Prosthetics Careers,” Northwestern University Prosthetics-Orthotics Center, Chicago, IL, Kansas State University, KSU IEEE EMBS Student Chapter, March 13, 2025.
- Griffin, Tanner. “Engenious,” Engenious Design, Prairie Village, Kansas City, KS, Kansas State University, KSU IEEE EMBS Student Chapter, February 27, 2025.
- Bledsoe, Heather, Melissa Frock, and Adam Frock. “Spinal Simplicity,” Spinal Simplicity, Overland Park, KS, Kansas State University, KSU IEEE EMBS Student Chapter, January 30, 2025.
- Dear, Adam. “Biosense Webster Opportunities,” Biosense Webster, Kansas City, MO, Kansas State University, *BME 575 – Clinical Systems Engineering*, December 5, 2024.
- Colling, Tom. “DEKA Projects and Life After College,” DEKA Research and Development Corporation, Kansas State University, *BME 575 – Clinical Systems Engineering*, November 22, 2024.
- Fischer, Ken. “Learn About Graduate School and KU Bioengineering,” The University of Kansas, Kansas State University, *BME 575 – Clinical Systems Engineering*, October 14, 2024.
- Prusener, Derek, Erin Piacetelli, and Camden Johnson. “Meet Medtronic,” Medtronic, Inc., Kansas State University, KSU IEEE EMBS Student Chapter, September 12, 2024.
- Torbord, Alan. “Boston Scientific,” Boston Scientific, Minneapolis, MN, Kansas State University, KSU IEEE EMBS Student Chapter, April 18, 2024.

- Kodanaz, Tyler and Megan Kinnane. “Opportunities with Ingenious Design,” Ingenious Design, Kansas State University, KSU IEEE EMBS Student Chapter, April 4, 2024.
- Bradley, Janae. “Engineering Efforts at Hill’s Pet Nutrition,” Hill’s Pet Nutrition, Kansas State University, KSU IEEE EMBS Student Chapter, March 21, 2024.
- Leet, Cory. “Medical Device Testing at NAMSA,” NAMSA, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Presentation, February 22, 2024.
- Frizzell, Bradie. “Xenon MRI,” Kansas University Medical Center, Kansas State University, KSU IEEE EMBS Student Chapter, February 8, 2024.
- Griffith, Craig. “IEEE: Who We Are and How We Can Help,” IEEE, Tucson, AZ, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, November 30, 2023.
- Warren, Kelsey. “Clinical Engineering at Children’s Mercy Hospital,” Children’s Mercy Hospital, Kansas City, MO, Kansas State University, KSU IEEE EMBS Student Chapter, November 9, 2023.
- Shane, Douglas. “Technology for Animal Health,” Kansas State University, Olathe, KS, Kansas State University, KSU IEEE EMBS Student Chapter, October 19, 2023.
- Dodge, Matt. “Electrophysiology” Biosense Webster, Kansas State University, KSU IEEE EMBS Student Chapter, October 5, 2023
- Jansen, Michael. “Remote Patient Monitoring,” Biointellisense, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, September 21, 2023
- Medtronic. “The Role of an Engineering in a Medical Device Company,” Kansas City, KS, Kansas State University, KSU IEEE EMBS Student Chapter, May 4, 2023.
- Reagan, Corbin and Jeff Chininis. “Complex Medical Devices,” RBC Medical Innovations, Olathe, KS, Kansas State University, KSU IEEE EMBS Student Chapter, April 6, 2023.
- Friedrichs, Dan. “Medical Device Industry Operations, Minnetronix, St. Paul, Minnesota, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, February 23, 2023.
- Synexsis. Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, January 26, 2023.
- Gould, Grant and Taylor Rogalia. “Biosense Webster,” Biosense Webster, Kansas City, KS, Topeka, KS, Kansas State University, KSU IEEE EMBS Student Chapter, November 10, 2022.
- Griffith, Craig. “IEEE: Who We Are and How We Can Help,” IEEE, Tucson, AZ, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, October 13, 2022.
- White, Rachelle. “TRIMEDX,” TRIMEDX, Manhattan, KS, Kansas State University, KSU IEEE EMBS Student Chapter, April 7, 2022.
- Golway, Michael. “3D Tissue Printing Systems,” President and CEO, Advanced Solutions, Louisville, KY, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, December 2, 2021.
- Baird, Keith. “You’ve Got a Product Idea. So What ...,” President, Biomedical Devices of Kansas, Tonganoxie, KS, Kansas State University, KSU IEEE EMBS Student Chapter, November 18, 2021.
- Dear, Adam. “Products and Software for Cardiac Ablation Surgery,” Regional Technology Specialist, Biosense Webster, Kansas City, KS, Kansas State University, KSU IEEE EMBS Student Chapter, October 7, 2021.
- Ybarra, Anthony. “KSU Career Center,” Kansas State University, BME 001 – New Student Assembly, Online, October 5, 2021.
- Chappell, Jake. “Infection Control Technology,” Mechanical Engineer, HEPACART, Lenexa, KS, Kansas State University, KSU IEEE EMBS Student Chapter, September 23, 2021.
- Weiss, Mark. “Machine Learning for Biologists,” Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, April 22, 2021.
- Glick, Stephen. “Emerging Technologies in X-Ray Imaging of Breast Cancer,” Office of Science and Engineering Laboratories, Center for Devices and Radiological Health, U.S. Food and Drug

Administration, Silver Spring, MD, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, April 8, 2021.

- Lee, Taehee. “Early Detection of Psychiatric Disorders,” iMediSync, Seoul, South Korea, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, March 11, 2021.
- Crosby, Petra, Katie Ostmeyer, and Gilbert Magallanes. “NSUTLC Services and Technology Needs,” No Stone Unturned Therapeutic Learning Center, Manhattan, KS, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, March 11, 2021.
- Griffith, Craig. “IEEE Resources, Activities, and Student Membership,” IEEE, San Francisco, CA, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, February 11, 2021.
- Ybarra, Anthony. “KSU Career Center,” Kansas State University, BME 001 – New Student Assembly, Online, December 6, 2020.
- Hatesohl, Brian. “Biotechnology and the Drive to Create,” Engenious Design, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, November 19, 2020.
- Steele, Brian and Mark Meyer. University of Kansas School of Medicine, “Engineering in Medicine and Biology Society,” Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, November 6, 2020.
- Keogh, Kelly. No Stone Unturned Foundation, “Welcome KSU EMBS Students,” Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, October 29, 2020.
- Lee, Taehee. “Pen-Type Electrodermal Activity Sensor,” Experimental Session, ECE 571 – Introduction to Biomedical Engineering, January 30, 2020.
- Hopper, Ben. “Building a Generation of Biomedical Engineers,” *K-Stater*, Winter 2019, pp. 26-29.
- Thompson, David. Kansas State University, “Neural Interfacing,” Kansas State University, BME 200 – Introduction to Biomedical Engineering, Fiedler Auditorium Room 1107, November 1, 2019.
- Carlson, Charles. Kansas State University, “Sensor Beds for Children with Severe Disabilities,” Kansas State University, BME 200 – Introduction to Biomedical Engineering, Fiedler Auditorium Room 1107, November 1, 2019.
- DeLuca, Anne. Kansas State University, “KSU Career Center,” BME 001 – New Student Assembly, Fiedler Auditorium Room 1107, September 3, 2019.
- Fattahi, Niloufar. Kansas State University, “Screening Diffusive, Antagonistic Bacterial Interactions Using Photoreleasable Hydrogels,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, April 25, 2019.
- Aceros, Juan. University of North Florida, “Adaptive Toys, Neurodevelopment, and Play,” Kansas State University, Electrical & Computer Engineering Distinguished Lecture Series, KSU IEEE EMBS Student Chapter, Fiedler Hall Room 1109, April 17, 2019.
- Aceros, Juan. University of North Florida, “Neural System Development and Play,” Kansas State University, *ECE 571 – Introduction to Biomedical Engineering*, Rathbone Hall Room 1052, April 17, 2019.
- Martinez, Andres. Vanderbilt University, “Control of Powered Lower-Limb Prostheses and Exoskeletons,” Kansas State University, *ECE 571 – Introduction to Biomedical Engineering*, Rathbone Hall Room 1052, March 29, 2019.
- Martinez, Andres. Vanderbilt University, “Center for Rehabilitation Engineering and Assistive Technology,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, March 28, 2019.
- Dear, Adam. Biosense Webster, Kansas State University, *BME 200 – Introduction to Biomedical Engineering*, Engineering Hall Room 3099, October 26, 2018.
- Lyle, John. “Experiences with an Implanted Defibrillator,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, October 25, 2018.

- Citron, Paul. National Academy of Engineering, “Medical Devices: Saving and Improving Lives,” Kansas State University, *BME 200 – Introduction to Biomedical Engineering*, Engineering Hall Room 0093, September 28, 2018.
- Griffith, Craig. IEEE Client Services Manager. “Introduction to IEEE,” Kansas State University, *BME 200 – Introduction to Biomedical Engineering*, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, September 26-27, 2018.
- Radke, Donald. Heartland O&P, Inc., “Life with a Prosthetic,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, September 12, 2018.
- Snyder, Larry. KSU Veterinary Medicine, “Stem Cell Therapy in Veterinary Medicine,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 3099, March 28, 2018.
- Monroe, Tyrone. Hangar Prosthetics and Orthotics, “Prosthetic Design and Manufacturing,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 3099, March 15, 2018.
- Dear, Adam. Biosense Webster, Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 3099, April 24, 2018.
- Ade, Carl. KSU Kinesiology, “How Human Spaceflight Has, and Will, Advance Medicine,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 3099, February 8, 2018.
- Zollman, Dean. “KSU Center for Research and Innovation in STEM Education (CeRISE),” Seminar sponsored by the KSU College of Engineering and the American Society for Engineering Education, Engineering Hall Room 0093 (Phase IV), April 26, 2016.
- Yang, Lydia. “Fundamentals of Education-Related Research,” Seminar sponsored by the KSU College of Engineering and the American Society for Engineering Education, Engineering Hall Room 0093 (Phase IV), March 29, 2016.
- Schwartz, Carole. “Ageism and Society,” Kansas State University, Introduction to Biomedical Engineering Class, Rathbone Hall, April 17, 2015. 2015 K-State Global Campus Alumni Fellow, Department of Gerontology.
- Grande-Allen, Jane. “Experimental Framework for Analysis of Heart Valve Mechanobiology,” Kansas State University, College of Engineering, February 11, 2015. Sponsored by the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society, the KSU Student Chapter of the IEEE, and the KSU Department of Electrical & Computer Engineering.
- Grande-Allen, Jane. “Heart Valve Mechanics,” Kansas State University, *ECE 571 – Introduction to Biomedical Engineering*, February 11, 2015.
- Cox, Jessica. “Think Outside the Shoe,” Kansas State University, College of Engineering, November 6, 2013. Sponsored by the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society, the KSU Student Chapter of the IEEE, and the KSU Department of Electrical & Computer Engineering.
- Gallagher, Richard. “The History of Biomedical Engineering,” Kansas State University, College of Engineering, October 9, 2013. Sponsored by the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Gay, Nicholas and Sam Bruner. Garmin Day speakers for *ECE 512 – Linear Systems*, Sept. 3, 2013.
- Prakash, Punit. “Technologies for Non- and Minimally-Invasive Image-Guided Surgery,” Kansas State University, College of Engineering, April 26, 2013. Sponsored by the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Singleton, Gary and Riley Dutton. “You Can Make a Difference: The Future of Engineering and Children with Special Needs,” Heartspring, Wichita, Kansas, Kansas State University, Paslay Lecture Hall, October 9, 2012. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.

- Fowler, Kim. “Bringing a Medical Device to Market,” Kansas State University, November 15, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Mayer, Carl and Chris Justice. “Medical Device Development: Engineering for Life,” RBC Medical Innovations, Kansas State University, October 27, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Thompson, David. “Neural Interfacing,” Kansas State University, October 14, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Nelson, Tommy and Phillip Schumm. “EMBC 2011 and Epidemic Modeling,” Kansas State University, September 26, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Gustafson, Tim. “Prosthetics Technology,” Kansas State University, September 9, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Larson, Brian. “Software Verification and Medical Device Regulation,” Kansas State University, February 11, 2011, *ECE 571 – Introduction to Biomedical Engineering*.
- Chikan, Viktor. “Magnetic Hyperthermia: Integrating Nanotechnology with Biology for Cancer Therapy,” Kansas State University, November 19, 2010. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Coetzee, Hans. “Use of Electronic Tools to Measure Pain in Cattle,” Kansas State University, September 16, 2010. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Weiss, Mark L. “Engineering Cells, Tissues, and Critters,” Kansas State University, February 15, 2010. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Singleton, Gary and Chris Vacek. “Technology to Improve the Quality of Life for Disabled Children,” Heartspring, Wichita, Kansas, ECE Graduate Seminar, Kansas State University, September 24, 2009. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Derakhshani, Reza. “Biometrics: An Introduction to Current Technologies and Future Trends,” EECE Graduate Seminar, Kansas State University, October 25, 2007. Sponsored by the KSU Department of Electrical & Computer Engineering, the Midwest Section of the IEEE Engineering in Medicine & Biology Society, and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Perkins, Leigh and David Lingerfelt. “Opportunities at Cerner Corporation,” KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society, Kansas State University, October 8, 2007.

GRANTS

Total award amount since 1999: **\$14,017,910**; ECE allocation: **\$6,344,523**

- (Accepted) Carlson, Charles, Steve Warren, and Colin Tobin. Nathan Undergraduate Research Experience Award, Student Nominee: Brinlynn Johnson, Kansas State University, 05/2026–05/2027, \$5,000.
- (Submitted) Kump, Paul (Marty), Charles Carlson, Steve Warren, and Amy Betz. “Research Initiation: Integrating Ethical Design Challenges Across Electrical, Computer, and Biomedical Engineering to Foster Professional Identity,” National Science Foundation, NSF 20-558: PFE: Research Initiation in Engineering Formation (PFE: RIEF), 06/01/2026–05/31/2028, \$199,919.

- (Accepted) Zhu, Ninghao, Steve Warren, Charles Carlson, Mohammad Hashemi, Scott Fan, Gabriel Converse, and Steve Warren. “Entrepreneurship in Healthcare Technology (BME Hub),” K-State Applied Learning Experiences (ALE) Incentive Grants, K-State Program Curriculum Exploration, Kansas State University, 08/01/2025–07/31/2027, \$17,000.
- (Declined) Zhu, Ninghao, Charles Carlson, and Steve Warren. “Biomedical Engineering Research Hub (BME Hub),” K-State Applied Learning Experiences (ALE) Incentive Grants, K-State Program Curriculum Exploration, Kansas State University, 08/01/2025–07/31/2027, \$20,000.
- (Declined) Carlson, Charles, Steve Warren, and Briana Nelson Goff. “Meeting the Needs of Children with Disabilities Through Community Engaged Applied Human Sciences and Biomedical Student Design Projects,” K-State Applied Learning Experiences (ALE) Incentive Grants, K-State Transdisciplinary Applied Learning, Kansas State University, 06/01/2026–05/31/2027, \$39,548.
- (Declined) Goff, Briana Nelson, Charles Carlson, and Steve Warren. “Meeting the Needs of Children with Disabilities Through Community Engaged Applied Human Sciences and Biomedical Student Design Projects,” Harold and Letha Reser Family Community Innovation Award, Kansas State University, College of Health and Human Sciences, 06/01/2025–05/31/2026, \$29,843.
- (Submitted) Carlson, Charles, Steve Warren, and Briana Nelson Goff. “Biomedical Design Projects and Community Engagement Experiences Informed by the Needs of Children with Disabilities,” National Science Foundation, NSF 23-510: Improving Undergraduate STEM Education: Directorate for STEM Education (IUSE: EDU), Engaged Student Learning (Level I), 08/01/2025–07/31/2027, \$399,738.
- (Submitted) Thompson, David, Carl Ade, Heather Bailey, Charles Carlson, and Steve Warren. “Bed-Based Sensors for Predictive Care Management of Daytime Behavior in Alzheimer's Disease Populations,” National Institutes of Health, R01 Research Project Grant, PAR-22-093: Research on Current Topics in Alzheimer's Disease and Its Related Dementias (R01 Clinical Trial Optional), 07/01/2025–06/30/2030, \$3,619,621.
- (Awarded) Robert and Becca Reichenberger Cornerstone Teaching Scholar, Kansas State University College of Engineering, May 1, 2024 – April 30, 2027, \$28,500.
- (Declined) Carlson, Charles, Steve Warren, Briana Goff, Punit Prakash, David Thompson, and Don Gruenbacher. “Biomedical Design Projects and Community Engagement Experiences Informed by the Needs of Children with Disabilities,” National Institutes of Health, R25 Education Project, PAR-19-215: Team-Based Design in Biomedical Engineering Education, 5/15/2022–5/14/2027, \$204,000.
- (Awarded) Rice, Dallas and Steve Warren. “Sanitation Drone for Large Indoor Venues,” OURCI Research Award, Kansas State University, Office of Undergraduate Research & Creative Inquiry, 5/3/2021–12/03/2021, \$500.
- (Declined) Carlson, Charles, Steve Warren, Punit Prakash, David Thompson, and Don Gruenbacher. “Biomedical Design Projects and Community Engagement Experiences Informed by the Needs of Children with Disabilities,” National Institutes of Health, R25 Education Project, PAR-19-215: Team-Based Design in Biomedical Engineering Education, 5/15/2021–5/14/2026, \$204,000.
- (Awarded) “Biomedical Engineering Teaching Laboratory,” Tong Foundation, 5/01/2020 – 4/30/2023, \$750,000, Kansas State University, Carl R. Ice College of Engineering, \$750,000 match. Total award: \$1,500,000.
- (Awarded) Doll, Gayle, Migette Kaup, and Laci Cornelison. “PEAK 2.0 Evaluation,” Kansas Department of Health and Environment, 9/01/2019 - 8/31/2021, \$560,292. Role: key participant and advisory council member.
- (Awarded) Kim, Jungkwun, Don Gruenbacher, Steve Warren, Charles Carlson, Dwight D. Day, and Yingying Wu. “Advanced Expert Training Program for Industrial Convergence of Wearable Smart Devices,” Korea Institute for Advanced Technology (KIAT), Korea Display Industry Association (KDIA), Project no. P0002397, 1/4/2020–1/11/2020, \$13,163.

- (Awarded) Robert and Becca Reichenberger Cornerstone Teaching Scholar, Kansas State University College of Engineering, September 1, 2019 – June 30, 2024, \$47,500.
- (Declined) Zollman, Dean, Lydia Yang, Lisa Wilken, and Amy Betz. (Warren role: senior personnel) “IGE: Providing Graduate Students with Interdisciplinary Skills to Succeed in Discipline-Based Education Research,” National Science Foundation, Innovations in Graduate Education (IGE) Program, 6/1/2019–5/31/2022, \$499,423.
- (Submitted) Zacharakis, Jeffrey, Charles Rice, and Steve Warren. “IGE Leadership Development for STEM Graduate Students,” National Science Foundation, Innovations in Graduate Education (IGE) Program, 7/1/2019–6/30/2022, \$493,446.
- (Declined) Diaz, Eduardo, Grant Anderson, Steve Warren, and Michael Sawyer. “Incorporating ‘Individual’ Functional Characteristics with Bio-Sensors to Support Optimal Human Health and Performance,” NASA TRISH BRASH1801 Program, 1/1/2019–12/31/2020, \$390,232, (KSU: \$199,274).
- (Awarded) Carlson, Charles, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, Balasubramaniam Natarajan, and Steve Warren. “A Pilot Study of an Unobtrusive Bed-Based Sleep Quality Monitor for Severely Disabled Autistic Children,” *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Hawaii Convention Center, Honolulu, HI, USA, July 17-21, 2018, KSU Graduate Student Council, July 2018, \$450.
- (Awarded) Suliman, Ahmad, Charles Carlson, Punit Prakash, Steve Warren, and David Thompson. “Performance Evaluation of Processing Methods for Ballistocardiogram Peak Detection,” *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Hawaii Convention Center, Honolulu, HI, USA, July 17-21, 2018, KSU Graduate Student Council, July 2018, \$400.
- (Declined) Zollman, Dean, Steve Warren, and Lydia Yang. “NRT: Crosscutting Research in STEM Education (CRiSE): Training Graduate Students to Succeed in Discipline-Based Education Research,” National Science Foundation, NSF Research Traineeship (NRT) Program, 8/1/2018–7/31/2021, \$2,999,185.
- (Awarded) Betz, Amy and Emily Dringenberg. “Research Initiation: The Formation of Engineering Students' Beliefs about Intelligence,” National Science Foundation, NSF EEC 1738209, Division of Engineering Education and Centers, 9/1/2017–8/31/2019, \$199,995. Role: evaluator.
- (Withdrawn) Zollman, Dean, Michael Herman, Steve Warren, and Lydia Yang. “IGE: Providing Graduate Students with Interdisciplinary Skills to Succeed in Discipline-Based Education Research,” National Science Foundation, Innovations in Graduate Education (IGE) Program, 6/1/2018–5/31/2021, \$498,579.
- (Declined) Zacharakis, Jeffrey, Charles Rice, Andrew Wefald, and Steve Warren. “IGE Leadership Development for STEM Graduate Students,” National Science Foundation, Innovations in Graduate Education (IGE) Program, 7/1/2018–6/30/2021, \$499,637.
- (Submitted) Warren, Steve, Punit Prakash, David Thompson, Bala Natarajan, and Don Gruenbacher. “Netsmart/KSU Sensor Bed Effort,” Netsmart Corporation, 1/15/2018–1/14/2019, \$50,000.
- (Declined) Rozell, Timothy, Ashley Rhodes, and Steve Warren. “USRG: Development of Wearable Sensors to Evaluate Heat Stress in Dairy Cattle,” Kansas State University, Office of the Vice President for Research, University Small Research Grant program, 7/1/2017–6/30/2018, \$4,500.
- (Declined) Zollman, Dean, Lydia Yang, Steve Warren, and Michael Herman. “NRT-IGE: Providing Graduate Students with Interdisciplinary Skills to Succeed in Discipline Based Education Research,” National Science Foundation, National Science Foundation Research Traineeship (NRT) Program, 6/1/2017–5/31/2020, \$486,608.
- (Declined) Zacharakis, Jeffrey, Charles Rice, Andrew Wefald, and Steve Warren. “NRT-IGE Leadership Development for STEM Graduate Students,” National Science Foundation, National Science Foundation Research Traineeship (NRT) Program, 7/1/2017–6/30/2020, \$498,073.

- (Awarded) Grother, Ethan, Mei He, and Steve Warren. “Cautious Use of Mobile Device Cameras for Color Scale Assays,” *IEEE-NIH Special Topics Conference on Healthcare Innovations and Point-of-Care Technologies (HI-POCT ‘16)*, Cancun, Mexico, November 9-11, 2016. Travel award from the KSU Johnson Cancer Research Center, \$870.
- (Awarded) Carlson, Charles, Ahmad Suliman, Punit Prakash, David Thompson, Shangxian Wang, Bala Natarajan, and Steve Warren. “Bed-Based Instrumentation for Unobtrusive Sleep Quality Assessment in Severely Disabled Autistic Children,” *38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Disney’s Contemporary Resort, Orlando, FL, USA, August 16-20, 2016, KSU Graduate Student Council, 4/28/2016, \$500.
- (Awarded) He, Mei, Steve Warren, and Matthew Campbell. “Self-Management of Ovarian Cancer Using Smartphone-Based, 3D Chip Diagnostic Strategy,” Johnson Cancer Research Center, Kansas State University, 12/1/2015–11/30/2016, \$30,000.
- (Declined) He, Mei, Marselina Tan, Steve Warren, Alisjahbana Bacht, Mukta Sharma, and Susan Brown. “Low-Cost iPOC^{3D} Diagnostic Technology for Promoting Wireless Healthcare of Anemia in Indonesia,” National Institutes of Health, NIH PAR-14-028, Mobile Health: Technology and Outcomes in Low and Middle Income Countries (R21), 10/15/2015–10/14/2017, \$375,000.
- (Declined) Zollman, Dean, et al. “Center for Research and Innovation in STEM Education: CeRISE @ KSU,” NSF Science and Technology Centers. Declined by KSU prior to submission to the NSF.
- (Awarded) Thompson, David (PI) and Steve Warren (Mentor). 2015-2016 K-State Mentoring Fellowship, Alfred E. Sloan Foundation, \$6,000.
- (Awarded) Warren, Steve, Punit Prakash, Bala Natarajan, David Thompson, and Wayne Piersel. “GARDE: Research to Quantify the Health and Development of Children with Disabilities Around the Clock,” National Science Foundation, General & Age-Related Disabilities Engineering (GARDE) Program, UNS 1512564, 8/15/2015–8/14/2018, \$399,962.
- (Declined) Warren, Steve, Jianchu Yao, Sunghan Kim, and William Hageman. “EXP: Collaborative Research: A Cyber-Physical System for Electrical Engineering Education that Generates and Grades Personalized Hands-On Assignments,” National Science Foundation, Cyberlearning and Future Learning Technologies (Cyberlearning) Program, 9/15/2014–9/14/2017, \$324,834 (KSU) + \$225,142 (ECU) = \$549,976.
- (Declined) Warren, Steve, Punit Prakash, Bala Natarajan, Minyoung Suh, and Wayne Piersel. “GARDE: Research to Quantify the Health and Development of Children with Disabilities Around the Clock,” National Science Foundation, General & Age-Related Disabilities Engineering (GARDE) Program, 6/15/2014–6/14/2017, \$299,999.
- (Declined) Warren, Steve, Jianchu Yao, and Sunghan Kim. “Collaborative Research: Cyber-Physical Tools for the MOOC Age: Personalized Hands-On Electrical Engineering Learning Experiences for Distributed Education,” National Science Foundation, EHR Core Research (ECR) Program, 1/1/2014–12/31/2016, \$149,936 (KSU) + \$150,063 (ECU) = \$299,999.
- (Awarded) Thomson, Daniel, Lindsey Hulbert, Michael Dikeman, Christopher Reinhardt, K. Olson, John Gonzales, Steve Warren, David Renter, Steve Bartle, and Daniel Frese. “Animal Health and Animal Welfare Implications of Beta Adrenergic Agonists (BAA) in Feedlot Cattle,” National Cattlemen's Beef Association (840738973), 5/1/2013–8/31/2014, \$190,455.
- (Awarded) Hatcliff, John, Patrice Chalin, Robby, and Steve Warren. “FDA SIR: Risk Assessment Techniques for Apps and Devices in Interoperable Medical Device Frameworks,” National Science Foundation, U.S. Food and Drug Administration, Scholars In Residence Program, 10/1/2013–9/30/2015, \$104,348.
- (Declined by KSU) Rebello, N. Sanjay, Andrew G. Bennett, Eleanor Sayre, Steve Warren and Dean Zollman. “IGERT: Integrated Experiences Preparing Future Faculty in Interdisciplinary STEM Education Research,” National Science Foundation, IGERT Program. Two-page summary.

- (Awarded) Hatcliff, John M., Patrice Chalin, Robby, Eugene Y. Vasserman, Steve Warren, Insup Lee, and Oleg Sokolsky. “CPS: Synergy: Collaborative Research: Trustworthy Composition of Dynamic App-Centric Architectures for Medical Application Platforms,” National Science Foundation, Cyber-Physical Systems Program, CNS-1239543, 9/1/2012 – 8/31/2016, \$800,000.
- (Declined) Warren, Steve, Donald Gruenbacher, Balasubramaniam Natarajan, and John Hatcliff. “SHB: Type I (EXP): Hierarchical Coordination and Computation Frameworks for Reconfigurable Everyday-Carry Medical Devices and Body Area Networks,” National Science Foundation, Smart Health and Wellbeing Program, IIS-1231767, 8/15/2012 – 8/14/2015, \$600,000.
- (Accepted) Ben-Arieh, David, John Wu, Caterina Scoglio, and Steve Warren. “Operations Systems Engineering Support for the Department of Veterans Affairs MidWest Mountain Veterans Engineering Resource Center”, Veterans Administration, Department of Veterans Affairs, VA Nebraska – Western Iowa Health Care System, 8/1/2011 – 7/31/2014. Inclusion on the bidding process for upcoming short-duration healthcare system tasks. Overall project amount: \$500,000.
- (Declined) Li, Kejia and Steve Warren. “Everyday Carry Wireless Health Monitor with Customizable Surface Components,” Center for Integration of Medicine and Innovative Technology (CIMIT), Student Prize for Technology in Primary Healthcare, submitted 5/30/2011, \$150,000.
- (Awarded) Singh, Gurdip, Steve Warren, Balasubramaniam Natarajan, John Hatcliff, and Virg Wallentine. “Falls Predictor Device”, Cerner Corporation, 5/1/2011 – 8/31/2012, \$93,992, \$139,108 with KSU match.
- (Awarded) Kuhn, William, Steve Warren, Balasubramaniam Natarajan, Don Gruenbacher, Dwight Day, Thomas Barstow, and Timothy Sobering. “Biosensor Networks and Telecommunication Subsystems for Long-Duration Missions, EVA Suits, and Robotic Precursor Scout Missions: Coordinated Research Spanning Early-Stage Innovations through Game Changing Technologies,” NASA EPSCoR Program, 10/1/2011 – 6/30/2015, \$1,224,635.
- (Awarded) Li, Kejia and Steve Warren. “Everyday Carry Wireless Health Monitor with Customizable Surface Components,” Center for Integration of Medicine and Innovative Technology (CIMIT), Student Prize for Technology in Primary Healthcare, submitted 1/17/2011, 6/2011–5/2012, \$10,000.
- (Pending) Teixeira, Rodrigo, XXXX. “Enhanced Pulse Oximetry for Neonates,” National Institutes of Health, Safe and Effective Instruments and Devices for Use in the Neonatal Intensive Care Units (SBIR [R43]), RFA-HD-10-012, 6/1/2010 – 5/31/2012, \$450,000. (Steve Warren – subcontractor)
- (Awarded) Hatcliff, John, Gurdip Singh, Robby, Virgil Wallentine, and Steve Warren. NSF CNS-1065887. “An Integrated Development and Certification Environment for a Medical Device Coordination Framework,” National Science Foundation, U.S. Food and Drug Administration, Scholars In Residence Program, 1/1/2011–12/31/2012, \$80,000.
- (Awarded) Warren, Steve, John Devore, and Balasubramaniam Natarajan. “Communication and Controls Development,” Sandia National Laboratories, 10/28/2008–2/28/2011, \$122,724.
- (Awarded) Hatcliff, John, Daniel Andresen, and Steve Warren. “REU: CPS: Medium: Collaborative Research: Infrastructure and Technology Innovations for Medical Device Coordination,” National Science Foundation, CNS Cyber-Physical Systems (CPS) Program, 9/15/2009–8/31/2012, \$12,000.
- (Declined) Warren, Steve, Jianchu Yao, and Tarek Abdel-Salam. “TUES: Interactive Technology to Facilitate Collaborative Learning in Engineering Courses,” National Science Foundation, Transforming Undergraduate Education in Science, Technology, Engineering and Mathematics (TUES) Program, 1/1/2011–12/31/2012, \$81,641 (KSU) + \$118,356 (ECU) = \$199,997.
- (Awarded) Warren, Steve, Punit Prakash, Balasubramaniam Natarajan, and Shelby Evans. “KSU Student Chapter of the IEEE EMBS as a Focal Point for Senior Design Projects to Aid

- Children with Disabilities,” National Science Foundation, Research to Aid Persons with Disabilities Program, CBET 1067740, 8/15/2011 – 7/31/2017, \$125,000.
- (Awarded) Teixeira, Rodrigo, Alton Reich, Jim Shaw, and Steve Malin. “Intelligent Data Extraction Algorithm for Pulse Oximetry,” National Science Foundation, SBIR Phase II, NSF08-548: Biotech and Chemical Technologies, 6/1/2010 – 5/31/2012, \$500,000. (Steve Warren – consultant)
 - (Awarded) Hatcliff, John, Virg Wallentine, Robby, Daniel Andresen, Gurdip Singh, and Steve Warren. “Development of a Prototype Healthcare Intranet for Improved Health Outcomes,” Massachusetts General Hospital through the National Institutes of Health NIBIB Quantum Program, 10/1/2010–8/31/2015, \$475,000.
 - (Awarded) Barstow, Thomas, Carl Ade, Steve Warren, and Chris Lewis. “Standardized “Pre-flight” Exercise Tests to Predict Performance During Extravehicular Activities in a Lunar Environment,” Step-2 proposal, Research and Technology Development to Support Crew Health and Performance in Space Exploration Missions, NASA Human Research Program, Exploration Systems Mission Directorate, Johnson Space Center, Houston, TX, 7/1/2010–5/30/2015, \$1,192,194.
 - (Awarded) Warren, Steve and Jianchu Yao. NSF DUE–0942425. “CCLI: Portable Cyber-Laboratories: Virtual Instruments and Affordable Prototyping Kits to Enhance Learning and Improve Access to Electrical Engineering Education,” National Science Foundation, Course Curriculum and Laboratory Improvement (CCLI) Program, 1/1/2010–12/31/2011, \$111,899 (KSU) + \$88,100 (ECU) = \$199,999.
 - Warren, Steve. Funding to support student travel to the 31st Annual Conference of the IEEE EMBS, Minneapolis, MN, Sept. 2-6, 2009, KSU Electrical & Computer Engineering (\$1000), KSU College of Engineering (\$1550), \$2550.
 - (Awarded) Hatcliff, John, Daniel Andresen, Robby, and Steve Warren. NSF CNS–0932289. “CPS: Medium: Collaborative Research: Infrastructure and Technology Innovations for Medical Device Coordination,” National Science Foundation, CNS Cyber-Physical Systems (CPS) Program, 9/1/2009–8/31/2012, \$851,548.
 - Zollman, Dean, Sanjay Rebello, Andrew Bennett, and Steve Warren. NSF DRL–0816207. “Investigating Trajectories of Learning & Transfer of Problem Solving Expertise from Mathematics to Physics to Engineering,” National Science Foundation REESE Program, DRL–0816207, 9/1/2008–8/31/2011, \$999,841.
 - (Declined) “Consortium for the Infusion of Technology into Rural Care Environments,” Kansas Congressional allocation for Kansas State University, 11/1/2007–10/31/2012, \$20,000,000.
 - (Declined) Smith, Gregory, Virgil Wallentine, Beth Unger, and Steve Warren. “Center of Innovation for Pervasive Health Information Technology,” Kansas Bioscience Authority, 1/1/2008–6/30/2008, \$200,000.
 - (Declined) Warren, Steve, Donald Gruenbacher, Caterina Scoglio, Bala Natarajan, Gayle Doll, Shing Chang, Margaret Rys, Scott DeLoach, John Hatcliff, and Daniel Andresen. “Pervasive Biomedical Technology Alliance,” Kansas State University, Targeted Excellence Program, 7/1/2008–6/30/2011, \$2,000,000.
 - (Declined) Bennett, Andrew, David Allen, LaVerne Bitsie-Baldwin, Kimberly Douglas, Sanjay Rebello, and Steve Warren. “Center for Quantitative Education,” Kansas State University, Targeted Excellence Program, 7/1/2008–6/30/2011, \$2,000,000.
 - (Declined) Scoglio, Caterina, and Todd Easton. “EPICENTER: The Epidemiological Modeling and Simulation Center at Kansas State University,” Kansas State University, Targeted Excellence Program, 7/1/2008–6/30/2011, \$1,000,000.
 - (Declined) Polson, Cheryl, Craig Beardsley, Lisa Freeman, Charlie Griffin, Curtis Kastner, Justin Kastner, Denis Medeiros, Gurdip Singh, Marty Vanier, Virgil Wallentine, and Steve Warren. “Homeland Security Studies: Development of an Interdisciplinary Collaborative Master’s Degree Program,” Kansas State University, Targeted Excellence Program, 7/1/2008–6/30/2013, \$2,166,588.

- Warren, Steve. Funding to support student travel to the 30th *Annual Conference of the IEEE EMBS*, Vancouver, British Columbia, Canada, August 20-24, 2008, KSU Electrical & Computer Engineering (\$500), KSE Chemical Engineering (\$400), KSU Mechanical & Nuclear Engineering (\$500), KSU College of Engineering (\$2272), KSU Office of Student Activities (\$300), \$3972.
- Hatcliff, John, Daniel Andresen, Robby, and Steve Warren. NSF CNS-0734204. "Development of an Open Test-Bed for Application of Formal Methods to Medical Plug-and-Play Architectures," National Science Foundation, U.S. Food and Drug Administration, Scholars In Residence Program, 8/1/2007–7/31/2008, \$64,098.
- (Declined) Erickson, Howard, Dale Blasi, Joann Kouba, and Steve Warren. "Evaluation of Bio-Thermo Radio Frequency Microchips in Resting and Exercising Horses," U.S. Department of Agriculture, Equine Species Working Group, 10/1/2006–9/30/2008, \$250,000.
- (Declined) Warren, Steve (PI), Scott Deloach, Mark Haub, Balasubramaniam Natarajan, and Caterina Scoglio. "EFRI-ARESCI Preliminary Proposal: iNtelligent Occupant-Aware Home (iNOAH)," National Science Foundation, Emerging Frontiers in Research Innovation (EFRI) Program, Autonomously Reconfigurable Engineered Systems Enabled by Cyberinfrastructure (ARES-CI), 8/1/07–7/31/11, \$1,954,047.
- (Declined) Bennett, Andrew, David Allen, LaVerne Bitsie-Baldwin, Kimberly Douglas, Sanjay Rebello, and Steve Warren. "Center for Quantitative Education," Kansas State University, Targeted Excellence Program, 7/1/2007–6/30/2011, \$2,000,000.
- Bennett, Andrew, Sanjay Rebello, and Steve Warren. "Center for Quantitative Education," Kansas State University, Targeted Excellence Program, 7/1/2006–6/30/2010, \$998,498. Reduced to \$100,000 for the period of 7/1/2006–6/30/2007.
- Martinez, Angel and Steve Warren. "Ingestible Pill for Heart Rate and Core Temperature Measurement in Cattle." Travel award for the 28th *Annual Conference of the IEEE EMBS*, Marriott at Times Square, New York City, New York, Kansas State University, August 30 – September 3, 2006, Graduate Student Council, 4/28/2006, \$90.
- Warren, Steve. Funding to support student travel to the 28th *Annual Conference of the IEEE EMBS*, Marriott at Times Square, New York City, New York, August 30 – September 3, 2006, KSU Electrical & Computer Engineering (\$2824), KSE Computing & Information Sciences (\$500), KSU Mechanical & Nuclear Engineering (\$500), \$3824.
- Warren, Steve and Suzanne Dubnicka. "Light Reflectance Signals as Biometric Authenticators for the Elderly," Galichia Center on Aging, 5/15/06–9/24/06, \$2,000.
- Singh, Gurdip, Daniel Andresen, Scott Deloach, Balasubramaniam Natarajan, and Steve Warren. NSF CNS-0551626. "CRI: An Experimentation Platform for Developing Customized, Large-Scale Sensor Systems," National Science Foundation, CNS Division, Computing Research Infrastructure, 4/1/06–3/31/09, \$420,110. Reduced by NSF to \$200,000.
- (Declined) Warren, Steve (PI). "Secure Infrastructure for Wearable Monitoring Systems Applied to Cardiac Rehabilitation," National Institutes of Health, Biomedical Research Partnerships Program, \$424,982. Subcontract within the project "Body Networks of Wearable Sensors for Patient Monitoring," with Emil Jovanov, Ph.D. (PI – University of Alabama in Huntsville), Piet de Groen, M.D. (CoPI – Mayo Clinic, Rochester, MN), Felicity Enders, Ph.D. (CoPI – Mayo Clinic, Rochester, MN), Bruce Johnson, Ph.D. (CoPI – Mayo Clinic, Rochester, MN), and Aleksandar, Milenkovic, Ph.D. (CoPI – University of Alabama in Huntsville). UAH 2004-566, 8/1/2006–7/31/2010, \$2,226,631.
- Warren, Steve (PI). "Laptop Computer for NI ELVIS Platform Development," EECE, Student Equipment Fee program, \$2,000, January 2006.
- Warren, Steve (PI). "SolidWorks with COSMOS – 10 Student Licenses," December 2005, \$1850.
- (Declined) Douglas, Kimberly, et al. "Battling the Quiet Crisis: Building the STEM Pipeline," Kansas State University, Targeted Excellence Program, 7/1/2006–6/30/2010. (Steve Warren – Senior Personnel)

- Singh, Gurdip, Douglas McGregor, and James Edgar. First year transition funds for the “Center for Sensors and Sensor Systems,” Kansas State University, Targeted Excellence Program, 1/1/2006–12/31/2007, \$132,000. (Steve Warren – Co-PI)
- (Declined) Sawan, Edwin, et al. (WSU); Bala Natarajan, Steve Warren, et al. (KSU); Ron Hui, et al. (KU), and Vasundara Varadan (UA). “IGERT: Graduate Student Training for Research and Healthcare,” Integrative Graduate Education and Research Traineeship (IGERT) program, National Science Foundation, 7/1/2006–6/30/2009. (Steve Warren – Co-PI)
- (Declined) Singh, Gurdip, Steve Warren, Daniel Andresen, Sudharman Jayaweera, and Mark Spire. “Human and Animal Health Monitoring: Enhancing Quality of Life and Security Through Research in Distributed, Reconfigurable Sensor Networks,” National Science Foundation, Kansas NSF EPSCoR Program, 4/1/2006–3/31/2009, \$2,449,319.
- Singh, Gurdip, Douglas McGregor, and James Edgar. “Center for Sensors and Sensor Systems,” Kansas State University, Targeted Excellence Program, 7/1/2006–6/30/2010, \$1,500,000. (Steve Warren – Co-PI)
- Starrett, Shelli, Anil Pahwa, and David Soldan. “Planning Learning Environments for a More Diversified Workforce,” National Science Foundation, EEC Division – Engineering Education, 1/1/06–12/31/07, \$100,000. Department Level Reform planning grant. (Steve Warren and William Kuhn – Senior Personnel)
- Andresen, Daniel, Howard Erickson, David Poole, Mark Spire, and Steve Warren. REU Supplement to ITR–0205487. “ITR: An Infrastructure for Veterinary Telemedicine – Proactive Herd Health Management for Disease Prevention from Farm to Market,” National Science Foundation, 10/1/03–9/30/09, \$12,000.
- Lebak, Jeff and Steve Warren. “HL7-Compliant Healthcare Information System.” Travel award for the *10th Annual Meeting of the American Telemedicine Association*, Colorado Convention Center, Denver, CO, April 17–20, 2005, Kansas State University, Graduate Student Council, 2/16/2005, \$200.
- Warren, Steve (PI). “Enhanced Learning Via the National Instruments LabVIEW and NI ELVIS Platforms,” 12/1/2004–8/31/2007, \$6,004.
- Yao, Jianchu and Steve Warren. “A Novel Algorithm to Separate Motion Artifacts from Photoplethysmographic Signals Obtained with a Reflectance Pulse Oximeter.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, Kansas State University, Graduate Student Council, 10/22/04, \$184.
- Warren, Steve (PI). NSF BES–0440183. REU Supplement to NSF BES–0093916. “CAREER: Beyond Telemedicine: Realizing a Proactive Home Health Care Delivery Model Through Research and Education in Plug-and-Play Components and Novel Sensors,” Faculty Early Career Development (CAREER) Program, National Science Foundation, Division of Bioengineering and Environmental Systems, 2/1/01–1/31/06, \$6,000.
- Nagl, Luke and Steve Warren. “Experiences Using WinCE PocketPCs as Computation and Data Acquisition Platforms for Ambulatory Telemonitoring.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, Whitaker Foundation, 7/19/04, \$145.
- Krishnamurthi, Balakumar and Steve Warren. “Experiences Using WinCE PocketPCs as Computation and Data Acquisition Platforms for Ambulatory Telemonitoring.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 7/19/04, \$500.
- Warren, Steve (PI). “iINTERFAWARE Chameleon Licenses,” 2004, \$12,000.
- Lebak, Jeffrey and Steve Warren. “HL7-Compliant Healthcare Information System for Home Monitoring.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 4/27/04, \$500.

- Schoenig, Scott and Steve Warren. “Ambulatory Instrumentation Suitable for Long-Term Monitoring of Cattle Health.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 4/27/04, \$500.
- Yao, Jianchu and Steve Warren. “A Novel Algorithm to Separate Motion Artifacts from Photoplethysmographic Signals Obtained with a Reflectance Pulse Oximeter.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 4/27/04, \$500.
- Nagl, Luke and Steve Warren. “Experiences Using WinCE PocketPCs as Computation and Data Acquisition Platforms for Ambulatory Telemonitoring.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 4/27/04, \$500.
- Andresen, Daniel, Howard Erickson, David Poole, Mark Spire, and Steve Warren. REU Supplement to ITR–0205487. “ITR: An Infrastructure for Veterinary Telemedicine – Proactive Herd Health Management for Disease Prevention from Farm to Market,” National Science Foundation, 10/1/03-9/30/09, \$12,000.
- Andresen, Daniel, Howard Erickson, David Poole, Mark Spire, and Steve Warren. ITR–0205487. “ITR: An Infrastructure for Veterinary Telemedicine – Proactive Herd Health Management for Disease Prevention from Farm to Market,” National Science Foundation, 10/1/03-9/30/09, \$899,996.
- Andresen, Daniel, Steve Warren, Howard Erickson, David Poole, and Jan Sargeant. ITR–0325921. “ITR: Veterinary Telemedicine – Proactive Herd Health Management for Disease Prevention from Farm to Market,” National Science Foundation, 10/1/02-9/30/04, \$107,025.
- Bennett, Andrew and N. Sanjay Rebello. NSF DUE–0206943 “Assessing Student Transfer and Retention of Learning in Mathematics, Physics, and Engineering,” National Science Foundation, 7/15/02-7/14/07, \$500,000. (Steve Warren – Senior Personnel)
- Warren, Steve (PI). “Software Components for Research and Education,” Spatial Corporation, 7/18/01–7/10/03, \$58,500.
- Warren, Steve (PI). NSF BES–0093916. “CAREER: Beyond Telemedicine: Realizing a Proactive Home Health Care Delivery Model Through Research and Education in Plug-and-Play Components and Novel Sensors,” Faculty Early Career Development (CAREER) Program, National Science Foundation, Division of Bioengineering and Environmental Systems, 2/1/01–1/31/06. Five-year continuing award totaling \$375,000.
- Warren, Steve (PI). NSF EPS–9874732. “Beyond Telemedicine: Realizing a Proactive Home Health Care Delivery Model Through Research and Education in Plug-and-Play Components and Novel Sensors,” National Science Foundation, Kansas EPSCoR Program, University of Kansas, 2/1/01–1/31/02. Originally \$31,750; adjusted to \$5,000 upon receipt of the National Science Foundation CAREER award.
- Warren, Steve (PI). “Explore Electron Transport in a Complex Solid Model Using the Adjoint Version of ITS-CAD,” Sandia National Laboratories, Albuquerque, NM, 1/20/2000–12/31/2004, \$66,993.
- Warren, Steve. IronCAD for Teaching and Research, Visionary Design Systems, 12/22/99, \$4,995.

PROGRAM DEVELOPMENT

- 2024 Kansas HOSA Fall Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, October 17, 2024
- Tour of Ascension Via Christi, Manhattan, KS, TRIMEDX, BME 575 – Clinical Systems Engineering, September 20, 2024

- 2024 Kansas HOSA Spring Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, March 27, 2024.
- Tour of Ascension Via Christi, Manhattan, KS, TRIMEDX, BME 575 – Clinical Systems Engineering, March 6, 2024
- 2023 Kansas HOSA Fall Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, October 4, 2023
- Discussion with Dr. Ryan Fischer, Dr. Danial Forsha, and other research faculty at Children’s Mercy Hospital, univentricular physiology and Fontan-associated liver disease, December 18, 2023
- Discussion with Rick Haigh, Sean Hayes, Alexx Frankos, Tim Duncan, and other Garmin engineers, canine biometrics, December 6, 2023
- Senior Experience Day, Kansas State University, Carl R. Ice College of Engineering, Various Dates, Fall 2023
- Powercat Preview, Kansas State University, Carl R. Ice College of Engineering, Various Dates, Fall 2023
- Tour of the Children’s Mercy Hospital clinical engineering facility, October 26, 2023
- Matthew Dodge, Biosense Webster, Electrophysiology Immersion Program, October 26, 2023
- Reformulation of the KSU Undergraduate Biomedical Engineering Program in response to the Kansas Board of Regents (KBoR) undergraduate core mandate, Fall 2023
 - Complete curricular overhaul for the BME emphasis areas, including new BME 390 and BME 470 courses
 - Updated BME emphasis areas: Sensors & Devices, Biomaterials, Computation, and Pre-Medicine
 - Nine CurricuLog proposals related to facets of this overhaul, November 5, 2023
- ABET Accreditation, Undergraduate Biomedical Engineering Program, Summer/Fall 2023
 - Course assessment collation
 - BME self-study, June 28, 2023
 - BME student transcript analysis, July 10, 2023
 - Interactions with the BME evaluator, Dr. Kim Anderson, University of Kentucky
 - Examples of student work
 - ABET evaluator visit, October 15-17, 2023
- DaVinci surgical robot tour, Ascension Via Christi, Manhattan, KS, KSU IEEE EMBS Student Chapter, April 15, 2023
- BME 575 tour of Ascension Via Christi, Manhattan, KS, Rachelle White, TRIMEDX, February 22, 2023; March 6, 2024
- Internship and Opportunity Discussions, Biosense Webster, Kansas City, MO, October 25, 2022, January 20, 2023.
- Internship and Opportunity Discussions, TRIMEDX, Salina, KS January 20, 2023.
- Internship and Opportunity Discussion, Children’s Mercy, Kansas City, KS, July 7, 2022.
- Internship and Opportunity Discussion, Garmin, Kansas State University, Carl R. Ice College of Engineering, Manhattan, KS, April 20, 2022
- K-State Group Travel, ECE Faculty and EMBS Officers, 21st Annual Design of Medical Devices Conference, University of Minnesota, April 11-14, 2022, Minneapolis, MN.
- Interactions with University of Wisconsin - Madison faculty/staff regarding elements of their BME curriculum, Madison, WI, Oct. 26, 2018
- Interactions with Marquette faculty/staff regarding elements of their BME curriculum, Milwaukee, WI, Oct. 25, 2018
- Undergraduate degree in Biomedical Engineering, KSU College of Engineering, 2017–present.
- “Master Agreement for Research and Service Projects,” Netsmart Technologies and Kansas State University, April 2017. Supplemented with a “Business Associate Subcontractor Agreement” and a “Special Intellectual Property Assignment and Confidentiality Agreement – Student Led Learning Experience.”

- Nondisclosure agreement with Streamline Automation, January 20, 2010. Subject: analysis of reflectance pulse oximeter data to assess hemodynamic parameters noninvasively.
- Nondisclosure agreement with Orbital Outfitters, November 24, 2009. Subject: creation of a customized space suit replica for anti-gravity mobility and physiology experiments.
- Nondisclosure agreement with Raytheon Corporation, October 8, 2008. Subject: co-development of an ingestible pill for cattle health monitoring

HONORS/SCHOLARSHIPS

- Robert and Becca Reichenberger Cornerstone Teaching Scholar, Kansas State University College of Engineering, May 1, 2024 – April 30, 2027
- 2022 Team Award of Excellence, Kansas State University, Carl R. Ice College of Engineering, Development of the Biomedical Education and Innovation Laboratory
- Faculty Recognition Award, XIX Chapter of the Mortar Board Senior Honor Society, Kansas State University, November 9, 2021
- Robert and Becca Reichenberger Cornerstone Teaching Scholar, Kansas State University College of Engineering, September 1, 2019 – June 30, 2024
- 2019 Larry E. and Laurel Erickson Public Service Award, Kansas State University College of Engineering
- KSU Medical Component Design Laboratory, KSU Student Chapter of the IEEE EMBS, Best ECE Department Display, 2019 Engineering Open House, KSU College of Engineering, April 5-6, 2019
- KSU Student Chapter of the IEEE EMBS, 2019 K-State College of Engineering Outstanding Professional Student Chapter Award, KSU College of Engineering, Engineering Student Council
- KSU Student Chapter of the IEEE EMBS, 2019 E-Week Best Booth Award, KSU College of Engineering, Engineering Student Council
- KSU Student Chapter of the IEEE EMBS, 2018 K-State College of Engineering Outstanding Professional Student Chapter Award, KSU College of Engineering, Engineering Student Council
- Quality Mentoring in Undergraduate Research Award, Kansas State University Developing Scholars Program, April 2018
- Outstanding Service Award, Midwest Section of the American Society for Engineering Education, September 2017
- Promotion to Professor, Kansas State University, February 2017
- Electrocardiogram Pill for Cattle – FY2007 CISE highlight for the National Science Foundation
- ‘Making a Difference’ Award, Kansas State University Women in Engineering and Science Program, December 2005, October 2007
- Tenure, Kansas State University, February 2002
- Promotion to Associate Professor, Kansas State University, February 2002
- Nominated for the 2002 James L. Hollis Memorial Award for Excellence in Undergraduate Teaching, KSU College of Engineering
- Award for Excellence, Kansas State University, Engineering Student Council, Dec. 1999
- Graduate Faculty, Kansas State University, Membership and Certification, Dec. 1999
- Promotion to Principal Member of the Technical Staff, Sandia National Laboratories, Nov. 1998
- Sandia National Laboratories Individual Performance Award, Telemedicine effort, August 1997
- Sandia National Laboratories Award For Excellence, Personal Status Monitor effort, July 1997
- Engineering Honors Program, Kansas State University
- Tau Beta Pi, Eta Kappa Nu National Engineering Honoraries, Kansas State University
- Phi Kappa Phi, Golden Key National Honoraries, Kansas State University
- 1st Place - Kansas State University Open House Poster Contest, 1992
- Special Citation for Outstanding Musicianship, International Association of Jazz Educators, Wichita Jazz Festival “20” College Big Band Competition, April 26, 1991
- Certificate of Recognition, University of Missouri, Kansas City, Conservatory of Music, Kansas City Jazz Commission, 1990 Kansas City Jazz Band Festival, March 9, 1990

- Graduate Research Assistantship, The University of Texas at Austin, 6 semesters, 1992-1994
- Graduate Teaching Assistantship, The University of Texas at Austin, 2 semesters, 1991-1992
- Graduate Research/Teaching Assistantship, Kansas State University, 4 semesters, 1989-1991
- Bandsman of the Year, Kansas State University, 1986
- Dean of Engineering Scholarship, Kansas State University, 1984-1989
- Dean's Honor Roll and National Dean's List, Kansas State University, 10 semesters, 1984-1989

COURSES TAUGHT

- **BME 001 – New Student Assembly**, Kansas State University, F18, F19, F20, F21, F22, F23
- **BME 200 – Introduction to Biomedical Engineering**, Kansas State University, F18, F19, F20, F21, F22, F23, S25, S26
- **BME 575 – Clinical Systems Engineering**, Kansas State University, S22, S23, S24, F24
- **DEN 398 – EMBS Learning Community**, Kansas State University, F05
- **ECE 499 – Honors Research**, Kansas State University, F99–present
- **ECE 510 – Circuit Theory I**, Kansas State University, S07, S08
- **ECE 511 – Circuit Theory II**, Kansas State University, F99, F00, F10, F16
- **ECE 512 – Linear Systems**, Kansas State University, F01, S02, F02, S03, S04, S05, F05, S06, F06, S07, F07, S08, F08, S09, F09, S10, S11, F11, S12, F12, S13, F13, F14, F15, F16, F17, F18, F19, F20, S21, F21, S22, F22, S23, F23, S24, F24, S25, S26
- **ECE 571 – Introduction to Biomedical Engineering**, Kansas State University, S00, S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15, S16, S17, S18, S19, S20, S21
- **ECE 636 – Introduction to Computer Graphics**, Kansas State University, F03, S05, S06, S09
- **ECE 690/890 – Problems in Electrical & Computer Engineering**, Kansas State University, F05, S07, S09, S12, F12, S13, F13, S14, F14
- **ECE 772 – Bioinstrumentation**, Kansas State University, F00, F01, F02, F03, F04, F05, F06, F08, F09, F11, F12, F17, F18
- **ECE 840 – Computer Engineering Methods for Analysis, Simulation, and Design**, Kansas State University, S00, S01, S03, S04

PROFESSIONAL DEVELOPMENT

- **James R. Coffman Leadership Institute**, August 9–11, 2006, Rock Springs 4-H Center
- **Feeding and Caring For Graduate Students**, The Graduate School, Kansas State University, December 8, 2001
- **What Matters Most**, Kansas State University, April 21, 2001. Time management and life leadership workshop sponsored by Raytheon Corporation
- **Kansas State University New Faculty Workshop**, Emporia, KS, February 17, 2001
- **Engineering LEA/RN Program**, Kansas State University, 1999-2004
- **Academic Advising**, Kansas State University, Campus Collaboration to Foster Retention, satellite teleconference, November 4, 1999
- **Management of Sandia Technical Projects**, Erica Jones Associates, Inc., Coronado Club, Albuquerque, NM, February 9-11, 1999
- **Marketing and Market Research Workshop**, sponsored by the Sandia New Ventures Program, Technology Ventures Corporation, and the Albuquerque Chamber of Commerce, Lockheed Martin Building, Albuquerque, NM, January 14, 1999
- **Structuring Group Relationships to Achieve Technical Success**. Workshop on high-performance teams and workgroups, sponsored by Sandia National Laboratories, January 12, 1999.
- **On the Fast Track: Recruit, Retain, Become a High-Potential Employee**, Advanced Technology and Management Programs, National Technological University, sponsored by the U.S. Chamber of Commerce, November 12, 1997
- **Seven Habits of Highly Effective People**, Covey Leadership Center, Coronado Club ZIA Room, Albuquerque, NM, January 24-26, 1995

- **Lessons in Leadership**, Covey Leadership Center, based on the book **First Things First**, Albuquerque Convention Center, Albuquerque, NM, November 16, 1994

TECHNICAL DEVELOPMENT

- **Hands-On Education: The Future of Engineering Education Made Possible by Portable Electronics**, U223 Workshop, *2012 Conference of the American Society for Engineering Education*, San Antonio Convention Center, San Antonio, TX, June 10, 2012.
- **Point-of-Care Medical Device Communications**, IEEE 1073 – Medical Information Bus, Westin Galleria, Houston, TX, October 22, 2002.
- **Enhancement and Revision of the IEEE 1451.2 and 1451.5 (Smart Sensors) Standards**, May 20-21, 2002, Sensors Expo 2002, San Jose McEnery Convention Center, San Jose, CA
- **Spatial User's Conference 2002**. Tutorial and application workshop in support of the ACIS solid modeling toolkit, May 13-15, 2002, Westminster, CO.
- **Spatial User's Conference 2000**. Three-day tutorial and application workshop in support of the ACIS solid modeling toolkit and customer applications (i.e., the Sandia/KSU work on radiation transport through complex solid models), June 21-23, 2000, Boulder, CO.
- **Medical Applications of Sensors, Transmitters, Devices, and Instrumentation**. 3-day short course on biomedical instrumentation offered through the UCLA Extension, Department of Engineering, Information Systems, and Technical Management, Los Angeles, CA, January 31 – February 2, 2000.
- **Introduction to CORBA Programming**. 4-day short course in CORBA/C++ implementation taught by Black & White Corporation, sponsored by Sandia National Laboratories, Research Park, September 8-11, 1998.
- **Development of ACIS-Based Applications**. Short course on the use of the ACIS geometric modeler, offered by Spatial Technology, Boulder, CO, November 25-26, 1996.
- **The Wavelet Transform and Applications**. Short course in applied optics jointly sponsored by the Department of Electrical and Computer Engineering and the Department of Physics at New Mexico State University, May 17, 1996.
- **Electrical Modeling, Simulation, and Design of Electronic Packages**. Class sponsored by the University of Illinois at Urbana-Champaign and North Carolina State University, Radisson Hotel, San Jose, CA, June 26-29, 1995.
- **Electrical Modeling with the Saber Circuit Simulator**. Class Sponsored by Analogy Corporation, Beaverton, OR, January 14-17, 1995.

SOFTWARE EXPERIENCE

- **Office/Productivity**: Microsoft Office, Microsoft Project, Adobe Acrobat, Lotus 123, Quattro-Pro, Graphics Workshop, Paint Shop Pro, Wordstar, TeX, LaTeX, PiCTeX, Graftool, and DesignCAD2D.
- **Computational Methods**: MATLAB/Simulink, Derive, Maple V, Numerical Recipes in C, ACIS 3D Modeler, IronCAD, Ansoft Maxwell Eminance, LINPACK, EISPACK, ODEPACK, IMSL, 3D/Eye Trispectives Technical, Microsoft Visual C++, Borland Turbo C/C++, Microsoft Fortran, DEC Fortran, SAS, SPICE/PSPICE, Analogy Saber Circuit Simulator, and NeXT Application Builder
- **Programming Languages**: C/C++, CORBA IDL, Pascal, Fortran, Forth, Objective C, and Basic
- **Operating Systems**: Microsoft Windows 3.1/95/98/NT, MS-DOS, Sun Solaris, NeXT, Hewlett-Packard, and Macintosh
- Software Reviewer for *Circuits and Devices Magazine*, 1994-1995

CONFERENCE SERVICE

- **Technical Program Committee Member**, *48th International Conference of the IEEE EMBS*, Toronto, Canada, July 26–30, 2006.

- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Charlotte, North Carolina, June 21–24, 2026.
- **Technical Program Committee Member**, *47th International Conference of the IEEE EMBS*, Copenhagen, Denmark, July 14–17, 2025.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Montreal, Canada, June 22–25, 2025.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Portland, OR, June 23–26, 2024.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *45th International Conference of the IEEE EMBS*, Sydney, Australia, July 24–28, 2023.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Baltimore, MD, June 24–27, 2023.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *44th International Conference of the IEEE EMBS*, Glasgow, Scotland, UK, July 11–15, 2022.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Minneapolis, Minnesota, June 26–29, 2022.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *43rd International Conference of the IEEE EMBS*, Guadalajara, Mexico, October 31 – November 4, 2021.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Long Beach, California, June 27–30, 2021.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *42nd International Conference of the IEEE EMBS*, in conjunction with the 43rd Annual Conference of the Canadian Medical and Biological Engineering Society, Montréal, Québec, Canada, July 20–24, 2020.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Montréal, Québec, Canada, June 21–24, 2020.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *41st International Conference of the IEEE EMBS*, CityCube Berlin, Messedamm 26, 14055 Berlin, Germany, July 23–27, 2019.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Tampa, FL, June 16–19, 2019.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *40th International Conference of the IEEE EMBS*, Hilton Hawaii Village Waikiki Beach Resort, Hawaii Convention Center, Honolulu, HI, July 17–21, 2018.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Salt Lake City, UT, June 24–27, 2018.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, *39th International Conference of the IEEE EMBS*, International Convention Center, Jeju Island, Korea, July 11–15, 2017.
- **Technical Program Committee Member**, *2017 ASEE Midwest Section Conference*, Stillwater, OK, September 24–26, 2017.
- **Chair**, *2016 ASEE Midwest Section Conference*, Manhattan, KS, September 25–27, 2016.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, New Orleans, LA, June 26–29, 2016.
- **Technical Program Committee Member**, *2016 ASEE Midwest Section Conference*, Manhattan, KS, September 25–27, 2016.
- Manuscript Review, *Annual Conference of the ASEE*, 2016
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, *38th International Conference of the IEEE EMBS*, Disney's Contemporary Resort, Orlando, FL, August 16–20, 2016.
- **Technical Program Committee Member**, *2015 ASEE Zone III Conference*, Springfield, MO, September 23–25, 2015.

- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, *37th International Conference of the IEEE EMBS*, Milano Conference Center, Milan, Italy, August 25–29, 2015.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, *36th International Conference of the IEEE EMBS*, Sheraton Chicago Hotel and Towers, Chicago, IL, August 26–30, 2014.
- **Invited Session Organizer (Co-Organizer: Punit Prakash)**. “Technologies for Around-the-Clock Health Assessment of Special Needs Children” ; Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies); *36th International Conference of the IEEE EMBS*, Sheraton Chicago Hotel and Towers, Chicago, IL, August 26–30, 2014.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies; Chair: Track 3.6 – Wearable Sensor Technology and Instrumentation**, *35th International Conference of the IEEE EMBS*, Osaka International Convention Center, Osaka, Japan, July 3–7, 2013.
- **Technical Program Committee Member**, *Frontiers in Education 2012 (FIE 2012)*, Seattle, WA, October 3–6, 2012.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, *34th International Conference of the IEEE EMBS*, IEEE EMBS Conference Editorial Board, San Diego, CA, August 28 – September 1, 2012.
- **Technical Program Committee Member**. *2nd International Conference on Computer Science and Automation Engineering (CSAE 2012)*, Zhangjiajie, China, May 25–27, 2012.
- **Co-Chair**: Session FrB19 – Wireless Sensors and Systems, September 2, 2011, *33rd Annual Conference of the IEEE EMBS*, Boston Marriott Copley Place, Boston, MA, Aug. 30 – Sept. 3, 2011.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, *33rd International Conference of the IEEE EMBS*, IEEE EMBS Conference Editorial Board, Boston, MA, August 30 – September 3, 2011.
- **Technical Program Committee Member**, *2009 International Conference on Computer Communications and Networks (ICCCN)*, San Francisco, CA, August 2–6, 2009.
- **Technical Program Committee Member**, *2009 International Conference on Communications (IEEE ICC)*, Dresden, Germany, June 14–18, 2009.
- **Technical Program Committee Member**, *2008 International Conference on Computer Communications and Networks (ICCCN)*, St. Thomas, U.S. Virgin Islands, August 4–7, 2008.
- **Panel Member**: Session SaD10 – Students: The Movers and Shakers of EMBS, September 2, 2006, *28th Annual Conference of the IEEE EMBS*, Marriott Times Square, New York City, NY, Aug. 30 – Sept. 3, 2006.
- **Chair**: Session 7.8.1 – Medical Information Systems I, September 4, 2004, *26th Annual Conference of the IEEE EMBS*, Westin St. Francis Hotel, San Francisco, CA, Sept. 1–5, 2004.
- **Workshop on Home Care Technologies for the 21st Century**. Sponsored by the National Science Foundation, the Catholic University of America, and the U.S. Food and Drug Administration. Department of Health and Human Services (DHHS), Rockville, MD, April 7–9, 1999. Session chair: Topic F – Smart Health Care Systems and the Home of the Future, pp. 58–63, ©1999, Catholic University of America.

PROFESSIONAL ACTIVITIES

Short Courses and Workshops

- “Wearable Sensors and Devices,” Winter Camp for Research Experience, sponsored by the Korea Institute for Advancement of Technology, Kansas State University, Carl R. Ice College of Engineering, Jan. 5–10, 2020.
- Warren, Steve. “The Health Care Facelift: How Telecommunications Technology, Distributed Computing, and Intelligent Medical Devices will Redefine the Patient-Physician Relationship,”

Four-hour short course on smart health care technology prepared for the Department of Biomedical Engineering, University of Minnesota, Twin Cities Campus, December 1, 1999.

Textbook, Proposal, and Manuscript Review

- Guest Editor, *Sensors*, Special Issue: “Recent Applications of Non-Invasive Biomedical Sensors,” ISSN 1424-8220, Manuscripts due March 10, 2022.
- Oxford University Press, Linear Systems and Signals Textbook (3rd edition) review, 2018
- National Science Foundation, Disability and Rehabilitation Engineering (DARE) Program, Proposal Review Panel B – *Biomechanics, Biomaterials, Computational Modeling, & Design*, January 22-23, 2018, Alexandria, VA
- CRC, Biomedical Engineering Textbook Feedback, 2017
- zyBooks Interactive Textbook Review, 2017
- Elsevier, Introduction to Biomedical Engineering Textbook Review, 2015
- Addison-Wesley Computer Graphics Textbook Review, 2004
- Science Foundation Ireland (SFI), SFI Investigators Programme, 2013.
- National Science Foundation, General and Age-Related Disabilities Engineering (GARDE) Proposal Panel Review, December 3–4, 2012, Arlington, VA
- National Science Foundation, Smart Health and Wellbeing (SHB) Proposal Panel Review, May 6-8, 2011, Arlington, VA
- National Institutes of Health, National Institute of Biomedical Imaging and Bioengineering (NIBIB) Panel Review, Point-of-Care (POC) Technology Research Network (U54) Proposals, July 13, 2007, Bethesda, MD
- National Science Foundation, Biomedical Engineering CAREER Proposal Panel Review, November 6–7, 2001, November 13–15, 2002, November 11–13, 2003, October 5–6, 2005, Arlington, VA
- Proposal Review, U.S. Department of Veterans Affairs, 2006
- Proposal Review, KSU Targeted Excellence Program, 2005
- Proposal Review, Louisiana Health Excellence Fund, 2000
- Proposal Review, Department of Energy Small Business Innovation Research (SBIR) Program, FY 1998/1999 Awards, Biological and Environmental Research – Medical Applications
- Proposal review, National Science Foundation, 1993
- Manuscript Review, *IEEE Transactions on Learning Technologies*, 2017
- Manuscript Review, *Computers and Electrical Engineering*, 2017
- Manuscript Review, *Annual Conference of the ASEE*, 2016–2020
- Manuscript Review, *ASEE Midwest Section Conference*, 2015–2018
- Manuscript Review, *ASEE Zone III Meeting, American Society for Engineering Education*, Springfield, MO, 2015
- Manuscript Review, *Healthcare Innovations and Point-of-care Technologies Conference of the IEEE Engineering in Medicine and Biology Society*, Seattle, Washington, 2014
- Manuscript Review, *IEEE-EMBS International Conferences on Biomedical and Health Informatics*, Valencia, Spain, 2014
- Manuscript Review, *2nd Middle East Conference on Biomedical Engineering*, Qatar, 2014
- Manuscript Review, *Frontiers in Education Conference*, 2012
- Manuscript Review, *IEEE ICC*, 2009
- Manuscript Review, *IEEE ICCCN*, 2008, 2009
- Manuscript Review, *Annual Conference of the IEEE EMBS*, 2007–2020
- Manuscript Review, *Biomedical Signal Processing and Control*, 2016
- Manuscript Review, *Expert Review of Medical Devices*, 2015
- Manuscript Review, *International Journal of Distributed Sensor Networks*, 2014
- Manuscript Review, *IEEE Microwave and Wireless Components Letters*, 2014
- Manuscript Review, *Sensors*, 2014
- Manuscript Review, *IEEE Transactions on Biomedical Circuits and Systems*, 2012–present

- Manuscript Review, *IEEE Communications Magazine*, 2011
- Manuscript Review, *EURASIP Journal on Wireless Communications and Networking*, 2011
- Manuscript Review, *Journal of Healthcare Engineering*, 2010
- Manuscript Review, *IEEE Transactions on Biomedical Engineering*, 2005–present
- Manuscript Review, *IEEE Journal of Biomedical and Health Informatics*, formerly *IEEE Transactions on Information Technology in Biomedicine*, 2000–2003, 2005–present
- Manuscript Review, *Journal of Neurorehabilitation Research*, 2007
- Manuscript Review, *Journal of Biomedical Optics*, 2005
- Manuscript Review, *Optical Spectroscopy*, 1994
- Manuscript Review, *Applied Optics* (Optical Society of America), 1997

Professional Societies and Industry Workgroups

- Institute of Electrical and Electronics Engineers (IEEE), 2001–present
- IEEE Engineering in Medicine & Biology Society, 2001–present
- American Society for Engineering Education (ASEE), 2001–present
- Founding chair, EMBS Chapter for the IEEE Kansas City section, 2006–2007
- IEEE 11073 Personal Health Devices Working Group, 2009–present
- IEEE 11073 Appeals Committee, 2010
- American Telemedicine Association, 1997–1998, 2005–2006
- National Instruments myDAQ Beta Program, 2010–2012
- Microsoft Health User’s Group, 1997–1998
- Health Level 7 message standards organization, 1997–1998
- Order of the Engineer, 2000–present
- Southwest Biomedical Industry Coalition, 1995
- Sandia National Laboratories Biomedical Engineering Focus Group, 1994–1996

Expert Witness

- Expert Witness, Physiological Monitoring Hardware and Software, 2013–present
- Expert witness engagement listing: provided upon request depending on need

Leadership Service

- Program Coordinator, Undergraduate Biomedical Engineering Degree Program, KSU College of Engineering, 2017–present
- Chair, Midwest Section of the American Society for Engineering Education, 2015–2016
- Chair Elect, Midwest Section of the American Society for Engineering Education, 2014–2015
- American Society for Engineering Education Representative for the Kansas State University College of Engineering, 2014–present

Committees

- Search committee, Dean of the College of Health and Human Sciences, Kansas State University, 2021–2022
- Curriculum subcommittee, B.S. in Public Health, Kansas State University, Department of Kinesiology, 2019–present
- Advisory Panel, Midwest Institute for Comparative Stem Cell Biology, Kansas State University, 2019–present
- Faculty Senator, Kansas State University, College of Engineering, 2019–present
- Academic Affairs Committee, Faculty Senate, Kansas State University, 2020–2026
- Faculty Affairs Committee, Faculty Senate, Kansas State University, 2019–2020
- College Assessment Review Committee, Kansas State University College of Engineering, 2018–present
- Honors & Awards Committee, Kansas State University College of Engineering, 2018–2020
- Chair, Biomedical Engineering Professor-of-Practice Search Committee, 2018–2019
- Chair, Biomedical Engineering Curriculum Task Force, Kansas State University, Electrical & Computer Engineering, 2016–2017

- KSU College of Engineering Phase IV Biomedical Suite and Resource Contributions, 2013–present
- Kansas State University Internal Review Board (Human Subjects Research), 2000–present
- Heartspring Board of Trustees, 2009–2015
- Kansas State University Goldwater Nominating Committee, 2009–present
- Women in Engineering and Science Program, Search Committee, 2003–2004
- Electrical and Computer Engineering, Education Research Committee, 2013–2015
- Electrical and Computer Engineering, Recruitment Committee, 2013–2015
- Electrical and Computer Engineering, Department Committee on Planning, 2011–2014, 2020–2022
- Electrical and Computer Engineering, College Committee on Planning, 2020–2023
- Electrical and Computer Engineering, Internship Program, 2000–2001
- Chair, Electrical and Computer Engineering, Quality of Life Committee, 2004–2007
- Electrical and Computer Engineering, Academic Progress Committee, 2001–2004
- Electrical and Computer Engineering, Advising Committee, 2009–2010
- Electrical and Computer Engineering, Department Head Search Committee, 2005–2006, 2006–2007
- Electrical and Computer Engineering, Faculty Search Committee, 2008, 2010–2011, 2011–2012
- College of Engineering, Senior Biomedical Researcher Search Committee, 2011–2012
- Electrical and Computer Engineering, Awards Committee, 2005–2007
- Electrical and Computer Engineering, Course and Curriculum Committee, 2002–2005, 2011, 2015–2016, 2020–2022
- Electrical and Computer Engineering, Department Appearance Committee, 1999–2001
- Electrical and Computer Engineering, Equipment Committee, 1999, 2003–2004, 2007–2010
- Gerontology Academic Affairs Committee, 2010
- College of Engineering Academic Reinstatement Committee, 2003
- KSU Research Excellence Award Committee, 2001–2003
- K-State Olathe Innovation Campus [KOIC] Faculty Advisory Council, 2009–2011
- K-State Olathe Innovation Campus [KOIC] Academic Planning Committee Member, 2009–2011
- Kansas State University Health IT Working Group, 2008–2011

Community Service

- Infrastructure Improvement and Vision Committee, Flint Hills Christian School, 2018–2019
- Manhattan High School Robotics Team Exchange, Medical Component Design Laboratory, KSU Electrical & Computer Engineering, November 30, 2017
- Laboratory demonstration, *DEN 161 – Engineering Problem Solving*, March 8, 2016
- Judge, Fall 2015 Engineering Undergraduate Research Poster Forum, December 3, 2015
- Bioengineering Session, E3 Workshop for Science Teachers Hosted by the KSU College of Engineering, Bette Grauer, Organizer, May 31, 2012; May 30, 2013
- Lecturer, Kansas State University Presidential Lecture Series, 2011–2013
- Girls Researching Our World (GROW) Workshops, 2001, 2002, 2007, 2008, 2016, 2018, 2019, 2020, 2022, 2023, 2024
- “Engineering the Body” Workshop, Medical Component Design Laboratory, Women in Engineering and Science Program, November 8, 2008
- Helping Hands Workshop Support, “Vital Signs Shirt,” Medical Component Design Laboratory, Women in Engineering and Science Program, November 10, 2007
- Judge, Flint Hills Christian School Creative Arts Fair, March 4–6, 2003, Manhattan, KS.
- Community Health Council, 2002–2004, Information Systems Management Committee
- Contributor, *Information for Tomorrow* home telehealth web site, <http://www.informationfortomorrow.com/>, 1999–2000
- Contributor, Expert Panel on Telemedicine Technology, Last Acts Web Site, <http://www.lastacts.org/>, 1999

- Judge, New Mexico Junior Academy of Sciences paper/oral competition, University of New Mexico, 1997–98
- Judge, 36th National Junior Science and Humanities Symposium, sponsored by the United States Army, Navy, and Air Force, Albuquerque, NM, May 3, 1998

Academic Advising/Mentoring

- Powercat Preview, Kansas State University, Carl R. Ice College of Engineering, Fall 2023
- Prospective Student Visits/Tours, KSU Biomedical Engineering Program, 2017–present
- Faculty Advisor, KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society (EMBS), 2005–present
- Faculty Advisor, KSU Student Ambassadors for Biomedical Research Enhancement (SABRE) Initiative, 2006–2008
- Faculty Advisor, KSU Future Engineers in Medicine and Law, 2005–2006
- Organizer, Learning Community for the KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society (EMBS), College of Engineering Learning Community Initiative, 2005–2006
- Faculty Liaison – KSU University Recognition Program, 2006 (Mark Freyenberger), 2007 (Heather Arnold)
- Faculty Advisor, KSU Tau Beta Pi student chapter, 2001–2004
- Faculty Advisor, Triangle Fraternity, 2000–2001
- KSU Engineering LEA/RN Program, 1999–2004
- College of Engineering – Advising for entering EE and CMPEN freshmen, Summer 2006/2007
- Teaching mentor for new faculty (Scoglio – 2005; Prakash – 2012)
- Kansas State University, Graduate Faculty, 1999–present

Other

- Dossier review, Texas Tech University, 2021
- Dossier review, University of Missouri Kansas City, 2018
- Dossier review, University of North Florida, 2017
- Dossier review, University of Alabama, Huntsville, 2015
- Academic Program Consultant, East Carolina University, General Engineering Department, 2009–2010
- USB pulse oximeter module shipment to Howard Apple, GlucoLight Corporation, Feb. 3, 2007
- No-fee consultant, Sandia National Laboratories, Telemedicine and Radiation Transport project teams, 1999–2001
- No-fee consultant, U.S. Food and Drug Administration, Center for Devices and Radiological Health, 2001–2004
- The University of New Mexico, Albuquerque, NM. Interim Instructor, *Engr. F-122, Introduction to Engineering*, 1994

PERSONAL ACTIVITIES

Physical Service/Missions

- Mission Trip, Santiago do Los Caballeros, Dominican Republic, University Christian Church, July 28 – August 4, 2017
- Mission Trip, Marouge, Haiti, Mole, Haiti, University Christian Church, July 19–26, 2013
- Disaster Response Team, Co-Organizer, University Christian Church, Manhattan, KS, 2009–present. Manhattan, KS tornado cleanup – June 2008. Reading, KS tornado cleanup – June May 2011; Oklahoma City tornado cleanup – June 2013
- Manhattan Emergency Shelter Meals (2006–present)
- Habitat for Humanity home construction – Kansas State University (2000,2002,2005), Lockheed Martin, Sandia National Laboratories (1998)
- Development Team, King Solomon Christian Service Camp, 2017–present
- Tree and field work, King Solomon Christian Service Camp, 2010–2013, 2015

- Inner-City Mission and Construction Work, University Christian Church, Manhattan, KS: Dallas, TX (2011), Cincinnati, OH (2012)
- Grow Our School Committee, Flint Hills Christian School, 2004–2005
- Grounds Maintenance, Flint Hills Christian School, Manhattan, KS, 2007–present
- Residential tree cleanup and firewood, 2008–present

Music

- University Christian Church Orchestra, Manhattan, KS. 1999–present
- Manhattan Municipal Band, 2000–2001, 2003
- Hoffmantown Orchestra, Albuquerque, NM. Trumpet player and soloist, 1994–1999. President, 1995–1997, 1999
- Trumpet instructor, Hoffmantown Academy of Music, 1995–1996
- Austin Community Jazz Ensemble. Lead trumpet, 1993–1994
- University of Texas Jazz Ensembles. Lead trumpet, 1991–1993
- Kansas State University Marching Band. Trumpet section leader (1986–1989), first chair trumpet/soloist (1985–1991)
- Kansas State University Men's Basketball Pep Band. First-chair trumpet/section leader, 1984–1991
- Kansas State University Concert Jazz Ensemble. Lead trumpet/soloist, 1984–1991
- Vice President, National Association of Jazz Educators Student Chapter (Kansas State University), 1988.

Church

- University Christian Church, Manhattan, KS. Elder, 2002–2004, 2009–2012
- Chinese Bible Study Meals (2009–present)
- University Christian Church, Manhattan, KS. Periodic classroom instructor, 1999–2000
- Promise Keepers Organization, 1995–1999
- Hoffmantown Church, Albuquerque, NM. Connection class care group leader, 1997–1999
- Hoffmantown Church, Albuquerque, NM. Ordained deacon, 1999

STUDENTS SUPPORTED AS MAJOR PROFESSOR

- Nazmun Khan, Ph.D. EE, December 2025 (GTA). Co-major professor: David Thompson. Dissertation title: *Advancing EEG-Based Brain-Computer Interfaces Through Performance Modeling, Decoding Validation, and Methodological Optimization: a Multi-Study Approach*
- Chase Harvey, Ph.D. EE anticipated December 2026 (GTA). Co-major professor: David Thompson
- Bitia Movazzafgharehbagh, M.S. EE, August 2025 (GTA) Thesis Title: *Integration of Wearable Sensors into Sensory Room Therapies for Children with Developmental Challenges*
- Dong Ren, M.S. EE August 2019 (GTA) Thesis Title: *Simultaneous Pulse Rate Estimation for Two Individuals that Share a Sensor-Laden Bed*
- Charles Carlson, Ph.D. EE May 2019 (NSF GARDE grant; GTA) Dissertation Title: *Development of a Bed-Based Nighttime Monitoring Toolset*
- Kim Fowler, Ph.D. EE December 2018 (GTA)
- Taehee Lee, M.S. EE August 2020 (GTA) Thesis title: *Pen-Type Electrodermal Activity Sensing System for Stress Detection Based on Likelihood Ratios*
- Ethan Grother, M.S. EE May 2017 (KSU JCRC grant; GTA) Thesis Title: *Mobile Device Reference Apps to Monitor and Display Biomedical Information*
- Tianyu Lin, M.S. EE August 2017 (NSF GARDE grant) Thesis Title: *Wireless Biomedical Sensor Network Reference Design Based on the Intel® Edison Platform*
- Phillip Kuehl, M.S. EE 5/2015 (NASA Crew Performance grant) Thesis Title: *Real-Time Processing of Electromyograms in an Automated Hand-Forearm Data Collection and Analysis System*
- Shiwei Luan, M.S. EE 8/2015 (NSF CPS Synergy grant) Thesis Title: *Modularized PCA Pump Design for an ICE-Informed Medical Device Coordination Framework*

- Xiongjie Dong, M.S. EE 8/2014 (NASA EPSCoR grant) Thesis Title: *A ZigBee-Based Wireless Biomedical Sensor Network as a Precursor to an In-Suit System for Monitoring Astronaut State of Health*
- Steve Rubenthaler, M.S. EE, 8/2014 (NSF RAPD grant; GTA) Report Title: *Unobtrusive Ballistocardiography Using an Electromechanical Film to Obtain Physiological Signals from Children with Autism Spectrum Disorder*
- Wen Song, M.S. EE 5/2013 (NASA Crew Performance grant) Thesis title: *Planetary Navigation Activity Recognition Using Wearable Accelerometer Data*
- Dana Gude, M.S. EE 8/2013 (NASA Crew Performance grant) Thesis title: *Automated Hand-Forearm Ergometer Data Acquisition and Analysis System*
- Devon Krenzel, M.S. EE anticipated 5/2013 (Cerner grant)
- Kejia Li, Ph.D. EE 12/2012 (NSF CPS grant) Dissertation title: *Custom Biomedical Sensors for Application in Wireless Body Area Networks and Medical Device Integration Frameworks*. ECE Outstanding GRA 2012
- Kejia Li, M.S. EE 5/2010 (KSU TE grant; GTA) Thesis title: *Wireless Reflectance Pulse Oximeter Design and Photoplethysmographic Signal Processing*
- Kentaro Takamatsu, M.S. EE 8/2009 (KSU Plant Pathology) Thesis title: *System for Greenhouse Climate Monitoring in Three Dimensions*
- William Hitchcock, M.S. EE 12/2008 (KSU TE grant) Report title: *Low-Power Wireless Sensor Network Data Collection*
- Chen Jia, Ph.D. EE 5/2015 (Grain Science; NSF REESE; GTA). ECE Outstanding GTA 2013. Dissertation title: *Higher-Level Learning in an Electrical Engineering Linear Systems Course*.
- Daniel Wilson, M.S. EE anticipated 12/2008 (NSF ITR grant)
- Tim Sobering, Ph.D. EE anticipated 12/2009 (KSU TE grant)
- Lara Pickel, M.S. EE 5/2008 (NIH grant – Vet Med) Thesis title: *Study of the Role of the Angiotensin II (Ang II) Type 2 Receptor (AT2) in Lung Tumorigenesis*
- Seth Hoskins, M.S. EE 5/2011 (NSF ITR grant) Thesis title: *A Physiological Sensor Network Supported by an Inductive Communication Link*
- Angel Martinez, M.S. EE 5/2007 (NSF ITR grant) Thesis title: *Acquisition of Heart Rate and Core Body Temperature in Cattle Using Ingestible Sensors*
- Kevin Smith, M.S. EE 8/2006 (NSF ITR grant) Thesis title: *A Wearable Cattle Health Monitoring System with an Emphasis on Motion-Based Behavior Assessment*
- Scott Schoenig, M.S. EE 12/2007 (NSF ITR grant) Final Project: *A Signal Processing System for Long-Term Acquisition of Heart-Rate in Cattle*
- Keil Regehr, M.S. EE 8/2006 (Anatomy & Physiology NIH grant (external)); Recipient – 2005 Benjamin Zweifach Student Award from the Microcirculatory Society. Thesis title: *Analysis of Ca^{2+} Sparks in the Gerbil Spiral Modiolar Artery*
- Jeffrey Lebak, M.S. EE 12/2005 (NSF CAREER grant; ECE matching funds) Thesis title: *The Design and Implementation of a Standards-Based, Home-to-Remote Health Care Information System*
- Yong Yang, M.S. EE 5/2005 (NSF math assessment grant) Thesis title: *Computer-Based Homework Generation and Assessment for Linear Systems*
- Luke Nagl, M.S. EE 12/2004 (NSF ITR grant) Thesis title: *The Design and Implementation of a Cattle Health Monitoring System*
- Jianchu Yao, Ph.D. EE 8/2005 (NSF CAREER grant) Dissertation title: *Design of Standards-Based Medical Components and a Plug-and-Play Home Health Monitoring System*
- Matthew Martin, M.S. EE 5/2004 (Sandia Monte Carlo grant) Thesis title: *Improved Geometry Representations for Monte Carlo Radiation Transport*
- Ryan Schmitz, M.S. EE 12/2003 (NSF CAREER grant) Thesis title: *Bluetooth Sensor Networking for Distributed Medical Systems*
- G. Edward Barnes, M.S. EE 2/2002 (NSF CAREER grant; startup funds) Thesis title: *Development of a Wearable, Bluetooth-Enabled System for Home Health Care*

STUDENTS SUPPORTED AS A COMMITTEE MEMBER OR CO-ADVISOR

- Taylor Gregory-Sweet, Ph.D. EE anticipated 5/2025. Major professor: David E. Thompson.
- Nooshin Zeinali, Ph.D. EE 8/2024. Major professor: Punit Prakash.
- Allison Adams, M.S. ME 8/2021. Major professor: Amy Betz.
- Nazmun Khan, Ph.D. EE anticipated 12/2025. Co-major professor: David E. Thompson.
- Md Rakibul Mowla, Ph.D. EE, 5/2020. Major professor: David E. Thompson.
- Farrukh Ali, Ph.D. EE 5/2024. Major professor: William Hsu.
- Ghina Zia, Ph.D. EE 12/2021. Major professor: Punit Prakash.
- Shuvo Islam, Ph.D. CE 8/2019. Major professor: Mustaque Hossain. (External Chair)
Dissertation title: *Implementation of AASHTOWare Pavement ME Design Software for Pavement Rehabilitation*
- Hojjat Fallahi, Ph.D. EE 12/2019. Major professor: Punit Prakash.
- Alaleh Alivar, Ph.D. EE 8/2019. Major professor: Bala Natarajan. Dissertation Title: *Inferencing with Motion Artifact-Bed Based BCG and MRI Case Studies*
- Reza Barazideh, Ph.D. EE 5/2019. Major professor: Bala Natarajan. Dissertation title: *Impulsive Noise Detection and Mitigation in Communication Systems*
- Tra Thi Thanh Huynh, Ph.D. Physics 12/2020. Major professor: Eleanor Sayre.
- Pegah Faridi, Ph.D. EE 8/2020. Major professor: Punit Prakash.
- Dinaz Alaei, Ph.D. Physics 5/2018. Major professor: Eleanor Sayre.
- Nandana Liyanage, Ph.D. Physics 8/2019. Major professor: Eleanor Sayre.
- Mengjiao Wu, Ph.D. Statistics, 5/2019. Major professor: Suzanne Dubnicka. Dissertation title: *Equivalence testing for identity authentication using pulse waves from photoplethysmograph*
- Ahmad Suliman, Ph.D. EE 8/2019. Major professor: David Thompson. Dissertation title: *Quality of Heart Rate Variability Features Obtained from Ballistocardiogram*
- Yubaraj Malakar, Ph.D. Physics 5/2018. Major professor: Artem Rudenko. (External Chair)
- Hao Qian, Ph.D. CIS 12/2015. Major professor: Daniel Andresen. Dissertation Title: *Extending the Battery Life of Mobile Devices by Computation Offloading*
- Mohammed Taj Eldin, Ph.D. EE 5/2015. Major professor: William Kuhn. Dissertation Title: *Wireless Body Area Networks for Intra-Space Suit Communications: Modeling, Measurements and Wearable Antennas*
- Kim Fowler, Ph.D. EE August 2017. Major professor: Stephen Dyer.
- Sam Procter, Ph.D. CIS 8/2016. Major professor: John Hatcliff. Dissertation Title: *A Development Environment for Medical Application Platform Apps*
- Phillip Schumm, Ph.D. EE 12/2013. Major professor: Caterina Scoglio. Dissertation Title: *Characterizing Epidemics in Metapopulation Cattle Systems Through Analytic Models and Estimation Methods for Data-Driven Model Inputs*
- Jordan Bever, M.S. IE 5/2013. Major professor: John Wu.
- Amelia Hodges, M.S. EE 5/2013. Major professor: William Kuhn. Thesis Title: *Investigation of Antennas and Energy Harvesting Methods for Use with a Microtransceiver in a Biosensor Network*
- Timothy Murlam, M.S. MNE 12/2012. Major professor: Dale Schinstock. Thesis Title: *Developing a Mobile Reduced Gravity Simulator*
- James Strieby, M.S. IE 12/2012. Major professor: Todd Easton. Thesis Title: *Heuristically Optimizing Crane Schedules on Multiple Bays in Conjunction with Continuous Time Simulation*
- Joshua Weese, M.S. CIS 12/2013. Major professor: William Hsu. Thesis Title: *A Convolutional Model for Polyphonic Instrument Identification and Pitch Detection Using Combined Classification*
- Aaron Chavez, Ph.D. CIS 5/2012. Major professor: David Gustafson. (External Chair)
Dissertation Title: *Image Classification with Dense Sift Sampling: An Exploration of Optimal Parameters*

- Scott Bell, Ph.D. CIS 8/2014. Major professor: Eugene Vasserman. Dissertation Title: *Low Overhead Methods for Improving Education Capacity and Outcomes in Computer Science*
- Robert Richards, M.S. IE 12/2011. Major professor: John Wu.
- Dehui Hu, Ph.D. Physics 12/2013. Major professor: Sanjay Rebello. Dissertation Title: *Understanding Introductory Students' Application of Integrals in Physics from Multiple Perspectives*
- Yu Jin Kim, Ph.D. CIS 5/2014. Major professor: John Hatcliff.
- Rui Zhang, Ph.D. CIS 5/2015. Major professor: Scott Deloach. Dissertation Title: *A Theory for Understanding and Quantifying Moving Target Defense*
- Nidhi Tare, M.S. EE, 12/2009 (KSU TE grant; GTA) Major professor: Caterina Scoglio. Thesis Title: *A Comparative Performance Analysis of GENI Control Framework Aggregates*
- Dong-Hai Nguyen, Ph.D. Physics 12/2011. Major professor: Sanjay Rebello. Dissertation Title: *Facilitating Students' Application of the Integral and the Area Under the Curve Concepts in Physics Problems*
- Daniel Allegre, M.S. EE 5/2009. Major professor: William Kuhn. Thesis Title: *Reducing Phase and Spurious Tones in Fractional-N Synthesizers*
- Matthew Miller, Ph.D. CIS 12/2010. Major professor: Scott Deloach.
- Rajet Krishnan, M.S. EE 12/2009. Major professor: Balasubramaniam Natarajan.
- Justin Dugger, M.S. CIS 5/2007. Major professor: Daniel Andresen.
- Joseph Lutz, M.S. CIS 5/2007. Major professor: Daniel Andresen.
- Santosh Bejjamshety, M.S. CIS 5/2008 (KSU TE grant) Major professor: Daniel Andresen.
- Scott Harmon, Ph.D. CIS 5/2012. Major professor: Scott Deloach. Dissertation Title: *MASSPEC: Multiagent System Specification Through Policy Exploration and Checking*
- Herve Oyenon, Ph.D. CIS 5/2010. Major professor: Scott Deloach. Dissertation Title: *An Algebraic Framework for Compositional Design of Autonomous and Adaptive Multiagent Systems*
- Oksana Tkachuk, Ph.D. CIS 12/2006. Major professor: John Hatcliff. Dissertation Title: *Domain-Specific Environment Generation for Modular Software Model Checking*
- Pavan Nayini, M.S. CIS, 5/2006. Major professor: Daniel Andresen. Report title: *Applying HL7 Messages for Bovine Data*
- Balakumar Krishnamurthi, M.S. CIS 8/2005 (NSF ITR grant). Major professor: Daniel Andresen.
- Dominic Gelinas, M.S. CIS, 5/2005 (NSF ITR grant). Major professor: Daniel Andresen. Report title: *Bovine Mobile Observation Operation Device Software Engineering Report*
- Shilpa Vaze, Ph.D. EE 12/2007. Major professors: Jim DeVault & Prakash Krishnaswami. Dissertation title: *Integrated Formulation-Solution-Design Scheme for Nonlinear Multidisciplinary Systems in the MIXEDMODELS Platform*
- Daniel Stevens, M.S. EE 12/2004. Major professor: Sanjoy Das (Partial Kinesiology support). Thesis title: *Biologically Inspired Algorithms for DS-CDMA Code Design*
- Colin Christy, Ph.D. EE, 12/2004. Major professor: Stephen Dyer. Dissertation title: *Estimation of Soil Properties Using a Combination of Spectral and Scalar Sensor Data*
- Rajaravi Kollarapu, M.S. CIS 12/2003 (NSF ITR grant). Major professor: Daniel Andresen. Report Title: *Windows CE Based Application for Telemonitoring Cattle*
- Chadd May, M.S. EE, 5/2003. Major professor: Kenneth Carpenter. Thesis title: *A Comparison of Equations of State for Use with Exploding Conductors*
- Jason Wertz, M.S. EE, 5/2003. Major professor: Don Gruenbacher. Thesis title: *Modeling Specialized High Speed Wide Area Networks*
- William Hageman, M.S. EE, 12/2002. Major professor: Andrew Rys. Thesis title: *Electrical Characterization of Aluminum Nitride on Silicon Carbide Metal Insulator Semiconductor Structures*
- Xin He, M.S. EE, 5/2002. Major professor: Stephen Dyer. Thesis title: *A Comparison of Various Methods of Approximation to Real and Complex Data by Rational Functions*

- Nesreen Alsobou, M.S. EE, 12/2001. Major professor: William Kuhn. Report title: *Verilog Design of Digital FM Demodulator*
- Brayden Wilbeck, M.S. EE, 5/2001. Major professor: Kenneth Carpenter. Thesis title: *Design and Implementation of a Graphical User Interface to Construct Input Files for a Hydrodynamics Code*

UNDERGRADUATE STUDENTS SUPPORTED

- Halle Eastling, B.S. BME anticipated May 2028 (Honors Program)
- Bailey Bliss, B.S. BME anticipated May 2028 (Honors Program)
- Damian Garcia, B.S. EE anticipated December 2024 (Developing Scholars Program)
- Dallas Rice, B.S. EE anticipated December 2021 (Developing Scholars Program)
- Sophie Braynock, B.S. BME anticipated May 2022 (KSU ECE U/G Research)
- Kelsey Warren, B.S. BME anticipated May 2022 (KSU ECE U/G Research)
- Kala Keusler, B.S. EE December 2020 (Women in Engineering Program)
- Derrius Washington, B.S. EE anticipated December 2021 (Developing Scholars Program)
- Kendra Schuette, B.S. BAE May 2017 (NSF GARDE Summer Research)
- Martin Ortega, B.S. EE May 2021 (KS-LSAMP RiPS U/G Research)
- Andrew McKittrick, B.S. EE May 2018 (NSF GARDE U/G Research)
- Li Wang, B.S. EE May 2017 (NSF GARDE U/G Research)
- Taishan Li, B.S. EE December 2016 (NSF GARDE U/G Research)
- Shangxian Wang, B.S. EE May 2018 (NSF GARDE U/G Research)
- Kaleb Anderson, B.S. EE 5/2014 (NSF RAPD Independent Study)
- Sarah Carr, B.S. EE 5/2014 (NSF RAPD Independent Study)
- Ethan LaForge, B.S. EE 12/2013 (NSF RAPD Independent Study)
- Ethan Grother, B.S. EE 12/2014 (NSF RAPD Independent Study)
- Yuwei Wang, B.S. EE 12/2014 (NSF RAPD Independent Study)
- Emily Schnell, B.S. EE 5/2014 (NSF RAPD, ECE 773)
- Larissa Hall, B.S. BSE 5/2013 (NSF RAPD, BAE 636)
- Yu He, B.S. BSE 5/2013 (NSF RAPD, BAE 636)
- Dakota Bixler, B.S. EE 5/2014 (NSF RAPD, ECE 773)
- Devin Rood, B.S. EE 5/2014 (NSF RAPD, ECE 773)
- Amanpreet Singh, B.S. EE 5/2014 (NSF RAPD, ECE 773)
- Sergio Ortiz, B.S. EE 5/2016 (Developing Scholars Program)
- Shuo Ouyang, B.S. EE 5/2013 (NCBA; Beef Cattle Institute)
- Tianyu Lin, B.S. EE 5/2014 (Entomology; NCBA; Beef Cattle Institute)
- James Thoennes, B.S. BIOL 12/2014 (Independent Study)
- Neelou Hadavandifard, B.S. EE 12/2014 (Honors Research, NSF RAPD Independent Study)
- Austin White, B.S. EE 5/2016 (Developing Scholars Program; NSF GARDE; ECE 690)
- Hannah Gray, B.S. BAE anticipated 5/2015 (Developing Scholars Program)
- Kevin Garman, B.S. BAE 5/2014 (K-INBRE Scholarship)
- Tommy Nelson, B.S. EE 12/2011 (NASA Crew Performance)
- Jacob Nagely, B.S. BAE 5/2012 (NASA Crew Performance)
- Chenyu Zheng, B.S. EE 12/2012 (NSF CCLI; NASA Crew Performance)
- Xiongjie Dong, B.S. EE 12/2011 (NSF CCLI)
- Ashvin Nagaraja, B.S. EE 5/2010 (NSF CCLI)
- Adam Casada, B.S. EE anticipated 5/2012 (NSF CCLI)
- Kekoa Mendenhall, B.S. EE antic. 5/2013 (NIH Bridges)
- Jim Groening, B.S. EE 5/2010 (Honors research)
- Derek Woodman, B.S. CMPEN 5/2009 (Honors research; NSF ITR REU)
- Neal Dettmer, B.S. EE 12/2007 (Honors research)
- Adriann Sullivan, B.S. EE 5/2010 (Honors research)

- Nicholas Rome, B.S. EE 12/2009 (CSI)
- Paul Edgington, B.S. EE 12/2009 (TE)
- Jeffrey Newlin, B.S. EE, 5/2007 (Honors research)
- Paul Creekmore, B.S. EE, 12/2006 (Honors research)
- David Thompson, B.S. EE 5/2006 (NSF CAREER grant; Galichia Mini-Grant; Honors research)
- Suzanne Newman, B.S. EE 5/2006 (NSF CAREER grant)
- Patrick York, B.S. EE 12/2006 (NSF ITR grant)
- Matthew Spexarth, B.S. EE 5/2006 (National Instruments; Honors research)
- Albert Zhang, B.S. EE 12/2005 (Honors research)
- Roland Craddolph, B.S. CIS 5/2006 (NSF ITR REU grant)
- Federico Lopez-Bertoni, B.S. EE 12/2005 (Honors research)
- Austin Wareing, B.S. EE 5/2005 (NSF CAREER REU grant; Honors research)
- Jonathan Hicks, B.S. EE, B.S. ME, 12/2004 (Honors research)
- Jonathon Noble, B.S. CIS 5/2005 (NSF ITR grant)
- Craig Nelson, B.S. EE 12/2003 (Honors research)
- Scott Schoenig, B.S. EE 12/2003 (NSF CAREER & ITR grants; work study)
- Matthew R. Martin, B.S. EE 5/2003 (Sandia Monte Carlo grant)
- Jeffrey W. Lebak, B.S. EE 5/2003 (NSF CAREER grant)
- John McConeghey, B.S. EE 12/2002 (NSF CAREER grant)
- John Turner, B.S. EE 5/2004 (NSF CAREER grant; work study)
- Benjamin D. Young, B.S. EE 5/2002 (Honors research)
- Shelly A. Allison, B.S. EE 12/2002 (Honors research)
- Matthew D. Ownby, B.S. EE, B.S. ME, 12/2002 (Honors research)

EXPERT WITNESS ENGAGEMENT SUMMARY

Total Engagements: 24
Total Cases: 26
Active Engagements: 15
Depositions: 9
Hearing/Trial Testimonies: 4
Expert Reports and Declarations: 35
Engagements on Behalf of Plaintiffs: 11
Engagements on Behalf of Defendants: 13

EXPERT WITNESS ENGAGEMENTS

<A given engagement was/is on behalf of the company or companies **identified in bold.**>

- Expert Witness, Confidential Matter, February 2026 – present.
- Expert Witness, Confidential Matter, February 2026 – present.
- Expert Witness, *Certain Wearable Devices*, O'Melveny & Myers, LLP, **Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc.** v. Ouraring, Inc. and Oura Health Oy, ITC Inv. No. 337-TA-1478, Filed Dec. 12, 2025, United States International Trade Commission, Washington, D.C., January 2026 – present.
- Expert Witness, Confidential Matter, November 2025 – present.
- Expert Witness, Confidential Matter, October 2025 – present.
- Expert Witness, *Infusion System Functionality*, Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), **Becton, Dickinson and Company, CareFusion 303, Inc., CareFusion Manufacturing, LLC, CareFusion Solutions, LLC, and BD Switzerland S.A.R.L.** v. Baxter International, Inc. and Baxter Healthcare Corporation, Case No. 1:25-cv-00591, Filed May 12, 2025, United States District Court for the District of Delaware, April 2025 – present.
- Expert Witness, Confidential Matter, March 2025 – present.
- Expert Witness, *Instantaneous Wave-Free Ratio*, Finnegan, Henderson, Farabow, Garrett & Dunner, LLP, **Koninklijke Philips N.V. and IP2IPO Innovations, Ltd.** v. OpSens Inc., OpSens Medical, OpSens Medical, Inc., and Haemonetics Corporation, Case No. 1:24-cv-00206-UNA, Filed February 15, 2024, United States District Court for the District of Delaware, March 2025 – September 2025.
- Expert Witness, Confidential Matter, October 2024 – present.
- Expert Witness, *Animal Management System*, Dechert LLP through IMS Expert Services, ST Reproductive Technologies, LLC d/b/a/ STinsight v. **smaXtec Inc. and smaXtec Animal Care GmbH**, Case No. 3:24-cv-00037-JDP, Filed January 18, 2024, United States District Court for the Western District of Wisconsin, October 2024 – present.
- Expert Witness, *Inter Partes Review of U.S. Patent No. 7,969,307*, Perkins Coie LLP through IMS Expert Services, Abbott Laboratories, Inc. and Abbott Diabetes Care, LLC v. **Gentag, Inc. and Altivera, LLC**, Case No. IPR2024-00425, Filed January 16, 2024, Instituted July 22, 2024, United States Patent and Trademark Office, Patent Trial and Appeal Board, August 2024 – present. Deposition – Manhattan, KS, December 19, 2024.
- Expert Witness, O'Melveny & Myers, LLP, **Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc.** in matters adverse to Ouraring, Inc., Oura Health Oy, or related entities. April 2024 – present.
- Expert Witness, *Smartwatch Wearer Physiological Functions*, Finnegan, Henderson, Farabow, Garrett & Dunner, LLP (Finnegan) through IMS Expert Services, The Research Foundation for the State University of New York, University of Connecticut, and Worcester Polytechnic Institute v. **Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., Samsung Austin**

Semiconductor, LLC, and Samsung Semiconductor, Inc., Case No. 2:23-cv-00141, Filed April 3, 2023, United States District Court for the Eastern District of Texas, Marshall Division, February 2024 – May 2025. Deposition – Washington, D.C., March 5-6, 2025.

- Expert Witness, *Wearable Devices*, Desmarais LLP and Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), **Apple, Inc.** v. Masimo Corporation and Sound United, LLC, Case Nos. 1:22-cv-01377-UNA (Filed October 20, 2022; Design Patents) and 1:22-cv-01378-UNA (Filed October 20, 2022; Utility Patents), United States District Court for the District of Delaware, May 2023 – present. Various reports. Deposition – New York City, NY, January 23, 2024. Various reports and declarations. Jury Trial Testimony – Wilmington, DE, October 24, 2024.
- Expert Witness, *Disposable Wireless Sensors*, Perkins Coie LLP through IMS Expert Services, **Gentag, Inc. and Altivera, LLC** v. Abbott Laboratories, Inc. and Abbott Diabetes Care, LLC, Case No. 1:23-cv-00310, Filed January 19, 2023, United States District Court, Northern District of Illinois, Eastern Division, May 2023 – present.
- Expert Witness, *Percutaneous Hair-Like Sensors*, Through Cahn Litigation, LLC, **Jane Doe** v. University of Southern CA, Case No. 20CV373489, Filed July 12, 2021, Superior Court of the State of California, County of Santa Clara, April 2022 – October 2022.
- Expert Witness, *Oscillometric Blood Pressure Monitor*, Foley & Lardner LLP, The Trustees of Purdue University v. **Omron Corporation, Omron Healthcare Company, Limited, and Omron Healthcare, Inc.**, Case No. 1:20-cv-05443, Filed December 4, 2020, United States District Court for the Northern District of Illinois, Eastern Division, September 2021 – May 2022.
- Expert Witness, *In the Matter of Certain Light-Based Physiological Measurement Devices and Components Thereof*, Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), Masimo Corporation and Cercacor Laboratories, Inc. v. **Apple, Inc.**, ITC Inv. No. 337-TA-1276, Filed June 30, 2021, United States International Trade Commission, Washington, D.C., August 2021 – present. Various reports. Deposition – Boston, MA, April 1, 2022. Hearing testimony – Washington, D.C., June 10, 2022.
- Expert Witness, *Peripheral Arterial Tonometry*, Foley & Lardner LLP through IMS Expert Services, **Itamar Medical Ltd.** v. Ectosense nv and Virtuox, Inc., Case No. 0:20-cv-60719-WPD, Filed April 8, 2020, United States District Court, Southern District of Florida, Fort Lauderdale Division, April 2021 – April 2022.
- Expert Witness, *Diabetes Management Technology*, Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), **Roche Diabetes Care, Inc.** v. Insulet Corporation, Case No. 1:20-cv-00825-MN, Filed June 17, 2020, United States District Court for the District of Delaware, November 2020 – August 2022.
- Expert Witness, *Wearable Pulse Oximetry*, Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), Gibson, Dunn & Crutcher LLP, Masimo Corporation and Cercacor Laboratories, Inc. v. **Apple, Inc.**, Case No. 8:20-cv-00048-DOC-DFM, Filed January 9, 2020, United States District Court for the Central District of California, Southern Division, March 2020 – present. Various reports. Deposition – Boston, MA, December 9-10, 2022. Jury Trial Testimony – Costa Mesa, CA, April 20, 2023. Direct Witness Testimony – October 23, 2024. Bench Trial Testimony – Costa Mesa, CA, November 13, 2024. Deposition – Kansas City, MO, July 19, 2025.
- Expert Witness, *Certain Wearable Monitoring Devices, Systems, and Components Thereof*, Foley & Lardner LLP through Rubin Anders Scientific, **Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V.** v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, Filed December 10, 2019, United States International Trade Commission, Washington, D.C., March 2020 – October 2020. Various reports and witness statements. Deposition – August 11, 2020 via Zoom.
- Expert Witness, *Certain Electrochemical Glucose Monitoring Systems and Components Thereof*, Goodwin Procter LLP through Scitemex LLC, Dexcom, Inc. v. **AgaMatrix, Inc.**, ITC Inv. No. 337-TA-1075, Civil Action No. 1:17-cv-01310-UNA, Filed September 18, 2017, United States International Trade Commission, Washington, D.C., January 2018 – June 2018. Various reports. Deposition – Boston, MA, May 10, 2018.

- Expert Witness, *Home Telemedicine Technology*, Merchant & Gould, P.C. through Expert Search Group, LLC, Robert Bosch Healthcare Systems, Inc. v. **CardioCom, LLC and Abbott Diabetes Care**, Civil Action No. 2:13-cv-00349-JRG, M&G No. 12771.0106USZA, Filed April 26, 2013, United States District Court for the Eastern District of Texas, Marshall Division, January 2014 – January 2015.
- Expert Witness, *Pulse Oximetry Signal Processing*, Mayor Brown LLP through IMS Expert Services, Masimo Corporation v. **Philips Electronics North America Corporation and Philips Medizin Systeme Boblingen GMBH**, Civil Action No. 1:09-cv-00080-LPS-MPT, Filed Feb. 3, 2009, United States District Court for the District of Delaware, August 2013 – November 2016. Various reports. Deposition – New York, New York, May 27, 2016.

EXPERT WITNESS REPORTS, DECLARATIONS, AND DIRECT TESTIMONY

- Warren, Steve. “Reply Expert Report of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEx), June 20, 2025, 233 pages.
- Warren, Steve. “Rebuttal Expert Report of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEx), May 30, 2025, 45 pages plus appendices.
- Warren, Steve. “Opening Expert Report of Steven Warren, Ph.D. Regarding U.S. Patent Nos. 8,457,703 and 10,433,776,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEx), April 11, 2025, 536 pages plus appendices.
- Warren, Steve. “Rebuttal Expert Report of Dr. Steven Warren Regarding Non-Infringement of U.S. Patent Nos. 8,417,326 and 9,986,921,” The Research Foundation for the State University of New York, University of Connecticut, and Worcester Polytechnic Institute v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., and Samsung Semiconductor, Inc., U.S District Court for the Eastern District of Texas, Marshall Division, Case No. 2:23-CV-00141-RWS-RSP, February 21, 2025, 113 pages plus exhibits.
- Warren, Steve. “Expert Report of Dr. Steven Warren Regarding Invalidity of U.S. Patent Nos. 8,417,326 and 9,986,921,” The Research Foundation for the State University of New York, University of Connecticut, and Worcester Polytechnic Institute v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., and Samsung Semiconductor, Inc., U.S District Court for the Eastern District of Texas, Marshall Division, Case No. 2:23-CV-00141-RWS-RSP, January 31, 2025, 189 pages plus exhibits.
- Warren, Steve. “Direct Testimony of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEx), October 23, 2024, 71 pages.
- Warren, Steve. “Declaration of Steven Warren, Ph.D.,” Abbott Laboratories and Abbott Diabetes Care Inc., Petitioner v. GenTag, Inc. and Altivera LLC, Patent Owner., United States Patent and Trademark Office, Patent Trial and Appeal Board, Case No. IPR2024-00425, Patent No. 7,969,307 B2, October 15, 2024, 99 pages.
- Warren, Steve. “Declaration of Steven Warren, Ph.D.,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, February 19, 2024, 512 pages.
- Warren, Steve. “Reply Expert Report of Steven Warren, Ph.D.,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, January 5, 2024, 564 pages.
- Warren, Steve. “Rebuttal Expert Report of Steven Warren, Ph.D.,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, December 15, 2023, 552 pages.
- Warren, Steve. “Expert Report of Steven Warren, Ph.D. Regarding Invalidity of U.S. Patent Nos. 8,190,223, 10,736,507, & 10,984,911,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, November 22, 2023, 1,011 pages.

Steve Warren, Ph.D.

Expert Witness Engagements and Reports, June 2026

- Warren, Steve. “Sur-Reply Declaration of Steven Warren, Ph.D. Regarding Claim Construction Brief,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, August 24, 2023, 24 pages.
- Warren, Steve. “Declaration of Steven Warren, Ph.D. Regarding Plaintiff Apple Inc.’s Reply Claim Construction Brief,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, August 17, 2023, 347 pages.
- Warren, Steve. “Declaration of Steven Warren, Ph.D. Regarding Claim Construction,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, August 10, 2023, 171 pages.
- Warren, Steve. “Supplemental Rebuttal Expert Report of Steve Warren, Ph.D. Regarding Certain of Plaintiffs’ Alleged Technical Trade Secrets,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEX), December 2, 2022, 11 pages.
- Warren, Steve. “Rebuttal Expert Report of Steve Warren, Ph.D. Regarding Certain of Plaintiffs’ Alleged Technical Trade Secrets,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEX), November 4, 2022, 395 pages. Appendices: 111 pages.
- Warren, Steve. “Expert Report of Steve Warren, Ph.D., Regarding Certain of Plaintiffs’ Alleged Technical Trade Secrets,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEX), September 23, 2022, 442 pages.
- Warren, Steve. “Supplement to Expert Report of Dr. Steven Warren Regarding Invalidity for U.S. Patent Nos. 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 25, 2022, 6 pages.
- Warren, Steve. “Rebuttal Expert Report of Dr. Steven Warren Regarding Non-Infringement of U.S. Patent Nos. 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 23, 2022, 177 pages.
- Warren, Steve. “Rebuttal Expert Report of Steven Warren, Ph.D. and Majid Sarrafzadeh, Ph.D. Regarding the Domestic Industry Technical Prong for U.S. Patent Nos. 10,687,745, 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 23, 2022, 401 pages.
- Warren, Steve. “Amendment to Expert Report of Dr. Steven Warren Regarding Invalidity for U.S. Patent Nos. 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 21, 2022, 6 pages.
- Warren, Steve. “Expert Report of Dr. Steven Warren Regarding Invalidity for U.S. Patent Nos. 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 4, 2022, 858 pages.
- Warren, Steve. “Rebuttal Expert Claim Construction Report of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, January 25, 2022, 14 pages.
- Warren, Steve. “Initial Expert Claim Construction Report of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, January 18, 2022, 100 pages.
- Warren, Steve. “Declaration of Dr. Steve Warren in Support of Apple’s Portion of the Joint Stipulation Regarding Plaintiffs’ Motion for Protective Order,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., Case No. 8:20-cv-00048-DOC-DFM, November 9, 2020, 79 pages.
- Warren, Steve. “Rebuttal Witness Statement of Steve Warren,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, September 18, 2020, 52 pages.

- Warren, Steve. “Direct Witness Statement of Steve Warren,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, September 8, 2020, 74 pages.
- Warren, Steve. “Rebuttal Expert Report of Steve Warren Regarding the Validity of U.S. Patent No. 9,820,698,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, August 7, 2020, 86 pages.
- Warren, Steve. “Supplemental Expert Report of Steve Warren Concerning Infringement by Respondents and the Technical Prong of the Domestic Industry Requirement,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, July 31, 2020, 42 pages.
- Warren, Steve. “Expert Report of Steve Warren Concerning Infringement by Respondents and the Technical Prong of the Domestic Industry Requirement,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, July 22, 2020, 350 pages.
- Warren, Steve. “Expert Report of Steven Warren, Ph.D. Concerning the Technical Prong of Domestic Industry of U.S. Patent Nos. 9,724,045 and 9,750,460,” Dexcom, Inc. v. AgaMatrix, Inc., ITC Inv. No. 337-TA-1075, April 26, 2018, 198 pages.
- Warren, Steve. “Reply Expert Report of Dr. Steve Warren Regarding Masimo’s Expert Report of Joe Kiani Dated April 29, 2016,” Masimo Corporation v. Philips Electronics North America Corporation and Philips Medizin Systeme Böblingen Gmbh, May 23, 2016, 11 pages.
- Warren, Steve. “Reply Expert Report of Dr. Steve Warren Regarding the Invalidity of U.S. Patent No. 7,509,154,” Masimo Corporation v. Philips Electronics North America Corporation and Philips Medizin Systeme Böblingen Gmbh, April 29, 2016, 48 pages.
- Warren, Steve. “Expert Report of Dr. Steve Warren Regarding the Noninfringement of U.S. Patent No. 7,509,154,” Masimo Corporation v. Philips Electronics North America Corporation and Philips Medizin Systeme Böblingen Gmbh, March 28, 2016, 72 pages.
- Warren, Steve. “Expert Report of Dr. Steve Warren Regarding the Invalidity of U.S. Patent No. 7,509,154,” Masimo Corporation v. Philips Electronics North America Corporation and Philips Medizin Systeme Böblingen Gmbh, February 12, 2016, 101 pages.

EXHIBIT 3

Julia Rowe

Vice President

MA, Accountancy
DePaul University

BA, Business Administration
University of Michigan

Julia Rowe brings a unique blend of public policy, valuation, and litigation expertise to the valuation of intellectual property. She has provided deposition and trial testimony and advises clients in a wide range of settings—including licensing negotiations, business strategy, financial reporting, the International Trade Commission, Federal Court, State Court and Arbitration. Julia has performed business and asset valuations, economic and financial analyses, and licensing-related consulting services. Her industry experience includes computer hardware and software, e-commerce, medical devices, telecommunications, and much more.

In addition to her deposition and testimony experience, Julia has also served as a consultant through trial and jury verdict in several high-profile cases involving patent infringement, trademark infringement, copyright infringement, and theft of trade secrets. She has extensive experience analyzing international trade issues (including public interest, domestic industry and bonding), lost profits, reasonable royalties, and unjust enrichment damages. Julia has also valued privately-held interests and has served in a variety of commercial litigation matters involving breach of contract, unfair competition, and corporate fraud.

Prior to joining CRA, Julia was a consultant at Kroll. She has a bachelor's degree in business administration from the University of Michigan and a master of accountancy degree from DePaul University. Julia is a registered CPA in Illinois.

Speeches, Presentations & Publications

"What Every IP Lawyer Needs to Know International Licensing," ACI's 5th Annual Summit for Women Leaders in IP Law, June 2025

"Valuing Intellectual Property and Intangible Assets." In *The Lawyer's Business Valuation Handbook* (chapter 28), August 2024 (co-author).

"Advanced Damages Techniques," 2023 ABA-IPL Section Annual Meeting, April 2023

"Strategies for Realizing Value from Patents Without Litigation," Continuing Legal Education Program for the Georgia Bar Patent Committee, January 2021

"ITC Investigations: Economic Issues" Continuing Legal Education Program for Faegre Drinker Biddle & Reath LLP, November 2020

"IP Value And the Benefits Of Cross-Department Collaboration," Duff & Phelps Webinar, November 2019

"IP Of the Future: Issues For Disruptive Technologies," Duff & Phelps 2018 IP Value Summit, November 2018

“The Growing Role Of Economics In IP-based Investigations At the ITC,” Annual Technology & Law Seminar, Kirkland Ellis LLP-Chicago, May 2018

“Hot Topics In Patent Licensing And Litigation,” Licensing Executives Society, Silicon Valley Chapter Meeting, March 2018

“Competitors And the Right To Exclude,” Duff & Phelps 2017 IP Value Summit, December 2017

“Connecting DC With Wall Street—How IP Investors View the Legal And Policy Landscape,” Intellectual Asset Management Patent Law and Policy Conference, November 2016

“Remedies In the Defend Trade Secrets Act,” Association of Corporate Counsel (Nationwide teleconference), July 2016

“Valuing Intellectual Property & Trade Secrets For Financial Reporting Purposes,” Michigan Association of Certified Public Accountants, May 2015

“Eastern District Of Texas (EDTX) General Order Regarding Alternative Procedure for Case Management,” Duff & Phelps IP Client Update, February 2014

“The Use Of Surveys For U.S. Patent Litigation,” Continuing Legal Education Program for Covington & Burling LLP (Nationwide teleconference), October 2013

“Recent Trends In IP Damages Law,” Licensing Executives Society, Inc. Chicago Chapter Meeting, June 2013

“Recent Trends In IP Damages Law and International Trade Commission Issues,” Continuing Legal Education Program For Duane Morris (Boston), October 2012

“Valuing Limited Partnership Interests,” Trusts & Estates, October 2010

Professional & Business History

2024–Present	<i>Vice President</i> , Charles River Associates, Chicago, IL
2011–May 2024	<i>Managing Director</i> , Kroll (formerly Duff & Phelps) <i>Managing Director</i> , 2021–2024 <i>Director</i> , 2013–2021 <i>Vice President</i> , 2011–2013 <i>Senior Associate</i> , 2011
2009–2011	<i>Associate</i> , CRA International, Chicago, IL <i>Analyst</i> , 2008–2009

Professional Associations and Certifications

Certified Public Accountant (CPA), Registered in Illinois

Certified Licensing Professional (CLP)

Member, Illinois CPA Society, Licensing Executives Society, Chicago Women in IP (“ChiWIP”)

EXHIBIT 4

REDACTED IN ITS ENTIRETY

CERTIFICATE OF SERVICE

Investigation No. 337-TA-1478

The undersigned certifies that on August 10, 2026, true and correct copies of the foregoing document were served upon the following parties as indicated below:

The Honorable Lisa R. Barton Secretary to the Commission U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, SW, Room 112 Washington, DC 20436	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☐ Via Electronic Mail
The Honorable Doris Johnson Hines Administrative Law Judge U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, S.W. Washington, D.C. 20436 Email: JohnsonHines1478@usitc.gov	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☒ Via Electronic Mail
<i>Counsel for Respondents Ouraring Inc. and Oura Health Oy</i> Jasjit S. Vidwan James A. Fussell, III Saqib J. Siddiqui Bryan C. Nese MAYER BROWN LLP 199 K. Street NW Washington, DC 20006 Tel: 202-263-5282 Robert G. Pluta Nicholas H. Ciulla MAYER BROWN LLP 71 S. Wacker Drive Chicago, IL 60606 Tel: 312-782-0600 Email: Oura-ITC@mayerbrown.com	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☐ Via Electronic Filing (EDIS) ☒ Via Electronic Mail

/s/ Brian A. Treggs

Brian A. Treggs
Senior IPT Specialist
O'Melveny & Myers LLP
400 South Hope Street
Los Angeles, CA 90071

CERTIFICATE OF SERVICE

Investigation No. 337-TA-1478

The undersigned certifies that on August 21, 2026, true and correct copies of the foregoing document were served upon the following parties as indicated below:

The Honorable Lisa R. Barton Secretary to the Commission U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, SW, Room 112 Washington, DC 20436	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☐ Via Electronic Mail
The Honorable Doris Johnson Hines Administrative Law Judge U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, S.W. Washington, D.C. 20436 Email: JohnsonHines1478@usitc.gov	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☒ Via Electronic Mail
<i>Counsel for Respondents Ouraring Inc. and Oura Health Oy</i> Jasjit S. Vidwan James A. Fussell, III Saqib J. Siddiqui Bryan C. Nese MAYER BROWN LLP 199 K. Street NW Washington, DC 20006 Tel: 202-263-5282 Robert G. Pluta Nicholas H. Ciulla MAYER BROWN LLP 71 S. Wacker Drive Chicago, IL 60606 Tel: 312-782-0600 Email: Oura-ITC@mayerbrown.com	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☐ Via Electronic Filing (EDIS) ☒ Via Electronic Mail

/s/ Brian A. Treggs

Brian A. Treggs
Senior IPT Specialist
O'Melveny & Myers LLP
400 South Hope Street
Los Angeles, CA 90071